

Physics Calc

Mathematical Foundations, Classical Theory, and Modern Physics

Physics and Programming

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Learning physics by building the argument

This book is meant to be read with a pencil nearby. An equation is most useful when you can explain what problem it answers, where its assumptions enter, and how its mathematical steps follow from a physical starting point. Our aim is to build that understanding gradually, from measurements and derivatives to fields, quantum states, and the universe as a whole.

As you work through a section, begin by picturing the system. Choose the quantity you want to predict and the variables that describe it. Then identify the principle that connects them: force balance, conservation, stationary action, or a stated quantum postulate. The algebra is the bridge from that starting point to a prediction. After crossing it, return to the physics: check dimensions, signs, limiting cases, and whether the assumptions still fit the situation.

The chapters retain compact equations so you can return to them as a reference, but the surrounding discussion develops the route into and out of the central results. You will see some methods repeatedly: Taylor expansion near equilibrium, integration of a local balance, separation into normal modes, and matching boundary conditions. Repetition is useful here. It reveals how much of physics can be understood through a small collection of well-chosen arguments.

Not every formula has the same status. Some are definitions, some are empirical or theoretical starting laws, and others are consequences we derive. Advanced topics occasionally require results whose full derivations belong to a longer course. Where that happens, the discussion identifies the starting structure, works through an accessible consequence, and indicates the additional work required. You should never have to mistake an unexplained quotation for a derivation.

Read a derivation once for the physical story, then cover the next line and try to supply it yourself. If a step seems mysterious, ask whether the missing ingredient is an algebraic identity, a boundary condition, or an assumption about the system. Those are different obstacles and call for different remedies. The review problems at the end provide another chance to reconstruct the argument without its intermediate steps already in front of you.

A slower rhythm for every unit

The units are intentionally layered. Each one begins with an observable physical situation and a picture, introduces only the notation needed for that situation, works through a small model in full,

and then returns to the same idea at greater depth. A reader should be able to pause after the first layer and still understand the physical question; later layers add precision rather than repairing an unexplained jump.

Use this four-step rhythm throughout the book:

1. **See it:** identify the objects, measurements, symmetries, and limiting cases in a visual model.
2. **Name it:** define the variables, units, mathematical operation, and assumptions.
3. **Derive it:** work from the local rule to a prediction, checking dimensions and boundary conditions.
4. **Test it:** change one parameter, compare with a simulation or experiment, and record what would falsify the model.

Chapters may therefore become longer as the subject becomes more advanced. Length is less important than giving each new abstraction enough physical and mathematical repetitions to become familiar.

A note on notation

Symbols are defined locally because different subjects traditionally reuse the same letters. For example, μ can denote reduced mass, linear density, or chemical potential; k can denote a spring constant or wave number. Context and units distinguish them. A dot usually denotes a time derivative, a prime a derivative of a one-variable function, and bold symbols denote spatial vectors. The upright d marks an ordinary differential. When an approximation is made, look for its stated regime before applying it elsewhere.

Every substantial equation should be read in three passes. First ask what physical relationship the equation states. Next identify what every symbol means, including its units, sign convention, and whether it is fixed or changing. Finally ask what happens when one parameter is increased, set to zero, or taken to a limiting value. The prose will define symbols before or immediately after they appear; a reader should never have to infer the meaning of a symbol from typography alone.

Where to begin and how to continue

You can begin this book knowing only basic algebra, graphs, and the idea of an angle. No calculus or previous physics course is assumed. If a symbol is unfamiliar, read its nearby definition and work through the example before trying to memorize it. The first four chapters use everyday motion to introduce measurement, slopes, areas, arrows, and forces. They are a complete first route through the subject: pause there and practice until you can explain each idea in words and with a sketch.

The next four chapters revisit measurement and build a mathematical toolkit. Measurement, derivatives, and spatial geometry support the classical mechanics chapters that follow. Modes, differential equations, and probability become increasingly useful in later parts of the book. On a first reading, you can leave the sections on tensors, Green functions, and entropy maximization until a later chapter points you back to them. The subject parts then grow from mechanics and waves to heat, electricity, relativity, quantum physics, and applications. The advanced chapters assume the earlier physical ideas even when their mathematics is new.

At the end of a chapter, try its short questions before reading on. If you cannot answer one, revisit the worked example and draw the situation yourself. The review problems near the end of the book combine ideas from several chapters. Each pass is meant to increase what you can explain and predict, rather than simply increase the number of formulas you have seen.

Part I

First Steps: Physics and Mathematics

Chapter 1

What a physical model is

Physics begins with a concrete question: how far did something move, how long did it take, or how much energy changed hands? We decide what to measure, what can be ignored for now, and what observation would tell us whether a model worked. Then we use mathematics to make a prediction and return to the measurement to test it. Every later chapter repeats this process with more powerful tools.

1.1 Quantities, units, and estimates

A physical quantity combines a number with a unit. Five meters and five seconds have different meanings even though both begin with the number five. A unit conversion changes the number and the unit together without changing the physical quantity:

$$3.0 \text{ m} = 300 \text{ cm}, \quad 2.0 \text{ min} = 120 \text{ s}. \quad (1.1)$$

Before calculating, estimate the answer. A person walks roughly one meter per second, a classroom is a few meters across, and a car travels tens of meters in a second on a highway. Estimates catch missing factors and impossible answers before detailed algebra hides them. We will use x for a measured position and t for time when we study motion.

Worked example 1.1 (A walk across a room). Suppose a room is about 6 m long and you cross it in about 5 s. Distance divided by time gives an average speed of $6/5 \approx 1.2 \text{ m/s}$. The result has

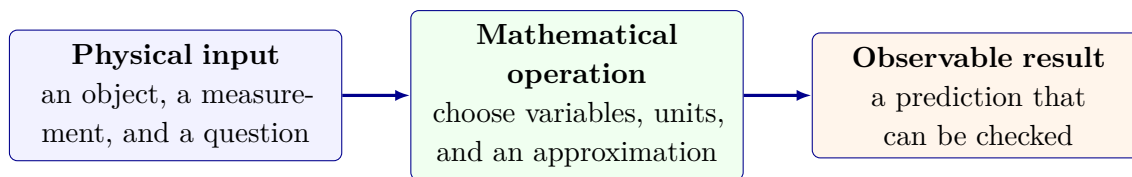


Figure 1.1: Visual model for the first physical model: the physical question becomes mathematics and returns as an observable prediction.

units of length per time, which is exactly what speed should have. Walking faster for part of the trip and slower for another part would leave this average unchanged, so the model does not tell us your speed at each moment.

1.2 Graphs and proportional reasoning

A graph is a picture of how one quantity changes with another. A straight line through the origin represents direct proportionality, $y = Cx$, where C is the constant that sets the line's steepness. Doubling x doubles y . A horizontal line means that the plotted quantity remains constant, while a curved line tells us that the rate of change itself is changing.

For a pendulum in the small-angle regime, dimensional reasoning suggests $T \propto \sqrt{\ell/g}$. This is not yet a complete derivation, but it already predicts that a longer pendulum swings more slowly and a stronger gravitational field makes it swing faster. Good physics often begins with this kind of reliable partial answer.

1.3 A first model and its limits

A model is a deliberately simplified description. Treating a thrown ball as a point particle ignores its size and spin; ignoring air resistance removes a force; assuming constant gravity limits the height range. These choices are useful when they are stated and tested. A result becomes unreliable when it is used outside the regime that supported it. For motion along a line, choose a positive direction before assigning signs: a negative displacement then means motion opposite to that choice.

At the end of every first calculation, ask four questions: Are the units correct? Is the sign sensible? Does the result behave correctly when a parameter becomes very large or very small? What effect did the model leave out?

Try it yourself

A friend walks 12 m in 10 s. Estimate the average speed and give its units. Then name one reason that this number might not describe the friend's speed at every instant. For a graph $y = Cx$, sketch what happens when C doubles.

Chapter 2

Change and accumulation

The previous chapter turned an observation into a model. To describe something that changes, we now need two ideas from graphs: slope tells us how fast a quantity changes, and area helps us add small changes into a total. We will learn these ideas through motion along a line before using them for fields and several coordinates later.

2.1 Average and instantaneous change

Let $x(t)$ be a cyclist's position at time t . If the position changes by Δx during a time Δt , the average velocity is $\Delta x/\Delta t$. To ask for the signed velocity $v(t)$ at one instant, examine smaller and smaller intervals around that instant:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{dx}{dt}. \quad (2.1)$$

On a graph, this is the slope of the tangent line. A steep slope means rapid motion; a negative slope means that the chosen coordinate is decreasing.

Worked example 2.1 (Reading a slope before taking a limit). If a position graph goes from $(t, x) = (0 \text{ s}, 2 \text{ m})$ to $(4 \text{ s}, 10 \text{ m})$ along a straight line, its slope is $(10 - 2)/(4 - 0) = 2 \text{ m/s}$. The line has the same slope everywhere, so its instantaneous velocity is also 2 m/s . A curved position graph can have the same average slope between these endpoints but a different slope at each instant.

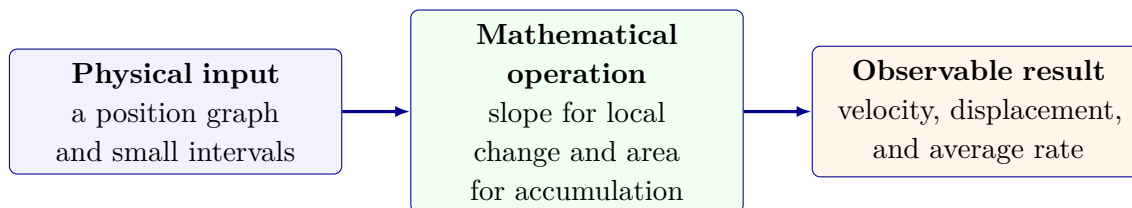


Figure 2.1: Visual model for change and accumulation: the physical question becomes mathematics and returns as an observable prediction.

Acceleration, written $a(t)$, is the rate at which velocity changes:

$$a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2}. \quad (2.2)$$

The second derivative is not “distance traveled twice.” It measures how the velocity changes. A car moving at constant speed around a bend has acceleration because its velocity direction changes even when its speed does not.

2.2 Area under a graph

If velocity is known, a short time interval contributes approximately $v(t)dt$ to the displacement. Adding all those signed contributions from an initial time t_0 gives

$$x(t) - x(t_0) = \int_{t_0}^t v(s)ds. \quad (2.3)$$

The integral is the signed area under the velocity-time graph. Motion in the negative direction contributes negative area, so displacement and total distance are different quantities. The starting position $x(t_0)$ tells us where to place the entire motion on the position axis.

2.3 The first differential equation

An equation such as $\dot{x} = v$ is a rule for how a quantity changes. It does not determine one unique motion until an initial value is supplied. If v is constant and $x(0) = x_0$, integration gives

$$x(t) = x_0 + vt. \quad (2.4)$$

This is the basic pattern of an initial-value problem: a local rule plus enough starting information determines a particular history.

Try it yourself

A cyclist’s position changes from 2 m to 14 m in 4 s. Find the average velocity. Sketch a position graph that has this average velocity but is steeper near the end. If $v(t) = -2$ m/s for three seconds, what is the signed displacement?

Chapter 3

Vectors and first motion models

Along a straight line, a signed number can tell us where an object is. On a map, we need to say how far east or west and how far north or south. An arrow, called a vector, carries both size and direction. We will use vectors to describe motion in a plane, then meet them again when we study forces, fields, and fluids.

3.1 Components and directions

Choose three perpendicular axes. The position vector $\mathbf{r}$ points from the origin to an object. Its components x , y , and z tell us how far to move along each axis:

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}. \quad (3.1)$$

Here $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are unit arrows along the axes. We add vectors component by component. The vector stays the same if we rotate the axes, although its components change. Its length is

$$|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}, \quad (3.2)$$

For two vectors, the dot product $\mathbf{a} \cdot \mathbf{b}$ measures how much they point along the same direction. It is zero when the vectors are perpendicular.

Worked example 3.1 (Two steps in different directions). Walk 3 m east and then 4 m north. Your displacement is the arrow $(3, 4)$ m. Its length is $\sqrt{3^2 + 4^2} = 5$ m, although you walked a total

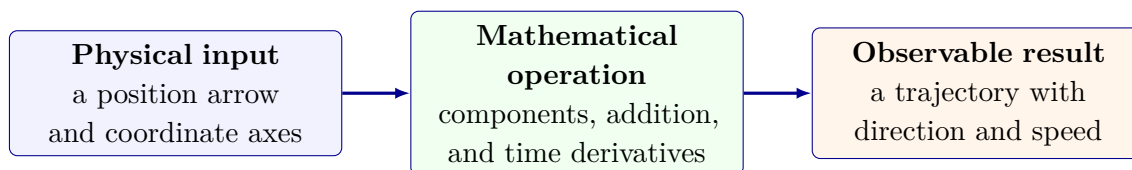


Figure 3.1: Visual model for vectors and motion: the physical question becomes mathematics and returns as an observable prediction.

distance of 7 m. Distance adds the lengths of the path segments; displacement joins the starting point to the ending point.

When the position vector changes with time, its rate of change is the velocity vector $\mathbf{v}$. The rate of change of velocity is the acceleration vector $\mathbf{a}$:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \mathbf{a} = \frac{d\mathbf{v}}{dt}. \quad (3.3)$$

The components can change independently. In projectile motion, horizontal velocity remains constant in the simplest model while vertical velocity changes because gravity acts downward.

3.2 Constant-acceleration motion

For constant acceleration, integrate $\dot{\mathbf{v}} = \mathbf{a}$ and then integrate velocity:

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}t, \quad (3.4)$$

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0t + \frac{1}{2}\mathbf{a}t^2. \quad (3.5)$$

These equations are not universal motion formulas. They apply only while acceleration is constant. Their value is that they let a beginner see how a local rule becomes a trajectory.

3.3 A first visual experiment

Launch a projectile with a chosen initial speed and angle. Display its position, velocity arrow, and acceleration arrow at the same time. The path curves because the velocity changes; the acceleration arrow remains constant in the ideal model. Then add drag and observe precisely which assumption stops working.

Try it yourself

An object begins with horizontal velocity 3 m/s and zero vertical velocity. With vertical acceleration -10 m/s^2 , find its velocity components after 2 s. Sketch the velocity and acceleration arrows. Explain why the horizontal component did not change in this model.

Chapter 4

First forces and oscillations

The earlier chapters described motion. We now ask what changes it. A force changes an object's momentum, and Newton's law connects the total force to its acceleration. A spring will give us a first example of how that local rule predicts an entire motion.

4.1 Newton's second law

Draw every force acting on a constant-mass object and add their arrows to get the net force $\mathbf{F}_{\text{net}}$. If its mass is m and acceleration is $\mathbf{a}$, Newton's second law says

$$\mathbf{F}_{\text{net}} = m\mathbf{a}. \quad (4.1)$$

The force diagram helps us choose directions before writing components. It also prevents us from assuming, without checking, that the normal force must equal the weight.

Worked example 4.1 (Pushing a cart). Push a 2 kg cart to the right with 8 N while friction acts to the left with 2 N. Choosing right as positive, the net horizontal force is $8 - 2 = 6$ N. Newton's law gives $a = 6/2 = 3 \text{ m/s}^2$ to the right. If friction grew to 8 N, the net force and acceleration would be zero, even though you would still be pushing.

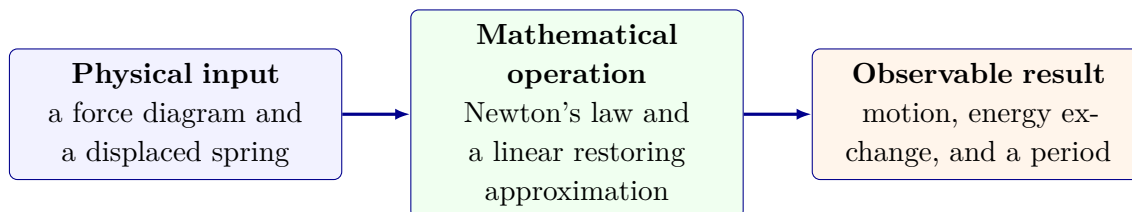


Figure 4.1: Visual model for first forces and oscillations: the physical question becomes mathematics and returns as an observable prediction.

4.2 Work and energy

Dot Newton's law with displacement velocity to obtain

$$\mathbf{F} \cdot \mathbf{v} = \frac{d}{dt} \left(\frac{1}{2} m v^2 \right). \quad (4.2)$$

After integration, work changes kinetic energy. If a force comes from a time-independent potential, $E = K + V$ is constant. Energy is useful because it can answer a speed question without solving the entire trajectory.

4.3 The spring as a first oscillation

Measure the displacement x from the spring's relaxed position and choose positive x to point away from it. The spring pulls back with force $F = -kx$, where $k > 0$ is its stiffness. The minus sign says the force points toward the relaxed position. Newton's law then becomes

$$m\ddot{x} + kx = 0, \quad \omega = \sqrt{k/m}, \quad x(t) = A \cos(\omega t + \phi). \quad (4.3)$$

The solution says that the displacement repeats: A is the greatest distance from equilibrium, ω sets the rate of oscillation, and ϕ sets where in the cycle the motion begins. Kinetic and potential energy trade places as the spring moves. Later, the same pattern will describe a circuit, a wave mode, and a molecular vibration.

4.4 Checkpoint before the deeper pass

At this point a reader should be able to interpret a graph's slope, calculate an area, add vectors, draw a force diagram, use Newton's law, and explain why a spring oscillates. The following foundations chapters develop these tools more deeply, but the first four chapters already give a complete introduction to the basic physical argument.

Try it yourself

A 2 kg cart experiences a net horizontal force of 6 N. Draw the force direction and find its acceleration. If the cart starts at rest and the force remains constant for 3 s, find its speed. Then explain in words why the formula for a spring force contains a minus sign.

Part II

Mathematical and Physical Foundations: Second Pass

Chapter 5

Scales, dimensions, and measurement

The first pass showed how a model grows from an observation. We now slow down and make the checks behind that process systematic: dimensions, scale, approximation, and uncertainty. These habits will let us recognize when the more advanced mathematics later in the book is being used responsibly. Before we describe how anything moves, let us ask what a physical prediction actually contains. A number alone is rarely enough: we need a unit, an estimate of uncertainty, and a statement of the conditions under which the number means something. These habits will guide every calculation in this book. We begin with dimensions because they let you question an equation even before you know how to solve it.

5.1 Physical quantities and dimensional analysis

Imagine predicting how long a pendulum takes to swing. You could begin by solving its equation of motion, but you already know something valuable: the answer must have the dimensions of time. Our first task is to use that requirement to narrow the possible answers. Keep the distinction between a dimension and a unit in mind: a distance can be measured in meters or kilometers, while remaining a length in either case. M, L, T denote the base dimensions mass, length, and time; square brackets such as $[v]$ mean “dimensions of.” Dimensionless ratios such as v/c or θ control approximations.

A physical quantity is a number times a unit. Its dimension is independent of the chosen unit system. The mechanical base dimensions are mass M , length L , and time T ; SI also includes

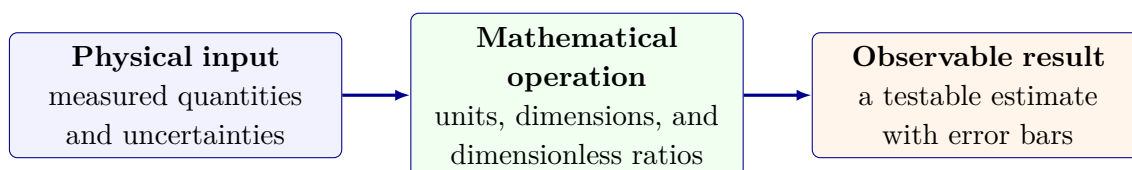


Figure 5.1: Visual model for scales and measurement: the physical question becomes mathematics and returns as an observable prediction.

electric current, temperature, amount of substance, and luminous intensity. For example,

$$[v] = LT^{-1}, \quad [a] = LT^{-2}, \quad [F] = MLT^{-2}, \quad [E] = ML^2T^{-2}. \quad (5.1)$$

Only quantities with equal dimensions may be added. Arguments of exponentials, logarithms, and trigonometric functions are dimensionless. Dimensional consistency is necessary but does not determine dimensionless coefficients.

If a problem contains n dimensional variables constructed from r independent base dimensions, Buckingham's Π theorem generally permits $n - r$ independent dimensionless combinations. These organize similarity experiments and identify competing effects.

Worked example 5.1 (Pendulum scaling). Assume a small-angle pendulum period depends only on length ℓ and gravitational acceleration g . Writing $\tau = C\ell^a g^b$ gives $a + b = 0$ and $-2b = 1$, so $\tau = C\sqrt{\ell/g}$. Dynamics supplies $C = 2\pi$. At finite amplitude, the release angle is another dimensionless variable, and C depends on it.

Let us unpack the pendulum argument rather than treating the exponents as a guess. Because $[\ell] = L$ and $[g] = LT^{-2}$, the proposed expression has dimensions

$$[\ell^a g^b] = L^a (LT^{-2})^b = L^{a+b} T^{-2b}. \quad (5.2)$$

Matching this with $[\tau] = L^0 T^1$ gives two equations, one for each independent dimension. From $-2b = 1$ we get $b = -1/2$; then $a + b = 0$ gives $a = 1/2$. We have learned that quadrupling the length doubles the period and quadrupling gravity halves it. What we have not learned is the numerical factor C . That missing information is exactly what a dynamical model must supply. Later, the small-angle equation of motion will complete the argument.

5.2 Orders of magnitude and approximation

Most useful physics begins by deciding which effects are small enough to set aside. The word “small” must refer to a comparison: a speed of one kilometer per second is large on a road but small compared with the speed of light. Form a dimensionless ratio first; then use an expansion to see what your approximation discards.

An approximation is usually controlled by a dimensionless parameter. Examples are $v/c \ll 1$ for Newtonian kinematics, $\theta \ll 1$ for $\sin \theta \simeq \theta$, and a wavelength much shorter than an apparatus for ray optics. Taylor expansion makes the neglected terms explicit:

$$f(x + \delta) = f(x) + \delta f'(x) + \frac{\delta^2}{2} f''(x) + O(\delta^3). \quad (5.3)$$

Keeping one correction can reveal when a leading model fails. For example, $\gamma = (1 - v^2/c^2)^{-1/2} = 1 + v^2/(2c^2) + 3v^4/(8c^4) + \dots$.

For a pendulum, the relevant expansion is

$$\sin \theta = \theta - \frac{\theta^3}{6} + O(\theta^5), \quad \frac{\theta - \sin \theta}{\theta} = \frac{\theta^2}{6} + O(\theta^4). \quad (5.4)$$

The second expression answers a more useful question than simply asking whether θ is small: it estimates the fractional error in replacing the restoring factor $\sin \theta$ by θ . Angles must be in radians. At $\theta = 0.1$, the leading relative correction is about 0.0017. This estimates the local restoring-force error; the accumulated error in a trajectory or period requires its own analysis.

5.3 Uncertainty and data

Suppose you repeat a timing measurement and obtain slightly different answers. Averaging helps with independent fluctuations, but it will not repair a clock that runs consistently slow. We therefore separate random variation from systematic error before using statistics. When measured quantities enter a formula, the next question is how sensitively the answer depends on each input. In uncertainty formulas, σ is a standard deviation and C_{ij} is a covariance; covariance can be positive or negative because two measured quantities may vary together or in opposition.

For N independent observations with finite variance, the sample mean is $\bar{x} = N^{-1} \sum_i x_i$ and its estimated standard error is $s/\sqrt{N}$, where

$$s^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2. \quad (5.5)$$

This does not remove calibration errors or other systematic biases. For a differentiable function $f(\mathbf{x})$ with small errors and covariance matrix C ,

$$\sigma_f^2 \simeq \sum_{ij} \frac{\partial f}{\partial x_i} C_{ij} \frac{\partial f}{\partial x_j}. \quad (5.6)$$

For independent inputs, the off-diagonal terms vanish. A least-squares fit with independent Gaussian errors minimizes $\chi^2 = \sum_i [y_i - f(x_i; \mathbf{a})]^2 / \sigma_i^2$. Good agreement is judged relative to uncertainties and degrees of freedom, not simply by how smooth a curve looks.

Common pitfall. Measurement uncertainty describes limited knowledge of a result; quantum uncertainty can remain even for an ideally prepared state and perfect apparatus. Neither means that physical predictions are arbitrary.

Write each input as its estimated value plus a small fluctuation, $x_i = \bar{x}_i + \delta x_i$. A first-order Taylor expansion gives

$$\delta f \simeq \sum_i f_i \delta x_i, \quad f_i = \left. \frac{\partial f}{\partial x_i} \right|_{\bar{\mathbf{x}}}. \quad (5.7)$$

Square this expression and average. With zero-mean fluctuations, $\langle \delta x_i \delta x_j \rangle = C_{ij}$, which produces the covariance formula above. Its cross terms describe inputs that move together. For example, if $v = d/t$ and the errors in d and t are independent,

$$\left(\frac{\sigma_v}{v} \right)^2 \simeq \left(\frac{\sigma_d}{d} \right)^2 + \left(\frac{\sigma_t}{t} \right)^2. \quad (5.8)$$

The minus sign in $\partial v / \partial t = -d/t^2$ disappears upon squaring. This is why independent fractional errors add in quadrature even when one quantity is in a denominator.

Try it yourself

A length of 2.0 m is measured with an uncertainty of 0.1 m. What is the fractional uncertainty? A timing device always reads 0.2 s too high. Explain why repeating the measurement does not remove that error.

Chapter 6

Calculus, vectors, and spatial geometry

The derivative and integral from the first pass now acquire companions. Vectors let change happen in space, while geometry tells us how to measure an area or volume. Read through coordinates on a first pass; the section on tensors prepares for later chapters on stress and spacetime. Calculus becomes much easier to remember when each operation answers a physical question. A derivative asks how something changes locally. An integral asks how many small contributions accumulate. Here we build that language for quantities spread through space, including the geometry needed when a system occupies more than a single line.

6.1 Derivatives, integrals, and conservation

Begin with a position record $x(t)$. Over a finite interval, $\Delta x/\Delta t$ is an average velocity. To define velocity at one instant, shrink the interval while keeping track of the limiting ratio:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}. \quad (6.1)$$

The derivative does not require the particle to move at constant speed. It gives the slope of the position curve precisely when that slope is changing. Once you know the slope at every instant, integration rebuilds the displacement, with one initial position needed to locate the entire trajectory.

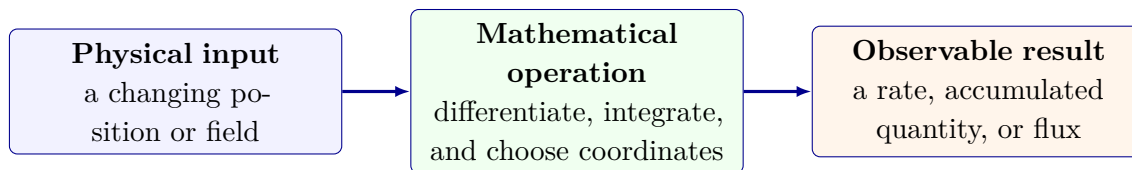


Figure 6.1: Visual model for calculus and spatial geometry: the physical question becomes mathematics and returns as an observable prediction.

f is a scalar field, $\mathbf{A}$ a vector field, and ∇ the spatial derivative operator. Divergence measures local outflow, curl measures local circulation, and $\nabla^2 = \nabla \cdot \nabla$ measures curvature or spreading. A field assigns a value to every point in the region being studied.

The derivative measures local change, while integration accumulates it:

$$v = \frac{dx}{dt}, \quad a = \frac{dv}{dt}, \quad x(t) = x(t_0) + \int_{t_0}^t v(s) ds. \quad (6.2)$$

For several variables, the total differential is $df = \sum_i (\partial f / \partial x_i) dx_i$. A derivative with some variables held fixed is physically different from one with other constraints; this matters especially in thermodynamics.

The gradient points toward fastest scalar increase; divergence measures local vector outflow; curl measures local circulation. In Cartesian coordinates,

$$\nabla f = (\partial_x f, \partial_y f, \partial_z f), \quad \nabla \cdot \mathbf{A} = \partial_i A_i, \quad (\nabla \times \mathbf{A})_i = \epsilon_{ijk} \partial_j A_k. \quad (6.3)$$

The Laplacian is $\nabla^2 f = \nabla \cdot \nabla f$. Useful identities are

$$\nabla \times \nabla f = 0, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0, \quad \nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (6.4)$$

The divergence and Stokes theorems turn local laws into global statements:

$$\int_V \nabla \cdot \mathbf{A} dV = \oint_{\partial V} \mathbf{A} \cdot d\mathbf{S}, \quad \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{A} \cdot d\boldsymbol{\ell}. \quad (6.5)$$

Orient the surface normal and boundary traversal consistently with the right-hand rule.

To see why divergence appears in conservation laws, imagine a tiny rectangular box. The net outward flow through its two x faces is approximately $[A_x(x + \Delta x) - A_x(x)]\Delta y\Delta z$. Divide by the box volume and let its size approach zero: the result is $\partial_x A_x$. Adding the three pairs of faces gives $\nabla \cdot \mathbf{A}$. The divergence theorem is the statement that these little balances add up; fluxes through internal shared faces cancel, leaving only the outer boundary.

In component notation, a repeated index is summed over the spatial coordinates: $\partial_i A_i$ means $\partial_x A_x + \partial_y A_y + \partial_z A_z$. The symbol ϵ_{ijk} is the antisymmetric Levi-Civita symbol, with $\epsilon_{123} = 1$. These compact symbols keep later field equations readable, but they represent ordinary sums of derivatives.

6.2 Coordinates and integration measures

Choose coordinates that follow the symmetry. A sphere is awkward to describe by many Cartesian slices and simple to describe by radius and two angles. The price of that convenience is that equal coordinate increments do not always enclose equal physical volumes. We must derive the little volume element before integrating.

Coordinate systems encode geometry, not new physics. Cylindrical volume is $dV = r dr d\phi dz$; spherical volume is $dV = r^2 \sin \theta dr d\theta d\phi$. The Jacobian cannot be omitted. For a spherically symmetric scalar,

$$\nabla^2 f(r) = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right). \quad (6.6)$$

A radial inverse-square field has zero divergence away from the origin, yet nonzero flux through an enclosing sphere. Its source at the origin is represented by a delta distribution.

In spherical coordinates, a small cell has side lengths dr , $r d\theta$, and $r \sin \theta d\phi$. Their product explains the factor $r^2 \sin \theta$. For a radial vector field $A_r(r) \hat{r}$, the outward flux through a sphere is $4\pi r^2 A_r$. The flux difference across a thin shell divided by its volume is therefore

$$\nabla \cdot \mathbf{A} = \lim_{\Delta r \rightarrow 0} \frac{4\pi[(r + \Delta r)^2 A_r(r + \Delta r) - r^2 A_r(r)]}{4\pi r^2 \Delta r} = \frac{1}{r^2} \frac{d}{dr} (r^2 A_r). \quad (6.7)$$

Putting $A_r = df/dr$ gives the radial Laplacian. The geometry has supplied the extra r^2 factors; they are not corrections to the physics.

6.3 Bases, inner products, and tensors

The components of a vector depend on the basis used to describe it, while the vector itself does not. In an orthonormal basis $\{\mathbf{e}_i\}$, write $\mathbf{v} = v^i \mathbf{e}_i$. If the basis changes by $\mathbf{e}'_a = R^i_a \mathbf{e}_i$, equality of the geometric vector requires

$$v'^a = (R^{-1})^a_i v^i. \quad (6.8)$$

This distinction becomes essential when comparing Cartesian, rotating, and curvilinear descriptions of the same physical quantity.

An inner product measures angles and lengths: $\mathbf{u} \cdot \mathbf{v} = u_i v_i$ in an orthonormal Euclidean basis. Orthogonal projection onto a unit direction $\hat{\mathbf{n}}$ is $(\mathbf{v} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}}$, and the residual is perpendicular to that direction. Least squares, Fourier coefficients, quantum amplitudes, and normal-mode decompositions all use this projection idea.

A tensor is a multilinear map whose components transform according to the transformations of its indices. For a change of basis, a $(1, 1)$ tensor obeys

$$T'^a_b = (R^{-1})^a_i T^i_j R^j_b. \quad (6.9)$$

The inertia tensor, stress tensor, and electromagnetic field tensor are different physical examples of this same transformation principle. Contracting one upper and one lower index produces a basis-independent quantity; this is why index notation is more than shorthand.

For a symmetric matrix, orthogonal eigenvectors provide principal directions. In a non-orthonormal coordinate basis, the metric $g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$ replaces the identity matrix, lengths become $\|\mathbf{v}\|^2 = g_{ij} v^i v^j$, and raising or lowering an index means contracting with g_{ij} or its inverse. These ideas will reappear as spacetime geometry rather than being introduced for the first time in relativity.

Try it yourself

If $f(x, y) = x^2 + y^2$, find $\partial f / \partial x$ and describe the direction in which f increases fastest at $(1, 0)$. Explain why a spherical volume element needs an r^2 factor. The tensor section can wait until you reach stress or relativity.

Chapter 7

Patterns, modes, and differential equations

The previous chapter described change in space. This one asks how a local rule determines a whole motion or field. Normal modes and complex numbers are useful shortcuts once the physical picture is clear. On a first pass, concentrate on the worked models; return to Green functions when later chapters need them.

7.1 Linear algebra and normal modes

When two objects are coupled, moving one generally changes the force on the other. Solving directly for every coordinate can hide a simpler description: some collective patterns keep their shape while their amplitudes oscillate. Finding those patterns is an eigenvector problem. An eigenvector gives a pattern that keeps its shape under a linear operation, and its eigenvalue gives the corresponding scale factor.

An eigenvector obeys $A\mathbf{v} = \lambda\mathbf{v}$. For real symmetric matrices, eigenvalues are real and an orthonormal eigenbasis exists. Hermitian complex matrices have the analogous property. Diagonalization separates coupled degrees of freedom.

For a quadratic mechanical system,

$$T = \frac{1}{2}\dot{\mathbf{q}}^T M \dot{\mathbf{q}}, \quad V = \frac{1}{2}\mathbf{q}^T K \mathbf{q}, \quad (7.1)$$

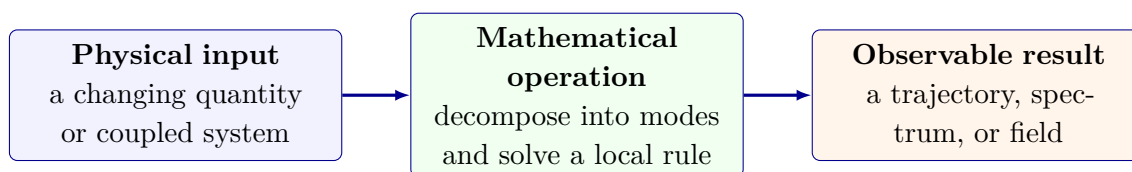


Figure 7.1: Visual model for patterns and equations: the physical question becomes mathematics and returns as an observable prediction.

the mode equation is $K\mathbf{a} = \omega^2 M\mathbf{a}$. Positive M and positive-definite K imply real nonzero frequencies and stable small oscillations. Zero eigenvalues often correspond to a symmetry; negative stiffness eigenvalues indicate instability.

For constant matrices M and K , the mechanical equations are $M\ddot{\mathbf{q}} + K\mathbf{q} = 0$. Try a motion in which every coordinate has the same frequency but a different amplitude:

$$\mathbf{q}(t) = \mathbf{a} \cos(\omega t + \phi) \implies (K - \omega^2 M)\mathbf{a} = 0. \quad (7.2)$$

A nonzero amplitude vector exists only when $\det(K - \omega^2 M) = 0$. Solve this equation for the allowed squared frequencies, then find the associated amplitude ratios. Finally, superpose the modes to meet the initial conditions. This sequence separates the properties of the system from the particular way you set it moving.

7.2 Complex numbers and Fourier analysis

A sine changes into a cosine when differentiated; a cosine changes back into a sine with a minus sign. Complex exponentials package that cycle into one operation. We use them because they simplify the work, and take the real part when reconstructing a real displacement or field. Fourier analysis extends the same idea: build a complicated shape from simple oscillating pieces. A complex exponential records oscillation compactly; the measured displacement or field is real.

Euler's identity $e^{i\theta} = \cos \theta + i \sin \theta$ converts sinusoidal differentiation into multiplication by $i\omega$. Physical fields are real parts of complex representations. With the convention

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk, \quad (7.3)$$

spatial differentiation becomes multiplication by ik . Fourier methods separate linear, translation-invariant differential equations into algebraic equations. Sharp localization requires a broad spectrum.

The delta distribution satisfies $\int f(x) \delta(x-a) dx = f(a)$ and $\delta(ax) = \delta(x)/|a|$ for nonzero real a . It represents an ideal point source or impulse, not an ordinary function with a finite height.

The Fourier differentiation rule follows from integration by parts. For a sufficiently decaying differentiable function,

$$\tilde{f}'(k) = \int f'(x) e^{-ikx} dx \quad (7.4)$$

$$= \left[f(x) e^{-ikx} \right]_{-\infty}^{\infty} + ik \int f(x) e^{-ikx} dx = ik \tilde{f}(k). \quad (7.5)$$

The boundary term vanishes under the stated assumption. A second derivative consequently becomes $-k^2 \tilde{f}$. That minus sign will turn a diffusion equation into exponential decay and a wave equation into oscillation. Keeping track of the transform convention makes these signs predictable.

7.3 Ordinary and partial differential equations

An equation of motion is a rule for generating a solution, not yet a complete solution. The equation $\ddot{x} = 0$, for example, permits every straight line $x = A + Bt$. Position and velocity at one time select the particular line. For a field, we must also specify what happens at the edges of the region. This is why initial data and boundary conditions belong beside the differential equation from the beginning. Initial data select a particular history, while boundary data constrain a field in space.

An n th-order ordinary differential equation generally requires n independent initial data. Boundary-value problems can instead select discrete eigenvalues. Three central PDEs are

$$\partial_t^2 u = v^2 \nabla^2 u \quad \text{wave propagation,} \quad (7.6)$$

$$\partial_t u = D \nabla^2 u \quad \text{diffusion,} \quad (7.7)$$

$$\nabla^2 \phi = s(\mathbf{r}) \quad \text{Poisson equation.} \quad (7.8)$$

The wave equation propagates signals; diffusion smooths gradients; Poisson's equation constrains a field at a given time in a static or quasistatic model. Boundary conditions are part of the problem, not optional additions.

Separation of variables, $u(x, t) = X(x)T(t)$, works when the operator and geometry permit it. Green functions solve $LG(x, x') = \delta(x - x')$ with specified boundary conditions, producing $u(x) = \int G(x, x')s(x')dx'$ plus any required homogeneous contribution.

Worked example 7.1 (Diffusion of a sinusoidal disturbance). For $u(x, 0) = u_0 + A \cos kx$ on an infinite or compatible periodic domain, try $u = u_0 + a(t) \cos kx$. The diffusion equation gives $\dot{a} = -Dk^2 a$, hence $a = Ae^{-Dk^2 t}$. Short spatial wavelengths relax faster, with time scale $1/(Dk^2)$.

Let us solve one separated problem explicitly. On $0 < x < L$, hold the ends of a diffusing field at zero, and put $u = XT$ into $\partial_t u = D\partial_x^2 u$. Wherever division is allowed,

$$\frac{\dot{T}}{DT} = \frac{X''}{X} = -k^2. \quad (7.9)$$

One side depends only on time and the other only on position, so both must be constant. We choose a negative constant because the zero-end boundary conditions call for sines. The two ordinary equations give

$$X_n(x) = \sin \frac{n\pi x}{L}, \quad T_n(t) = e^{-D(n\pi/L)^2 t}, \quad n = 1, 2, \dots \quad (7.10)$$

Thus $u(x, t) = \sum_n b_n X_n(x) T_n(t)$, with b_n found by expanding the initial profile in a sine series. We did not guess an entire complicated solution. We solved for simple allowed pieces and used the initial condition to decide how much of each piece to include.

7.4 Green functions and boundary-value problems

Separation of variables gives a basis of homogeneous solutions, while a Green function gives the response to a localized source. The boundary condition is part of its definition: two operators with the same differential expression but different boundaries have different Green functions.

For the one-dimensional operator $L = -d^2/dx^2$ on $0 < x < L$ with $G(0, x') = G(L, x') = 0$, solve

$$-\frac{\partial^2 G}{\partial x^2} = \delta(x - x'). \quad (7.11)$$

Away from $x = x'$, the solution is linear. Continuity at x' and integration across the source give the derivative jump

$$G_x(x'^+, x') - G_x(x'^-, x') = -1. \quad (7.12)$$

The result is

$$G(x, x') = \frac{x_{<}(L - x_{>})}{L}, \quad x_{<} = \min(x, x'), \quad x_{>} = \max(x, x'). \quad (7.13)$$

Therefore the solution of $-y'' = s(x)$ with zero endpoint values is

$$y(x) = \int_0^L G(x, x') s(x') dx'. \quad (7.14)$$

The kink in G is not a defect: it is exactly what produces the delta-function source. This pattern generalizes to electrostatics, diffusion, forced oscillators, and quantum propagators.

Try it yourself

A string is fixed at both ends. Sketch its lowest standing-wave pattern and explain why a pattern with a nonzero end displacement is not allowed. For an equation that describes changing motion, state what additional information selects one particular history.

Chapter 8

Probability as a model of uncertainty

Even a perfect equation for one idealized event cannot remove uncertainty from measurements or from a system with many possible microscopic arrangements. This chapter introduces probability as a tool for making testable predictions from those possibilities. It will return in thermal and quantum physics.

8.1 Probability and distributions

A density is probability per unit of its variable, rather than the probability of one exact value. For a continuous position, the probability of an interval of width dx is $p(x)dx$. This distinction is the reason normalization requires an integral and the reason a probability density can carry units. p_i is the probability of outcome i , with $p_i \geq 0$ and $\sum_i p_i = 1$. An average is a probability-weighted sum.

A probability density satisfies $p(x) \geq 0$ and $\int p(x)dx = 1$. Expectations and variances are

$$\langle f \rangle = \int f(x)p(x)dx, \quad \sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2. \quad (8.1)$$

The Gaussian density is $(2\pi\sigma^2)^{-1/2} \exp[-(x - \mu)^2/(2\sigma^2)]$. The central limit theorem explains its appearance for sums of many weakly dependent contributions under appropriate conditions. A Poisson count has $P(n) = e^{-\lambda}\lambda^n/n!$ and variance equal to its mean λ .

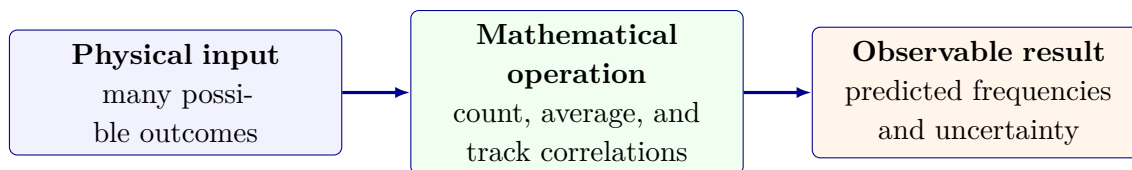


Figure 8.1: Visual model for probability and uncertainty: the physical question becomes mathematics and returns as an observable prediction.

The variance measures the mean squared departure from the mean $\mu = \langle x \rangle$. Expanding the square shows why the convenient formula works:

$$\langle (x - \mu)^2 \rangle = \langle x^2 \rangle - 2\mu \langle x \rangle + \mu^2 = \langle x^2 \rangle - \langle x \rangle^2. \quad (8.2)$$

For independent steps ξ_i with zero mean and variance s^2 , the displacement $X_N = \sum_i \xi_i$ has $\langle X_N^2 \rangle = Ns^2$: cross terms vanish by independence. Its typical magnitude is therefore proportional to $\sqrt{N}$, not N . When steps occur at regular time intervals, this becomes the $\sqrt{t}$ spreading characteristic of diffusion.

8.2 Covariance and entropy maximization

For random variables X_i , the covariance matrix is

$$C_{ij} = \langle (X_i - \langle X_i \rangle)(X_j - \langle X_j \rangle) \rangle. \quad (8.3)$$

It is positive semidefinite, and its diagonal entries are variances. Under a linear transformation $\mathbf{Y} = \mathbf{A}\mathbf{X}$, the covariance transforms as $C_Y = \mathbf{A}C_X\mathbf{A}^\top$. This is the probabilistic version of a tensor transformation and explains why correlated measurement errors cannot be combined by adding variances independently. Variance measures spread; covariance measures how two quantities vary together. Entropy summarizes how probabilities are distributed among outcomes.

The least-biased distribution subject to known constraints can be found by maximizing Shannon entropy. For discrete probabilities, maximize

$$S = - \sum_i p_i \ln p_i \quad (8.4)$$

subject to $\sum_i p_i = 1$ and $\sum_i p_i E_i = U$. With multipliers α and β , stationarity of $S - \alpha(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)$ gives

$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad Z = \sum_i e^{-\beta E_i}. \quad (8.5)$$

Thus the canonical distribution is not an arbitrary rule: it is the maximum-entropy distribution at fixed mean energy. The same calculation with additional constraints produces chemical potentials and other generalized ensembles.

Try it yourself

For two outcomes with probabilities 1/4 and 3/4, check that the probabilities add to one and find the mean of values 0 and 4. Explain in words why repeated independent random steps spread out even when the average step is zero.

Part III

Classical Mechanics

Chapter 9

Newtonian motion and conservation laws

The spring in the first pass gave us a force law simple enough to solve by inspection. We now widen the question: how do momentum, work, and energy organize motion when the force is not so simple? The answers will prepare us for the more flexible languages of Lagrange and Hamilton. We now put the mathematical language to work. The central problem of mechanics is to predict a trajectory from forces and initial conditions. Sometimes the most direct route is to integrate Newton's law. Sometimes energy or momentum lets us bypass most of that integration. Learning mechanics means learning to recognize which question each method answers most efficiently.

9.1 Kinematics and Newton's laws

First choose the object, the reference frame, and the positive coordinate directions. Draw the forces acting on that object before writing any equations. Kinematics describes the resulting motion; dynamics explains why the acceleration has that value. Keeping those jobs distinct prevents you from assuming a familiar trajectory before checking the forces that would produce it.

Position, velocity, and acceleration are $\mathbf{r}$, $\mathbf{v} = \dot{\mathbf{r}}$, and $\mathbf{a} = \ddot{\mathbf{r}}$. With constant acceleration,

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}t, \quad \mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0t + \frac{1}{2}\mathbf{a}t^2. \quad (9.1)$$

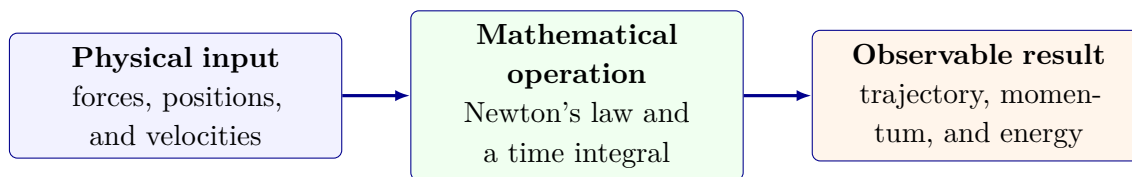


Figure 9.1: Visual model for Newtonian motion: the physical question becomes mathematics and returns as an observable prediction.

Newton's first law defines inertial frames. The second states $\mathbf{F}_{\text{net}} = \dot{\mathbf{p}}$, reducing to $m\mathbf{a}$ for a constant Newtonian mass. The third gives equal and opposite pair forces in standard Newtonian particle models; in field theories, field momentum must also be included in total momentum balance.

Common ideal force laws include gravity near Earth, $\mathbf{F} = -mg\hat{\mathbf{z}}$; Hooke's law, $F_x = -kx$; kinetic dry friction, $|F_f| = \mu_k N$ opposite sliding; and static friction, $|F_f| \leq \mu_s N$, adjusted to prevent slip. A normal force is fixed by constraints and is not automatically mg .

For constant acceleration, begin with $d\mathbf{v}/dt = \mathbf{a}$ and integrate between the initial time 0 and time t :

$$\mathbf{v}(t) - \mathbf{v}_0 = \int_0^t \mathbf{a} \, ds = \mathbf{a}t, \quad (9.2)$$

$$\mathbf{r}(t) - \mathbf{r}_0 = \int_0^t (\mathbf{v}_0 + \mathbf{a}s) \, ds = \mathbf{v}_0 t + \frac{1}{2} \mathbf{a}t^2. \quad (9.3)$$

The factor 1/2 comes from integrating a velocity that grows linearly. If the force depends on position or velocity, these constant-acceleration formulas no longer apply over the whole motion; return to Newton's differential equation. The initial values are not decorative constants: without them the force law selects a family of trajectories rather than one trajectory.

9.2 Work, energy, and power

Suppose the question asks for the speed after a fall, rather than the position at every instant. Solving for the full trajectory gives more information than we need. Energy is useful because it converts the vector force equation into a scalar relation between speeds and positions. The key step is to project force onto the direction of motion.

Dot $m\dot{\mathbf{v}} = \mathbf{F}$ with $\mathbf{v}$ to obtain

$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \mathbf{F} \cdot \mathbf{v}, \quad W = \int \mathbf{F} \cdot d\mathbf{r} = \Delta K, \quad P = \frac{dW}{dt}. \quad (9.4)$$

For $\mathbf{F} = -\nabla V$ with no explicit time dependence, $E = K + V$ is conserved. In a simply connected domain, a smooth curl-free force is conservative. If the potential depends explicitly on time, mechanical energy need not be constant.

Worked example 9.1 (Projectile range). Launch from and land at the same elevation with speed v_0 and angle θ , neglecting air resistance and Earth's curvature. Then $y = v_0 \sin \theta t - gt^2/2$ gives $t_f = 2v_0 \sin \theta / g$, and $R = v_0 \cos \theta t_f = v_0^2 \sin(2\theta)/g$. The maximum at 45° depends on these equal-height, no-drag assumptions.

Here is the algebra behind that projection. Since $v^2 = \mathbf{v} \cdot \mathbf{v}$, the product rule gives

$$\frac{d}{dt}(v^2) = 2\mathbf{v} \cdot \dot{\mathbf{v}}. \quad (9.5)$$

Multiplying Newton's law by $\mathbf{v}$ therefore produces $\dot{K} = \mathbf{F} \cdot \mathbf{v}$. Integrating in time and using $d\mathbf{r} = \mathbf{v}dt$ produces work. If $\mathbf{F} = -\nabla V$ and V has no explicit time dependence, the chain rule gives

$$\frac{dV}{dt} = \nabla V \cdot \mathbf{v} = -\mathbf{F} \cdot \mathbf{v}, \quad \frac{d}{dt}(K + V) = 0. \quad (9.6)$$

For a vertical fall through height h , this becomes $\frac{1}{2}m(v_f^2 - v_i^2) = mgh$. The mass cancels, giving $v_f^2 = v_i^2 + 2gh$. Notice what this method does and does not determine: it gives speed directly, but a direction of motion or travel time may require additional information.

9.3 Momentum, impulses, and collisions

During a collision, forces can be large, brief, and difficult to measure. Their detailed time dependence is often unnecessary. Choose a system containing both colliding bodies and compare its total momentum just before and just after impact. Internal forces then cancel in pairs; only the external impulse can change the total. $\mathbf{p} = m\mathbf{v}$ is momentum, $\mathbf{J} = \int \mathbf{F} dt$ is impulse, $K = mv^2/2$ is kinetic energy, and V is potential energy. The signs of work and impulse follow the chosen direction. For a collision, m_1, m_2 are the two masses, v_1, v_2 their initial velocities, and $\mu = m_1 m_2 / (m_1 + m_2)$ the reduced mass.

For particles of total mass M , the center of mass is $\mathbf{R} = M^{-1} \sum_i m_i \mathbf{r}_i$. Internal pair forces cancel, giving $M\ddot{\mathbf{R}} = \mathbf{F}_{\text{ext}}$. Impulse is $\mathbf{J} = \int \mathbf{F} dt = \Delta \mathbf{p}$.

In an isolated collision, total momentum is conserved. Kinetic energy is also conserved only for an elastic collision. In a perfectly inelastic one-dimensional collision, two masses stick:

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}, \quad K_i - K_f = \frac{1}{2} \mu (v_1 - v_2)^2, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}. \quad (9.7)$$

The lost kinetic energy becomes internal energy, deformation, or radiation; total energy remains conserved.

For two masses in one dimension, set $M = m_1 + m_2$, $V = (m_1 v_1 + m_2 v_2)/M$, and $u = v_1 - v_2$. Solving for the individual velocities gives

$$v_1 = V + \frac{m_2}{M} u, \quad v_2 = V - \frac{m_1}{M} u. \quad (9.8)$$

Substitute these into $K = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$. The cross terms cancel, leaving

$$K = \frac{1}{2} M V^2 + \frac{1}{2} \mu u^2. \quad (9.9)$$

An isolated sticking collision preserves V but makes the final relative velocity zero. The kinetic energy lost is exactly the initial relative-motion term. This explains both the reduced mass and the energy-loss formula without assuming that kinetic energy is conserved during impact.

9.4 Angular momentum and torque

Linear momentum tracks translation. To track rotation about a chosen point, we also need to know how far the line of motion passes from that point. The cross product supplies that information: motion directly toward the origin contributes no angular momentum, while transverse motion does.

About a fixed inertial origin,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad \boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \quad \dot{\mathbf{L}} = \boldsymbol{\tau}. \quad (9.10)$$

A central force has zero torque about its center. Angular momentum conservation implies planar motion and equal swept areas in equal times.

Differentiate the definition instead of memorizing the torque law separately:

$$\dot{\mathbf{L}} = \dot{\mathbf{r}} \times \mathbf{p} + \mathbf{r} \times \dot{\mathbf{p}} = \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times \mathbf{F} = \boldsymbol{\tau}. \quad (9.11)$$

For a central force the torque vanishes. During a short interval the radius vector sweeps a triangle with area $dA = \frac{1}{2}|\mathbf{r} \times \mathbf{v}|dt$, so

$$\frac{dA}{dt} = \frac{L}{2m}. \quad (9.12)$$

Constant angular momentum therefore means constant areal speed. A planet must move faster near the Sun because a shorter radius must sweep the same area in the same time.

9.5 Drag and variable mass

A falling object with drag gives us a useful first example in which acceleration changes during the motion. Start by finding the velocity at which the net force vanishes. Then solve for the departure from that velocity; the resulting equation is simpler and makes the approach to equilibrium visible.

Linear drag, $\mathbf{F}_d = -b\mathbf{v}$, approximates some slow viscous flows. With downward positive, a falling mass satisfies $m\dot{v} = mg - bv$, yielding

$$v(t) = \frac{mg}{b} + \left(v_0 - \frac{mg}{b}\right)e^{-bt/m}. \quad (9.13)$$

Quadratic drag $\mathbf{F}_d = -\frac{1}{2}C_D\rho A|\mathbf{v}|\mathbf{v}$ is often a better model at larger Reynolds number; C_D depends on shape and flow regime.

For an ideal rocket with exhaust speed u relative to the rocket and no external force, momentum flux gives $m dv = -u dm$, hence $\Delta v = u \ln(m_i/m_f)$. Naively applying $d(mv)/dt = F_{\text{ext}}$ to an open subsystem without its mass flux gives the wrong result.

Set $v_\infty = mg/b$ and $w = v - v_\infty$. The falling-body equation becomes $\dot{w} = -(b/m)w$, so separation gives $dw/w = -(b/m)dt$. Integration yields $w = w_0 e^{-t/\tau}$ with $\tau = m/b$. The time scale τ is when the departure from terminal speed has decreased by a factor e .

The rocket needs a different bookkeeping choice. During a short interval its mass changes by $dm < 0$, so the expelled mass is $-dm > 0$. In the original inertial frame the exhaust moves at approximately $v - u$. Momentum conservation, retaining only first-order small terms, gives

$$mv = (m + dm)(v + dv) + (-dm)(v - u), \quad (9.14)$$

$$0 = m dv + u dm, \quad (9.15)$$

$$v_f - v_i = -u \int_{m_i}^{m_f} \frac{dm}{m} = u \ln \frac{m_i}{m_f}. \quad (9.16)$$

The exhaust term is essential because the rocket alone is an open system. The logarithm follows from the changing mass that must be accelerated at each stage.

Try it yourself

A moving object receives a short impulse opposite its velocity. Explain what happens to its momentum. State one condition under which its mechanical energy is conserved.

Chapter 10

Lagrangian and Hamiltonian mechanics

Newton's laws taught us to follow forces component by component. That method remains valuable, but constraints and changing coordinates can make it cumbersome. The action reformulates the same mechanics as a statement about paths; Hamiltonian mechanics will then reorganize those paths into a phase-space flow. Newton's laws remain valid, but constrained systems can make their component form cumbersome. A pendulum moves along a circle; an angle describes that motion more naturally than two coordinates and an unknown tension. Analytical mechanics lets us choose such coordinates first and obtain their equations systematically. It also reveals why symmetries and conservation laws appear together.

10.1 Generalized coordinates and stationary action

Imagine comparing the actual motion with nearby trial paths that begin and end at the same configurations. The action assigns a number to each path. The physical path makes the first-order change in this number vanish. This is a variational principle: its input is an entire path, and its output is a differential equation along that path. q_i are generalized coordinates, $\dot{q}_i$ their time rates, and $L = T - V$ the Lagrangian. The action $S = \int L dt$ is a number assigned to an entire path.

Coordinates q_i describe independent degrees of freedom after holonomic constraints have been

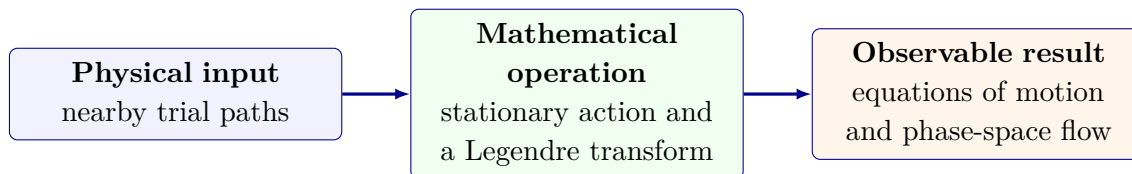


Figure 10.1: Visual model for analytical mechanics: the physical question becomes mathematics and returns as an observable prediction.

imposed. For standard conservative systems, $L = T - V$. The action is

$$S[q] = \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t) dt. \quad (10.1)$$

Varying the path while keeping its endpoints fixed and integrating by parts gives

$$\delta S = \int \sum_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \delta q_i dt = 0. \quad (10.2)$$

Since the variations are independent, the Euler–Lagrange equations follow:

$$\boxed{\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0.} \quad (10.3)$$

The action is stationary, not necessarily a minimum. Nonconservative generalized forces can be added on the right-hand side.

Worked example 10.1 (Pendulum from an action). For a mass on a rigid rod of length ℓ , $L = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell(1 - \cos\theta)$. Euler–Lagrange gives $\ddot{\theta} + (g/\ell)\sin\theta = 0$. Linearizing near $\theta = 0$ produces simple harmonic motion with $\omega_0 = \sqrt{g/\ell}$. Near the upright position, the linearized restoring coefficient changes sign and perturbations grow.

To supply the missing intermediate step, consider one coordinate and write $q_\epsilon = q + \epsilon\eta$, where $\eta(t_1) = \eta(t_2) = 0$. Differentiate the action with respect to ϵ at zero:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} \right) dt, \quad (10.4)$$

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \dot{\eta} dt = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta dt. \quad (10.5)$$

The boundary term vanishes because the endpoints are fixed. The remaining integral must vanish for every smooth interior variation η , forcing its coefficient to vanish. That is the Euler–Lagrange equation.

For the pendulum, carry out the derivatives explicitly:

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}, \quad \frac{\partial L}{\partial \theta} = -mg\ell \sin\theta, \quad m\ell^2\ddot{\theta} + mg\ell \sin\theta = 0. \quad (10.6)$$

Near the bottom, replace $\sin\theta$ by θ . Comparing with $\ddot{\theta} + \omega_0^2\theta = 0$ gives $\omega_0^2 = g/\ell$, so $\tau = 2\pi/\omega_0 = 2\pi\sqrt{\ell/g}$. We have now obtained the numerical factor that dimensional analysis could not determine.

10.2 Constraints and multipliers

Not every variation is physically allowed. A bead constrained to a surface must satisfy $f(q) = 0$, so its trial paths cannot move freely in every coordinate direction. Introduce a multiplier $\lambda(t)$ and vary the augmented action

$$S_*[q, \lambda] = \int [L(q, \dot{q}, t) + \lambda(t)f(q, t)] dt. \quad (10.7)$$

Variation with respect to λ returns the constraint. Variation with respect to q_i gives

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \lambda \frac{\partial f}{\partial q_i}. \quad (10.8)$$

The multiplier is the generalized constraint force required to keep the path on the allowed surface. It should be solved from the constraint and the equations of motion, rather than discarded as an algebraic nuisance. In constrained problems, λ is a multiplier whose value becomes the generalized constraint force.

The same idea handles constrained equilibrium. To maximize entropy subject to normalization and fixed mean energy, add multipliers to the objective; to minimize a potential subject to a geometric condition, add a multiplier to the potential. In each case the multiplier measures the response of the optimum to a small change in the imposed constraint. This common structure links mechanics, thermodynamics, and optimization.

10.3 Symmetry and Noether's theorem

A symmetry means that a change in our description or configuration leaves the relevant action unchanged. If sliding the whole experiment in space changes nothing, there is no preferred location. If shifting the starting time changes nothing, there is no preferred clock reading. Conservation laws give these statements a precise dynamical form.

A continuous symmetry of the action implies a conserved quantity, with appropriate boundary qualifications. Time translation yields energy, spatial translation momentum, and rotation angular momentum. A cyclic coordinate obeys $\partial L / \partial q_i = 0$, so its canonical momentum $p_i = \partial L / \partial \dot{q}_i$ is constant.

If L has no explicit time dependence, $E_L = \sum_i \dot{q}_i p_i - L$ is conserved. This equals $T + V$ for usual time-independent mechanical coordinates and velocity-independent potentials; more general Lagrangians require using the definition.

The energy statement can be checked directly. Write $p_i = \partial L / \partial \dot{q}_i$ and differentiate $E_L = \sum_i \dot{q}_i p_i - L$:

$$\dot{E}_L = \sum_i \dot{q}_i \left(\dot{p}_i - \frac{\partial L}{\partial q_i} \right) - \frac{\partial L}{\partial t} = - \frac{\partial L}{\partial t}. \quad (10.9)$$

The Euler–Lagrange equations remove the bracketed terms. Thus absence of explicit time dependence makes E_L constant, even though L usually changes along the motion through $q(t)$ and $\dot{q}(t)$. Distinguishing explicit dependence from dependence through the trajectory is the key to using this result correctly.

10.4 Hamiltonian dynamics and phase space

The Lagrangian describes a state using coordinates and velocities. Hamiltonian mechanics replaces velocities by momenta. This is useful because position and momentum form natural pairs, each

with its own first-order evolution equation. Instead of following a curve through configuration space, we follow a flow through phase space. Later, $p_i = \partial L / \partial \dot{q}_i$ is canonical momentum and H is the Hamiltonian, not automatically identical to kinetic plus potential energy in every theory.

When velocities can be expressed in terms of momenta, define

$$H(q, p, t) = \sum_i p_i \dot{q}_i - L, \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \quad (10.10)$$

A state is a point in phase space (q, p) . The Poisson bracket is

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right), \quad \dot{f} = \{f, H\} + \frac{\partial f}{\partial t}. \quad (10.11)$$

Hamiltonian flow preserves phase-space volume (Liouville's theorem). Canonical transformations preserve the Poisson structure. For periodic one-dimensional motion, an action variable $J = (2\pi)^{-1} \oint p dq$ is approximately conserved under sufficiently slow parameter changes away from separatrices.

The transformation explains Hamilton's equations. Begin with $H = \sum_i p_i \dot{q}_i - L$ and take its differential:

$$dH = \sum_i (\dot{q}_i dp_i + p_i d\dot{q}_i) - \sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) - \frac{\partial L}{\partial t} dt \quad (10.12)$$

$$= \sum_i (\dot{q}_i dp_i - \dot{p}_i dq_i) - \frac{\partial L}{\partial t} dt. \quad (10.13)$$

The velocity differentials cancel because $p_i = \partial L / \partial \dot{q}_i$. Comparing coefficients with the differential of $H(q, p, t)$ gives Hamilton's equations. For $H = p^2/(2m) + kx^2/2$, they read $\dot{x} = p/m$ and $\dot{p} = -kx$; differentiating the first recovers the spring equation. The new formalism has reorganized the same motion.

10.5 Fields and continuum action

A field gives a dynamical variable at each position. You can picture it as the limit of a very large set of coupled coordinates, with the sum over sites replaced by an integral. The variation now includes changes in both the field and its derivatives, but integration by parts performs the same job as in mechanics.

For fields $\phi_a(x)$ with Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a, x)$,

$$S = \int \mathcal{L} d^4x, \quad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0. \quad (10.14)$$

The mechanical sum over coordinates becomes a continuum of local degrees of freedom. This is the bridge from oscillators to elastic media, electromagnetism, and quantum fields.

For a stretched string, take the action $S = \int dt dx \mathcal{L}$ with

$$\mathcal{L} = \frac{1}{2}\mu(\partial_t y)^2 - \frac{1}{2}\mathcal{T}(\partial_x y)^2. \quad (10.15)$$

The first term is kinetic energy per length; the second is the increase of elastic energy at small slope. Varying y with the appropriate boundary terms fixed gives

$$\partial_t(\mu\partial_t y) + \partial_x(-\mathcal{T}\partial_x y) = 0, \quad \mu\partial_t^2 y = \mathcal{T}\partial_x^2 y. \quad (10.16)$$

This is our first example of a wave equation arising from an action. Each point supplies inertia, while spatial bending supplies a restoring force.

Try it yourself

For a simple pendulum, identify the coordinate that describes its motion and the two energies in its Lagrangian. Why must the endpoints of the trial paths be fixed when deriving its equation?

Chapter 11

Gravitation, orbits, and rigid bodies

The conserved quantities from Newtonian and analytical mechanics now meet geometry. Orbits use energy and angular momentum to reduce a two-body problem, while rigid bodies show why a single rotational coordinate is sometimes not enough. These are the first places where the shape of a system becomes part of its dynamics. Gravity and rotation bring the conservation laws together. An orbit is a continuously falling trajectory, shaped by both attraction and angular momentum. A rotating body is a collection of moving pieces whose geometry matters as much as its total mass. We will simplify both problems by identifying what is conserved and choosing coordinates that expose it.

11.1 Newtonian gravity and potentials

Begin with the direction of the force: gravity pulls a test mass toward the source. The potential packages the work needed to move against that attraction. Choosing zero potential at infinity is convenient for an isolated mass, because it makes the energy needed to escape especially easy to interpret. G is Newton's gravitational constant, M and m are source and test masses, and r is separation.

Two point masses attract with $\mathbf{F} = -GMm\hat{\mathbf{r}}/r^2$, and $V = -GMm/r$ with zero at infinity. For a continuous density,

$$\Phi(\mathbf{r}) = -G \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r', \quad \mathbf{g} = -\nabla\Phi, \quad \nabla^2\Phi = 4\pi G\rho. \quad (11.1)$$

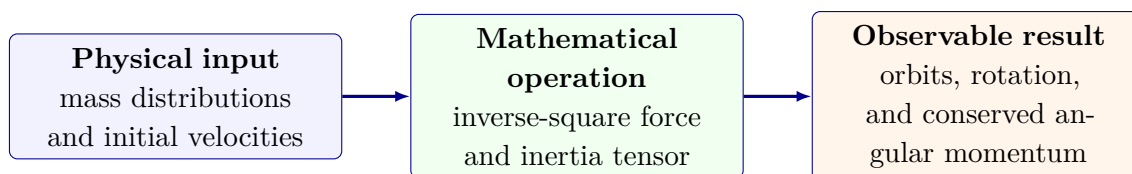


Figure 11.1: Visual model for gravity and rigid bodies: the physical question becomes mathematics and returns as an observable prediction.

A spherically symmetric body produces the same exterior field as a point mass at its center. Inside a spherical shell the net gravitational field is zero.

For a point source, $F_r = -dV/dr = -GMm/r^2$, hence $dV/dr = GMm/r^2$. Integrating from infinity to r gives

$$V(r) - V(\infty) = \int_{\infty}^r \frac{GMm}{s^2} ds = -\frac{GMm}{r}. \quad (11.2)$$

Define $\Phi = V/m$ to describe the source independently of the test mass. A small source element $dM = \rho(\mathbf{r}')d^3r'$ contributes $-GdM/|\mathbf{r} - \mathbf{r}'|$; adding these contributions produces the potential integral. Potential is a scalar, so this sum is often easier than summing force vectors directly.

11.2 The two-body problem and effective potential

The two masses both move, but their separation is what determines their mutual force. Separate out the center of mass and describe the relative motion in its orbital plane using r and ϕ . Conservation of angular momentum then lets us remove $\dot{\phi}$ from the energy, reducing the radial problem to motion in an effective one-dimensional potential. In the two-body reduction, μ is reduced mass, ℓ angular-momentum magnitude, E total mechanical energy, and e orbital eccentricity.

Separate center-of-mass motion from relative motion. The relative particle has reduced mass μ and potential $-Gm_1m_2/r$. With angular momentum magnitude ℓ ,

$$E = \frac{1}{2}\mu\dot{r}^2 + V_{\text{eff}}(r), \quad V_{\text{eff}} = -\frac{Gm_1m_2}{r} + \frac{\ell^2}{2\mu r^2}. \quad (11.3)$$

A circular orbit is an extremum of V_{eff} ; a local minimum gives radial stability. The orbit is a conic,

$$r(\phi) = \frac{p}{1 + e \cos \phi}, \quad p = \frac{\ell^2}{\mu Gm_1m_2}, \quad e^2 = 1 + \frac{2E\ell^2}{\mu(Gm_1m_2)^2}. \quad (11.4)$$

Bound orbits have $E < 0$ and $0 \leq e < 1$. For semimajor axis a ,

$$E = -\frac{Gm_1m_2}{2a}, \quad \tau^2 = \frac{4\pi^2 a^3}{G(m_1 + m_2)}. \quad (11.5)$$

These are Kepler's laws in dynamical form. For a light satellite around mass M , $v_c = \sqrt{GM/r}$ and $v_{\text{esc}} = \sqrt{2GM/r}$.

Worked example 11.1 (Why an orbiting astronaut is weightless). For a circular orbit, $v^2/r = GM/r^2$. Both spacecraft and astronaut accelerate gravitationally together. A scale reads the contact force, which is nearly zero; gravity itself is not zero. Small differences in gravity across the spacecraft produce tidal accelerations.

Write $\kappa = Gm_1m_2$. The relative kinetic energy is $T = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\phi}^2)$ and the angular momentum is $\ell = \mu r^2\dot{\phi}$. Substitution gives the centrifugal contribution $\ell^2/(2\mu r^2)$ in V_{eff} . It accounts for transverse kinetic energy, not an additional gravitational interaction.

To obtain the orbit shape, use $u = 1/r$ and primes for derivatives with respect to ϕ . Angular momentum gives $\dot{\phi} = \ell u^2/\mu$, so $\dot{r} = -(\ell/\mu)u'$ and $\ddot{r} = -(\ell^2/\mu^2)u^2u''$. The radial force equation becomes

$$\mu(\ddot{r} - r\dot{\phi}^2) = -\kappa u^2, \quad (11.6)$$

$$u'' + u = \frac{\mu\kappa}{\ell^2}, \quad (11.7)$$

$$u(\phi) = \frac{\mu\kappa}{\ell^2}[1 + e \cos \phi]. \quad (11.8)$$

We chose $\phi = 0$ at closest approach. Taking the reciprocal produces the conic equation. Substituting u and u' into

$$E = \frac{\ell^2}{2\mu}(u'^2 + u^2) - \kappa u = \frac{\mu\kappa^2}{2\ell^2}(e^2 - 1) \quad (11.9)$$

explains the relation between energy and eccentricity. Negative energy means $e < 1$, a bound ellipse. For a circular satellite orbit, equating mv^2/r to GMm/r^2 gives v_c ; setting total energy to zero gives v_{esc} . These two speeds answer different physical questions.

11.3 Rigid-body rotation

Two objects with equal mass can resist angular acceleration differently because their mass is distributed differently around the axis. Begin with each piece's speed, add its kinetic energy, and let the coefficient of $\omega^2/2$ define the moment of inertia. In three dimensions the axis itself can change, which is why a tensor replaces a single number. In rotation, I_{ij} is the inertia tensor, $\boldsymbol{\omega}$ angular velocity, and $\mathbf{L}$ angular momentum; tensor components depend on axes, while the physical rotation does not.

The inertia tensor about an origin is

$$I_{ij} = \int \rho(\mathbf{r})(r^2\delta_{ij} - x_i x_j) dV, \quad \mathbf{L} = I\boldsymbol{\omega}, \quad T_{\text{rot}} = \frac{1}{2}\boldsymbol{\omega} \cdot I\boldsymbol{\omega}. \quad (11.10)$$

Along a principal axis, $L = I\omega$. The parallel-axis theorem is $I = I_{\text{CM}} + Md^2$. In body-fixed principal axes, Euler's equations include

$$I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 = \tau_1, \quad (11.11)$$

with cyclic permutations. Torque-free rotation about the largest or smallest principal inertia is stable to small perturbations; rotation about the intermediate one is unstable.

Rolling without slipping imposes $v_{\text{CM}} = R\omega$. A body descending height h satisfies $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I_{\text{CM}}v^2/R^2$ if dissipative losses vanish. Static friction can enforce rolling without dissipating mechanical energy on a stationary rigid surface.

For rotation around a fixed axis, a mass element at perpendicular distance $r_{\perp}$ moves with speed $v = \omega r_{\perp}$. Thus

$$T = \int \frac{1}{2}v^2 dm = \frac{1}{2}\omega^2 \int r_{\perp}^2 dm, \quad I = \int r_{\perp}^2 dm. \quad (11.12)$$

For a rolling object write $I_{\text{CM}} = \alpha m R^2$. Conservation of energy then gives

$$mgh = \frac{1}{2}mv^2(1 + \alpha), \quad v = \sqrt{\frac{2gh}{1 + \alpha}}. \quad (11.13)$$

A larger α leaves more of the available energy in rotation and less in center-of-mass translation. This lets you compare shapes before inserting their particular moments of inertia.

11.4 Accelerated frames

If you describe motion from a turning platform, the coordinate axes move while the object moves. An apparent acceleration can therefore come from the frame itself. The safe way to find the extra terms is to differentiate vectors while explicitly accounting for the changing basis.

For axes rotating with $\boldsymbol{\Omega}$ and origin acceleration $\mathbf{A}$, the equation of motion includes fictitious forces

$$\mathbf{F}_{\text{fict}} = -m\mathbf{A} - 2m\boldsymbol{\Omega} \times \mathbf{v}' - m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) - m\dot{\boldsymbol{\Omega}} \times \mathbf{r}. \quad (11.14)$$

These are translational, Coriolis, centrifugal, and Euler forces. They account for using a noninertial frame; they do not describe additional interactions.

For any vector $\mathbf{A}$, the rotating-frame derivative obeys

$$\left(\frac{d\mathbf{A}}{dt}\right)_{\text{inertial}} = \left(\frac{d\mathbf{A}}{dt}\right)_{\text{rotating}} + \boldsymbol{\Omega} \times \mathbf{A}. \quad (11.15)$$

Apply this rule first to position and then to velocity. The result is

$$\mathbf{a} = \mathbf{A} + \mathbf{a}' + 2\boldsymbol{\Omega} \times \mathbf{v}' + \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}). \quad (11.16)$$

Insert this into $m\mathbf{a} = \mathbf{F}$ and move every frame term to the force side. That produces the fictitious forces above, including their signs. The factor two in the Coriolis term comes from two differentiations involving the rotating basis.

Try it yourself

If an orbiting body moves farther from the central mass, how do the gravitational force magnitude and potential energy change? Explain which conserved quantity helps reduce the two-body problem.

Chapter 12

Oscillations, coupled systems, and chaos

The spring has appeared as a first model, an energy system, and a rotating body's local approximation. We now ask how the pattern changes when several oscillators interact or when the restoring law is nonlinear. The normal-mode idea introduced here will later reappear in waves, molecules, circuits, fields, and quantum theory. Oscillations recur throughout physics because a smooth potential looks approximately quadratic near a stable minimum. This makes the spring much more than one mechanical example: it is a model for small departures from equilibrium. We will first solve that model, then ask what changes when energy leaks away, a driver supplies energy, or nonlinearity becomes too large to ignore.

12.1 Simple and damped oscillators

Locate the equilibrium before solving the motion. At x_0 , the force vanishes, so $V'(x_0) = 0$. Taylor expansion leaves $V''(x_0)(x - x_0)^2/2$ as the leading changing part of the potential. A positive curvature makes the force point back toward equilibrium, which is the physical origin of harmonic motion. In the equations below, x denotes displacement from that equilibrium. ω_0 is the undamped natural angular frequency, β the damping rate, F_0 the drive amplitude, and ω the drive frequency.

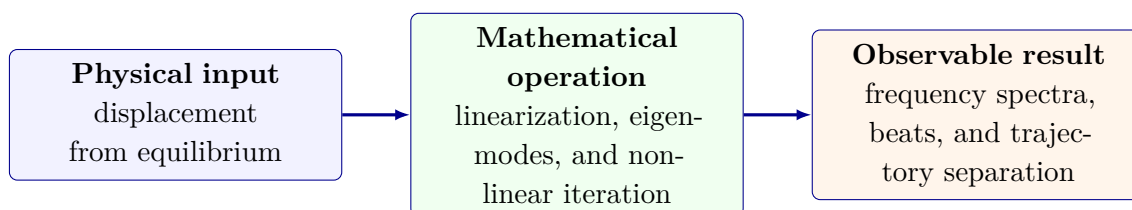


Figure 12.1: Visual model for oscillations and chaos: the physical question becomes mathematics and returns as an observable prediction.

Near a smooth stable equilibrium, $V(x) \simeq V(x_0) + \frac{1}{2}k(x - x_0)^2$. Thus

$$\ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{k/m}, \quad x = A \cos(\omega_0 t + \phi). \quad (12.1)$$

With linear damping and a sinusoidal drive,

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \frac{F_0}{m} \cos \omega t. \quad (12.2)$$

The free underdamped solution has envelope $e^{-\beta t}$ and frequency $\omega_d = \sqrt{\omega_0^2 - \beta^2}$. Critical damping occurs at $\beta = \omega_0$; overdamping has two real decay rates.

The steady-state displacement is $A(\omega) \cos(\omega t - \delta)$, where

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}, \quad \delta = \text{atan2}(2\beta\omega, \omega_0^2 - \omega^2). \quad (12.3)$$

For weak damping, $Q = \omega_0/(2\beta)$ and the resonance width is approximately 2β . The displacement peak occurs at $\sqrt{\omega_0^2 - 2\beta^2}$ if that quantity is real; it is not exactly ω_0 for finite damping.

For the free damped problem try $x = e^{st}$. Each time derivative supplies a factor s , turning the differential equation into

$$s^2 + 2\beta s + \omega_0^2 = 0, \quad s = -\beta \pm \sqrt{\beta^2 - \omega_0^2}. \quad (12.4)$$

Complex roots describe oscillation with decay, repeated roots mark critical damping, and distinct real roots give overdamping. This is why comparing β with ω_0 classifies the motion.

For a steady sinusoidal drive, try $x = \text{Re}(Xe^{i\omega t})$. Substitution gives

$$(\omega_0^2 - \omega^2 + 2i\beta\omega)X = \frac{F_0}{m}, \quad X = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\beta\omega}. \quad (12.5)$$

The magnitude of X is the displacement amplitude, while its argument is the phase relative to the drive. Taking the magnitude and angle of this one complex fraction gives both formulas above. The full solution also contains a transient determined by initial conditions; damping removes that transient at late times.

12.2 Coupled modes and beats

Watch what the coupling spring does in two special motions. If the masses move together by the same amount, its length does not change. If they move in opposite directions, it stretches twice as much as either displacement. These observations suggest adding and subtracting the equations before attempting a general solution. In coupled systems, x_1, x_2 are displacements and q_+, q_- are collective coordinates. An eigenvector gives a mode shape; its eigenvalue gives the associated squared frequency.

Two equal masses attached to fixed walls by springs k and coupled by a spring κ obey

$$m\ddot{x}_1 = -(k + \kappa)x_1 + \kappa x_2, \quad m\ddot{x}_2 = \kappa x_1 - (k + \kappa)x_2. \quad (12.6)$$

The coordinates $q_+ = x_1 + x_2$ and $q_- = x_1 - x_2$ oscillate with $\omega_+ = \sqrt{k/m}$ and $\omega_- = \sqrt{(k + 2\kappa)/m}$. General motion superposes these modes. Nearby frequencies yield beats through $\cos \omega_1 t + \cos \omega_2 t = 2 \cos[(\omega_1 - \omega_2)t/2] \cos[(\omega_1 + \omega_2)t/2]$.

Adding gives $m\ddot{q}_+ = -kq_+$; subtracting gives $m\ddot{q}_- = -(k + 2\kappa)q_-$. Each collective coordinate is now an independent oscillator. Once their initial values are known, solve them and reconstruct

$$x_1 = \frac{q_+ + q_-}{2}, \quad x_2 = \frac{q_+ - q_-}{2}. \quad (12.7)$$

The normal coordinates are useful because they follow motions the system can sustain without continually changing their shape. Beats arise when nearby mode frequencies slowly move into and out of phase.

12.3 Nonlinearity and deterministic chaos

The small-angle pendulum allowed us to replace $\sin \theta$ by θ . Retaining the full sine changes more than a numerical coefficient: sums of solutions no longer generally solve the equation. For a driven system, we therefore study trajectories and their stability rather than expecting every motion to be a sum of independent sinusoids.

Nonlinear systems need not obey superposition. A driven pendulum, $\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2 \sin \theta = f \cos \omega t$, can show periodic, quasiperiodic, or chaotic motion. Chaos is sensitive dependence on initial conditions, often quantified locally by $|\delta x(t)| \sim |\delta x(0)|e^{\lambda t}$ with positive largest Lyapunov exponent.

A phase portrait displays trajectories; a Poincare section samples them once per driving period. Determinism does not guarantee long-term practical predictability. Exponential separation is valid only while perturbations remain small; bounded trajectories eventually saturate the separation.

A useful starting point for stability is to write a nearby trajectory as $\theta + \delta\theta$. Subtract the original pendulum equation and retain terms linear in the separation:

$$\delta\ddot{\theta} + 2\beta\delta\dot{\theta} + \omega_0^2 \cos \theta(t) \delta\theta = 0. \quad (12.8)$$

The coefficient now changes along the reference trajectory. This equation explains how two solutions obeying the same deterministic law can separate. Chaos is not implied merely by nonlinearity; it requires analyzing the actual dynamics, parameters, and long-time perturbation growth.

Try it yourself

Two identical oscillators are coupled by a spring. Sketch an in-phase and an out-of-phase motion. Explain why their frequencies can differ.

Part IV

Continuum Physics, Waves, and Optics

Chapter 13

Elasticity, fluids, and transport

Until now, the objects were a few particles or coordinates. We now let the number of degrees of freedom become enormous and replace the list with fields. The same conservation logic survives: what changes inside a region must be explained by what crosses its boundary or is created within it. When tracking every atom is unnecessary, we describe matter using fields such as density, velocity, and temperature. The guiding method is to balance what enters and leaves a small region, then relate the fluxes and forces to local fields. Conservation laws supply the bookkeeping; material properties supply the response.

13.1 Stress and strain

Move every point in a solid by the same amount and you have translated it without deforming it. Deformation must therefore depend on spatial differences in displacement. Even those differences can describe a rigid rotation, so we separate the symmetric part, which changes shape, from the antisymmetric part, which describes local rotation. In elasticity, ϵ_{ij} is strain and σ_{ij} stress; repeated indices are summed. The symbol λ in the elastic law is a Lamé constant and should not be confused with wavelength elsewhere.

For a small displacement field $\mathbf{u}$, the infinitesimal strain tensor is

$$\epsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i). \quad (13.1)$$

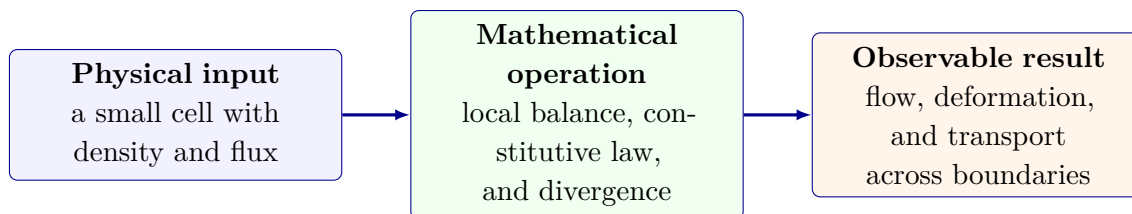


Figure 13.1: Visual model for continuum physics: the physical question becomes mathematics and returns as an observable prediction.

Stress σ_{ij} relates a surface's orientation to force per area. For a homogeneous isotropic linear elastic solid,

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}, \quad \rho \partial_t^2 u_i = \partial_j \sigma_{ij} + f_i. \quad (13.2)$$

Here λ, μ are Lamé constants and μ is the shear modulus. Longitudinal and transverse wave speeds are $v_L = \sqrt{(\lambda + 2\mu)/\rho}$ and $v_T = \sqrt{\mu/\rho}$. One-dimensional stretching gives $\sigma = Y\epsilon$. Linear elasticity fails for large strain, plastic deformation, or fracture.

For a uniform rod of length L extended by ΔL , the displacement changes linearly and $\epsilon = \Delta L/L$. With axial stress $\sigma = F/A$, the law $\sigma = Y\epsilon$ gives

$$F = \frac{YA}{L} \Delta L. \quad (13.3)$$

The rod acts like a spring with stiffness YA/L . This connects a bulk material property to the spring constant of a particular object. For transverse displacement $u_y(x, t)$, the isotropic field equation reduces to $\rho \partial_t^2 u_y = \mu \partial_x^2 u_y$; comparing with the wave equation explains $v_T = \sqrt{\mu/\rho}$.

13.2 Hydrostatics and buoyancy

Picture a thin horizontal slab of stationary fluid. Pressure below pushes upward; pressure above pushes downward; gravity supplies the slab's weight. Equilibrium requires the upward pressure excess to support that weight. Pressure must therefore decrease as height increases.

A static fluid cannot sustain shear stress. Its pressure satisfies $\nabla p = \rho g$. For constant density with upward z , $p(z) = p(z_0) - \rho g(z - z_0)$. Integrating pressure over a submerged object's surface gives a buoyant force equal to the weight of displaced fluid.

Surface tension γ is energy per area and force per length. Across a curved interface,

$$\Delta p = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right). \quad (13.4)$$

A spherical liquid droplet has $\Delta p = 2\gamma/R$; a thin soap bubble has two interfaces and approximately $4\gamma/R$.

For slab area A and thickness dz , vertical balance gives

$$p(z)A - p(z + dz)A - \rho g A dz = 0, \quad \frac{dp}{dz} = -\rho g. \quad (13.5)$$

Integrate only after deciding whether ρ can be treated as constant. To understand buoyancy, imagine replacing the submerged object by fluid of the same shape. That fluid would be in equilibrium, so the surrounding pressure forces must balance its weight. The same pressure forces act on the object, giving the weight of displaced fluid as the buoyant force.

For a spherical droplet, a small increase dR requires pressure work $\Delta p dV$ and increases surface energy by γdA . Since $dV = 4\pi R^2 dR$ and $dA = 8\pi R dR$, equality gives $\Delta p = 2\gamma/R$. Geometry explains the inverse-radius dependence.

13.3 Mass and momentum conservation

There are two ways to watch a fluid: remain at one location and observe what passes, or ride with a fluid element. The first gives partial time derivatives; the second gives the material derivative. We need both viewpoints because conservation is easy to state for a fixed volume, while Newton's law follows moving material. ρ is mass density, $\mathbf{v}$ fluid velocity, p pressure, and η dynamic viscosity.

A material derivative follows a fluid element: $D/Dt = \partial_t + \mathbf{v} \cdot \nabla$. Mass conservation is

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (13.6)$$

For an incompressible Newtonian fluid of constant density and viscosity,

$$\rho (\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v}) = -\nabla p + \eta \nabla^2 \mathbf{v} + \rho \mathbf{g}, \quad \nabla \cdot \mathbf{v} = 0. \quad (13.7)$$

These are the Navier–Stokes equations in a restricted but important form. Inviscid flow sets $\eta = 0$. In steady, inviscid, incompressible flow under gravity, integration along a streamline gives Bernoulli's equation:

$$p + \frac{1}{2} \rho v^2 + \rho g z = \text{constant along a streamline.} \quad (13.8)$$

Different streamlines need not share the same constant unless additional conditions such as irrotationality hold.

For a fixed volume, mass changes only through its boundary:

$$\frac{d}{dt} \int_V \rho dV = - \oint_{\partial V} \rho \mathbf{v} \cdot d\mathbf{S}. \quad (13.9)$$

Use the divergence theorem and bring both terms into one volume integral. Because the region can be arbitrarily small, its integrand must vanish, giving continuity. Along a path $\mathbf{r}(t)$ carried by the fluid, the chain rule gives

$$\frac{d}{dt} \mathbf{v}(\mathbf{r}(t), t) = \partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}. \quad (13.10)$$

This explains why a steady flow can still accelerate: velocity may change from one location to the next. In steady inviscid flow, project Newton's equation along a streamline. For constant density it becomes

$$d\left(\frac{1}{2} v^2\right) = -\frac{dp}{\rho} - g dz. \quad (13.11)$$

Integration gives Bernoulli's equation. The assumptions used during those steps are the conditions under which the result applies.

13.4 Viscosity and dimensionless parameters

Viscosity transfers momentum between neighboring layers moving at different speeds. To decide whether it dominates a flow, estimate the inertial term as $\rho U^2/L$ and the viscous term as $\eta U/L^2$. Their ratio is the Reynolds number. This estimate is a practical way to choose which terms deserve attention before solving a flow equation.

The Reynolds number $\text{Re} = \rho UL/\eta$ compares inertia with viscosity. Low Reynolds number implies creeping flow; large Reynolds number permits inertial instabilities but does not by itself specify a universal transition threshold. For steady fully developed laminar flow through a circular pipe,

$$v(r) = \frac{\Delta p}{4\eta L}(R^2 - r^2), \quad Q_V = \frac{\pi R^4 \Delta p}{8\eta L}. \quad (13.12)$$

These results assume a Newtonian incompressible fluid and no slip at the wall. Stokes drag on an isolated sphere at very low Reynolds number is $F = 6\pi\eta Rv$.

Worked example 13.1 (Pipe radius matters). Holding pressure drop, length, and viscosity fixed, doubling a pipe's radius multiplies laminar volume flow by $2^4 = 16$. This strong dependence is a viscous-flow result; applying it after the flow becomes turbulent is unjustified.

For a straight pipe, assume steady axial flow $v(r)$ with pressure decreasing uniformly by Δp over length L . The axial momentum equation reduces to

$$0 = \frac{\Delta p}{L} + \eta \frac{1}{r} \frac{d}{dr} \left(r \frac{dv}{dr} \right). \quad (13.13)$$

Multiply by r and integrate. Finite slope at the center removes the singular integration term, giving $v' = -\Delta p r / (2\eta L)$. Integrate again and impose the no-slip condition $v(R) = 0$ to obtain the parabolic profile. The volume flow is the sum over annular areas:

$$Q_V = \int_0^R v(r) 2\pi r dr = \frac{\pi \Delta p}{2\eta L} \int_0^R (R^2 r - r^3) dr = \frac{\pi R^4 \Delta p}{8\eta L}. \quad (13.14)$$

Two powers of radius come from area and two from the increased speed allowed by the gentler viscous gradient.

13.5 Diffusion, heat conduction, and random walks

A concentration gradient drives a particle flux from high concentration toward low concentration. Conservation then tells us how that flux changes the concentration. Heat conduction uses the same mathematical structure with a different stored quantity and a different transport coefficient.

Fick's law $\mathbf{j} = -D\nabla n$ together with number conservation yields $\partial_t n = D\nabla^2 n$ for constant D . Fourier's heat law is $\mathbf{q} = -\kappa\nabla T$, giving

$$\partial_t T = \alpha \nabla^2 T, \quad \alpha = \frac{\kappa}{\rho c_p} \quad (13.15)$$

when material properties are constant and the appropriate mechanical conditions permit use of c_p . A diffusion Green function in d dimensions is

$$n(\mathbf{r}, t) = \frac{N}{(4\pi Dt)^{d/2}} \exp\left(-\frac{r^2}{4Dt}\right), \quad \langle r^2 \rangle = 2dDt. \quad (13.16)$$

Diffusion distance grows as $\sqrt{t}$, whereas ballistic travel grows as t .

Combine $\partial_t n + \nabla \cdot \mathbf{j} = 0$ with $\mathbf{j} = -D\nabla n$:

$$\partial_t n = \nabla \cdot (D\nabla n) = D\nabla^2 n \quad (13.17)$$

for spatially constant D . The minus sign in the flux law makes peaks flatten rather than sharpen. In one dimension, an independent random walk of step length a every Δt has $\langle x^2 \rangle = Na^2 = (a^2/\Delta t)t$. Comparing with $2Dt$ identifies $D = a^2/(2\Delta t)$. The continuum coefficient thus measures the cumulative effect of microscopic steps, while the diffusion equation describes their smooth large-scale distribution.

Try it yourself

Draw a small box within a flowing fluid. Label the mass entering and leaving it. What extra information is needed beyond mass conservation to predict the flow?

Chapter 14

Waves, sound, and interference

The continuum description gave us fields that change from place to place. A wave is what happens when a local change communicates with neighboring regions. The oscillator language returns, spread across space; later, the same mathematics will describe light, quantum amplitudes, and numerical grids. A wave lets a disturbance travel without requiring the material to travel with it. Each part of the medium responds locally and influences its neighbors. We will derive the wave equation from that interaction, then use boundary conditions and superposition to explain standing waves, packets, and interference.

14.1 From a string to a wave equation

Take a short string segment of length Δx . Its mass is $\mu\Delta x$. At small slope, the vertical component of tension is approximately $\mathcal{T}\partial_x y$. The two endpoints generally have different slopes, so their vertical forces do not cancel. That slope difference is the source of transverse acceleration. $y(x, t)$ is displacement, μ is linear mass density, $\mathcal{T}$ tension, k wave number, ω angular frequency, and $v = \omega/k$ phase speed.

For small transverse displacement $y(x, t)$ of a string with constant tension $\mathcal{T}$ and linear mass density μ , transverse force balance on a short segment yields

$$\mu\partial_t^2 y = \mathcal{T}\partial_x^2 y, \quad v = \sqrt{\mathcal{T}/\mu}. \quad (14.1)$$

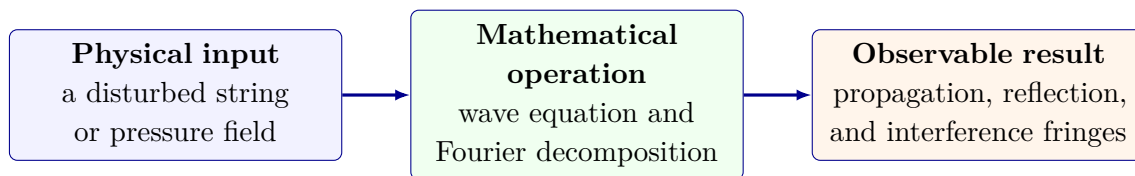


Figure 14.1: Visual model for waves and interference: the physical question becomes mathematics and returns as an observable prediction.

The general one-dimensional solution is $f(x-vt)+g(x+vt)$. A sinusoidal wave has phase $kx-\omega t+\phi$, wavelength $\lambda = 2\pi/k$, period $\tau = 2\pi/\omega$, and phase speed ω/k .

Newton's law for the segment reads

$$\mu\Delta x \partial_t^2 y = \mathcal{T}[\partial_x y(x+\Delta x, t) - \partial_x y(x, t)] \simeq \mathcal{T} \partial_x^2 y \Delta x. \quad (14.2)$$

Cancel Δx and take its limit to obtain the wave equation. To find the speed, try $y = A \cos(kx - \omega t)$. Its second derivatives are $-\omega^2 y$ and $-k^2 y$, so $\omega^2 = (\mathcal{T}/\mu)k^2$. A fixed phase obeys $kx - \omega t = \text{constant}$ and moves at $\omega/k = \sqrt{\mathcal{T}/\mu}$. More tension speeds up communication along the string; more inertia slows it down.

14.2 Energy, impedance, and boundaries

To find transported energy, look at the work one string segment does on its neighbor. To find allowed standing waves, look at what the endpoints permit. These are different questions about the same displacement field, and both can be answered without introducing a new wave law.

For a sinusoidal traveling string wave of displacement amplitude A ,

$$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v. \quad (14.3)$$

Waves carry energy even when material elements only oscillate locally. Reflection and transmission are fixed by boundary conditions. At a fixed string end the displacement changes sign on reflection; at a free end it does not.

Fixed ends at $x = 0, L$ allow $k_n = n\pi/L$, so $f_n = nv/(2L)$ for positive integers n . Standing waves are superpositions of counterpropagating waves; their nodes are permanently at zero displacement.

The instantaneous power passing to the right is $P = -\mathcal{T}(\partial_x y)(\partial_t y)$. For $y = A \cos(kx - \omega t)$, this becomes $\mathcal{T}k\omega A^2 \sin^2(kx - \omega t)$. Average the squared sine over a cycle and use $k = \omega/v$ and $\mathcal{T} = \mu v^2$ to recover $\langle P \rangle = \mu v \omega^2 A^2 / 2$.

For fixed endpoints, a separated spatial mode is $X = C \sin kx + D \cos kx$. The condition $X(0) = 0$ sets $D = 0$; $X(L) = 0$ then requires $\sin kL = 0$. Nontrivial modes therefore have $kL = n\pi$. Discrete frequencies come from the boundaries, even though a freely traveling wave could have any wavelength.

14.3 Dispersion and wave packets

A single infinite sinusoid has no identifiable beginning or localized center. To describe a pulse, combine nearby wave numbers. Their phases remain close in some regions and cancel in others, creating an envelope. The speed of that envelope need not equal the speed of the individual crests inside it. A positive k and ω define the direction of the phase $kx - \omega t$; reversing that sign reverses propagation.

A dispersion relation specifies $\omega(k)$. Phase and group speeds are

$$v_p = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk}. \quad (14.4)$$

The group velocity tracks a narrow packet envelope under suitable weak-distortion conditions. If $\omega''(k) \neq 0$, a packet generally spreads. In strongly absorbing or anomalously dispersive media, the group velocity need not be the information velocity.

Add two equal-amplitude waves with nearby frequencies and wave numbers. The trigonometric sum becomes

$$2A \cos\left(\frac{\Delta k x - \Delta \omega t}{2}\right) \cos(\bar{k}x - \bar{\omega}t). \quad (14.5)$$

The slowly varying first factor moves at $\Delta\omega/\Delta k$. Taking a narrow-band limit gives $v_g = d\omega/dk$, while the rapid carrier moves at $\bar{\omega}/\bar{k}$. If all wave numbers have $\omega = vk$, both speeds equal v and an arbitrary packet can retain its shape in the ideal model.

14.4 Sound and the Doppler effect

Sound is a sequence of small compressions and rarefactions. A compression raises the pressure, pressure gradients accelerate the fluid, and the resulting motion changes the density. The relevant stiffness is how much the pressure changes when a fluid element is compressed under the conditions of the wave. In sound, ρ_0 is background density and v_s sound speed.

Linear pressure disturbances in a fluid travel at

$$v_s^2 = \left(\frac{\partial p}{\partial \rho}\right)_s = \frac{\gamma p}{\rho} \quad (14.6)$$

for an ideal gas with locally adiabatic compression. Plane-wave acoustic impedance is $Z = \rho v_s$, and intensity is $I = p_{\text{rms}}^2/Z$. The sound intensity level is $10 \log_{10}(I/I_0)$ decibels for a specified reference I_0 .

For collinear motion through a stationary medium, using positive observer speed toward the source and positive source speed toward the observer,

$$f_{\text{obs}} = f_{\text{src}} \frac{v_s + v_o}{v_s - v_{\text{src}}}. \quad (14.7)$$

This classical formula depends on motion relative to the medium and differs from relativistic light Doppler shifts.

Around a uniform background ρ_0, p_0 , use small perturbations $\delta\rho, \delta p$ and velocity u in one dimension. Linearized continuity and momentum balance give

$$\partial_t \delta\rho = -\rho_0 \partial_x u, \quad \rho_0 \partial_t u = -\partial_x \delta p. \quad (14.8)$$

With $\delta p = v_s^2 \delta\rho$, take a time derivative of the first equation and substitute the second. You obtain $\partial_t^2 \delta\rho = v_s^2 \partial_x^2 \delta\rho$. For an adiabatic ideal-gas change, $p \propto \rho^\gamma$, so the derivative is $v_s^2 = \gamma p_0/\rho_0$.

For Doppler shifts, count crests. In one source period, a source moving toward the observer advances by $v_{\text{src}}/f_{\text{src}}$, shortening the forward wavelength to $(v_s - v_{\text{src}})/f_{\text{src}}$. The observer meets those crests at relative speed $v_s + v_o$. Dividing that speed by the wavelength gives the stated frequency ratio.

14.5 Superposition and coherence

Fields add before intensities do. Represent two monochromatic contributions by complex amplitudes A_1 and $A_2 e^{i\delta}$. The detector measures a quantity proportional to the squared magnitude of their sum. Expanding that square is the whole origin of the interference term.

For two coherent contributions with intensities I_1, I_2 and phase difference δ ,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta. \quad (14.9)$$

The interference term averages away when relative phase varies rapidly over the measurement. Coherence describes stable phase correlations, not merely equal wavelength.

With real nonnegative A_1, A_2 and a common intensity normalization,

$$|A_1 + A_2 e^{i\delta}|^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos \delta. \quad (14.10)$$

For equal individual intensities I_0 , constructive interference gives $4I_0$ and destructive interference gives zero. Energy is redistributed across the interference pattern rather than created at the bright fringes. If the phase wanders, the measured cross term contains $\langle \cos \delta \rangle$; a uniformly wandering phase makes that average vanish.

Try it yourself

A string is made twice as long while its wave speed stays fixed. What happens to its lowest standing-wave frequency? Sketch the nodes of that mode.

Chapter 15

Geometrical and physical optics

Waves have shown us why phase matters. Optics gives us a useful change of scale: when wavelength is small, the wave description becomes a ray picture; when apertures become comparable to wavelength, the full interference structure returns. This will be our first sustained example of one theory yielding different approximations. Optics offers two complementary descriptions. Rays are useful when wavelength is small compared with the features that shape the light. Wave amplitudes are necessary when phase differences, apertures, or polarization matter. We will connect these descriptions by asking which information each approximation retains.

15.1 Rays, refraction, and imaging

A ray between two points does not generally choose the shortest geometric distance when it crosses different media. Its travel time depends on the refractive index along the route. Let the crossing point move slightly along the interface and require the first-order change in travel time to vanish. n is refractive index, θ an angle measured from the normal, f focal length, and s, s' object and image distances. The sign of an image distance records whether the image is real or virtual; it is not an arbitrary algebraic sign.

In the short-wavelength limit, Fermat's principle makes optical travel time stationary. Reflection has equal incidence and reflection angles; refraction obeys

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (15.1)$$

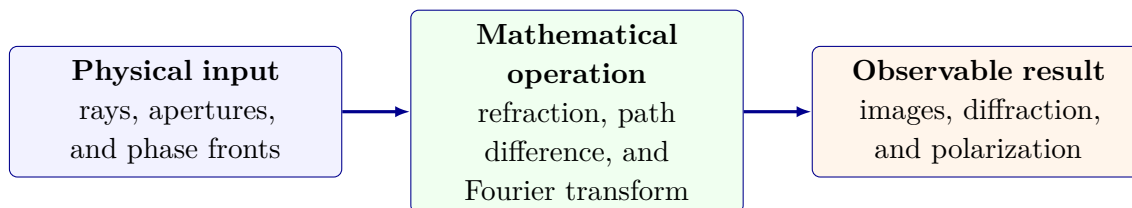


Figure 15.1: Visual model for optics: the physical question becomes mathematics and returns as an observable prediction.

For $n_1 > n_2$, total internal reflection occurs above $\theta_c = \arcsin(n_2/n_1)$. The transmitted side still contains an evanescent field near the interface.

For a thin paraxial lens in air, with real object distance $s > 0$ and real image distance $s' > 0$,

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}, \quad m = -\frac{s'}{s}. \quad (15.2)$$

A virtual image has $s' < 0$. With radii positive when their centers of curvature lie to the right for left-to-right incident light, the thin lensmaker equation is $1/f = (n - 1)(1/R_1 - 1/R_2)$. Large angles and finite apertures introduce aberrations.

Place the endpoints horizontal distance D apart and perpendicular distances h_1, h_2 from a flat interface. If the crossing point is at horizontal coordinate x , the travel time is

$$t(x) = \frac{n_1}{c} \sqrt{x^2 + h_1^2} + \frac{n_2}{c} \sqrt{(D - x)^2 + h_2^2}. \quad (15.3)$$

Setting its derivative to zero gives

$$n_1 \frac{x}{\sqrt{x^2 + h_1^2}} = n_2 \frac{D - x}{\sqrt{(D - x)^2 + h_2^2}}. \quad (15.4)$$

The ratios are the sines of the angles to the normal, giving Snell's law. For imaging, the paraxial thin-lens construction uses a central undeviated ray and a ray initially parallel to the axis. Similar triangles give $y'/y = -s'/s$ and $y' = y(1 - s'/f)$. Equating these expressions yields $1/f = 1/s + 1/s'$. The minus sign in magnification records an inverted image.

15.2 Diffraction as Fourier analysis

To understand a slit, add the amplitudes from every illuminated point across its width. At a distant observation angle, contributions from different points travel slightly different distances. Those path differences become phase differences. The diffraction pattern is the result of that continuous interference. In diffraction, λ is wavelength, a slit width, and d slit separation.

In Fraunhofer diffraction, the far-field complex amplitude is proportional to the Fourier transform of the aperture field. For a uniformly illuminated slit of width a ,

$$I(\theta) = I(0) \left(\frac{\sin \beta}{\beta} \right)^2, \quad \beta = \frac{\pi a \sin \theta}{\lambda}. \quad (15.5)$$

The first minima satisfy $a \sin \theta = \pm \lambda$. Two narrow slits separated by d yield maxima at $d \sin \theta = m\lambda$, modulated by each slit's diffraction envelope.

For a circular aperture of diameter D , the first dark ring is at approximately $1.22\lambda/D$ radians for small angles. The Rayleigh criterion is a useful resolution convention, not an absolute limit independent of noise or measurement method. Grating maxima obey $d \sin \theta = m\lambda$ at normal incidence.

Let x run across the slit and set $q = k \sin \theta$. Apart from common factors, the far-field amplitude is

$$\mathcal{A}(\theta) \propto \int_{-a/2}^{a/2} e^{iqx} dx = \frac{2 \sin(qa/2)}{q} = a \frac{\sin \beta}{\beta}. \quad (15.6)$$

At $\theta = 0$, take the limit $\sin \beta / \beta \rightarrow 1$. Divide by that central amplitude and square the magnitude to obtain the intensity profile. The first zero occurs when $\beta = \pi$, giving $a \sin \theta = \lambda$. A narrower slit spreads the pattern more widely because greater angular change is needed to accumulate the same phase difference across it.

15.3 Polarization and interfaces

An analyzer transmits the electric-field component along its axis. Projecting a field amplitude through angle θ multiplies it by $\cos \theta$; squaring to obtain intensity gives Malus's law. At a dielectric interface, reflection instead follows from matching the electromagnetic fields on both sides.

Polarization specifies the transverse electric field's direction and phase trajectory. Two equal orthogonal components with a $\pi/2$ relative phase produce circular polarization. An ideal linear analyzer obeys Malus's law, $I = I_0 \cos^2 \theta$, for linearly polarized input.

At normal incidence between transparent nonmagnetic media,

$$R = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2, \quad T = 1 - R. \quad (15.7)$$

The transmitted intensity includes the impedance factor; it is not simply the square of the electric-field amplitude ratio. At oblique incidence, the Fresnel coefficients differ for the two linear polarizations. Brewster's angle satisfies $\tan \theta_B = n_2/n_1$ for the same ideal media.

Worked example 15.1 (Interference fringe spacing). For slit separation d and screen distance $L \gg d$, with small angles, $y_m/L \simeq \sin \theta_m = m\lambda/d$. Thus adjacent bright fringes are separated by $\Delta y = \lambda L/d$. Decreasing slit width broadens the envelope; decreasing slit separation broadens the fringe spacing.

For normal incidence in transparent nonmagnetic media, tangential field continuity gives

$$E_i + E_r = E_t, \quad n_1(E_i - E_r) = n_2 E_t. \quad (15.8)$$

The reflected magnetic field has the opposite sign because the wave travels in the opposite direction. Solving these two equations gives $r = E_r/E_i = (n_1 - n_2)/(n_1 + n_2)$ and $t = E_t/E_i = 2n_1/(n_1 + n_2)$. The energy fractions are $R = |r|^2$ and $T = (n_2/n_1)|t|^2$. Their sum is one, providing an immediate check on the amplitude calculation.

15.4 Lasers and optical amplification

A laser requires more than emission of light. Amplification must favor the same optical mode strongly enough to overcome losses, and pumping must replenish the energy given to that mode.

Think of the resonator as repeatedly returning part of the field through the gain material: below threshold it fades; above threshold growth continues until saturation changes the gain.

Stimulated emission generates a photon in the stimulating mode. A laser combines gain, feedback, and a pumping process that sustains nonequilibrium populations. In a simple gain description, intensity grows as $I(z) = I(0)e^{gz}$ before saturation. Laser threshold occurs when round-trip gain balances round-trip loss. An isolated two-level system cannot maintain population inversion under steady resonant pumping alone; practical gain schemes use additional levels or other mechanisms.

If a short distance dz increases intensity by a fraction gdz , then $dI = gIdz$ and integration gives $I(z) = I(0)e^{gz}$. For a cavity of length L , mirror intensity reflectivities R_1, R_2 , and distributed intensity-loss coefficient α , a round trip multiplies intensity by

$$R_1 R_2 e^{2(g-\alpha)L}. \quad (15.9)$$

Threshold sets this factor to one, giving $g_{\text{th}} = \alpha + (2L)^{-1} \ln[1/(R_1 R_2)]$. This elementary model shows how material gain, cavity length, and mirror losses enter the same balance.

Try it yourself

Draw a ray crossing from air into a slower medium and show how it bends. Explain when a wave description is needed to predict an optical effect.

Part V

Thermal and Statistical Physics

Chapter 16

Thermodynamics

The conservation laws of mechanics and fluids tracked energy locally. Thermodynamics asks what can be said when the microscopic details are deliberately hidden. We will keep the same discipline of defining a system and its boundary, but now add temperature, entropy, and the direction of spontaneous change. Thermodynamics asks what we can predict without following every molecule. Its strength comes from carefully defining a system, the energy transfers across its boundary, and the constraints on its states. Much of the mathematics is ordinary calculus; the subtlety is knowing which variables are held fixed and which processes can actually occur.

16.1 Equilibrium, temperature, and the first law

Begin by drawing a boundary around the system. Energy can cross it as heat, through a temperature difference, or as work, through an organized interaction such as a piston moving against a force. Internal energy belongs to the system's state. Heat and work describe how it changes, so they cannot generally be inferred from the endpoints separately. U is internal energy, Q heat transferred into the system, W_{by} work done by the system, S entropy, T absolute temperature, p pressure, V volume, and N particle number. A differential such as dU belongs to a state function; δQ and δW depend on the path. Positive Q enters the system, while positive W_{by} leaves it as mechanical work.

Thermodynamics describes macroscopic states through variables such as U, S, V, N, T, p . Equilib-

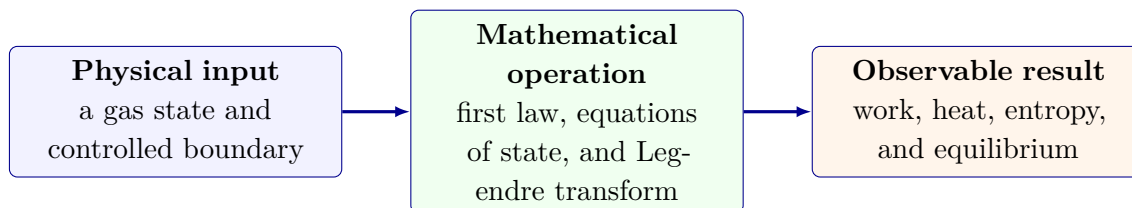


Figure 16.1: Visual model for thermodynamics: the physical question becomes mathematics and returns as an observable prediction.

rium means no net macroscopic flow under the allowed constraints. The zeroth law makes temperature a consistent indicator of mutual thermal equilibrium.

The first law is $dU = \delta Q - \delta W_{\text{by}}$. Heat and work are path-dependent transfers, hence the notation δ ; internal energy is a state function. For reversible pressure-volume work in a simple compressible system, $\delta W_{\text{by}} = p dV$. An ideal gas obeys

$$pV = Nk_{\text{B}}T = nRT, \quad U = \frac{f}{2}Nk_{\text{B}}T \quad (16.1)$$

Here f counts the thermally active quadratic degrees of freedom, assuming equipartition. A monatomic nonrelativistic classical gas has $f = 3$. Rotational and vibrational modes can freeze out quantum mechanically at low temperature.

For a gas pushing a piston of area A through distance dx , the reversible force is pA and the volume increase is $dV = Adx$. Therefore $\delta W_{\text{by}} = pAdx = p dV$. Expansion makes this work positive and removes energy from the gas unless heat replaces it. Over a complete cycle, $\Delta U = 0$, so net heat absorbed equals net work done. Always settle this sign convention before integrating a process; otherwise a correct integral can be assigned the wrong physical meaning.

16.2 Entropy and the second law

Energy conservation permits many processes that never occur spontaneously, such as heat flowing from a cold object into a hotter one without compensation. Entropy supplies the missing restriction. To calculate an entropy change between equilibrium states, you may choose a reversible reference path even when the actual process was irreversible, because entropy itself is a state function.

For a reversible heat transfer, $dS = \delta Q_{\text{rev}}/T$. More generally,

$$\Delta S \geq \int \frac{\delta Q}{T_{\text{boundary}}}, \quad \Delta S_{\text{isolated}} \geq 0. \quad (16.2)$$

Entropy can decrease locally if enough entropy is transferred elsewhere. The second law constrains total entropy production, not every subsystem's entropy separately.

For equilibrium neighboring states of a simple one-component system,

$$\boxed{dU = TdS - pdV + \mu dN.} \quad (16.3)$$

This fundamental relation identifies temperature, pressure, and chemical potential as derivatives of $U(S, V, N)$. It is a relation among state variables; it does not imply every actual process is reversible.

For an ideal gas with fixed particle number and constant C_V , use a reversible path and combine $dU = C_V dT$ with the first law:

$$TdS = dU + pdV = C_V dT + \frac{Nk_{\text{B}}T}{V} dV. \quad (16.4)$$

Divide by T and integrate to find

$$\Delta S = C_V \ln \frac{T_f}{T_i} + Nk_B \ln \frac{V_f}{V_i}. \quad (16.5)$$

The formula depends only on endpoints. For insulated free expansion of an ideal gas, $T_f = T_i$ but $V_f > V_i$, so entropy increases even though no heat enters. The actual heat transfer is not equal to TdS along an irreversible process.

16.3 Heat capacities and ideal-gas processes

Heating at fixed volume and heating at fixed pressure ask different things of the added energy. At fixed volume, pressure-volume work vanishes. At fixed pressure, a gas usually expands as it warms, so some of the supplied energy leaves as work. That distinction explains why the two heat capacities differ.

Define total heat capacities $C_V = (\partial U / \partial T)_{V,N}$ and $C_p = (\partial H / \partial T)_{p,N}$, where $H = U + pV$. An ideal gas has $C_p - C_V = Nk_B$ and $\gamma = C_p / C_V$. With constant heat capacities,

$$\text{isothermal: } \Delta U = 0, \quad W_{\text{by}} = Nk_B T \ln(V_f / V_i), \quad (16.6)$$

$$\text{reversible adiabatic: } pV^\gamma = \text{constant}, \quad TV^{\gamma-1} = \text{constant}. \quad (16.7)$$

An adiabatic process has no heat transfer; it is isentropic only if entropy production also vanishes.

Worked example 16.1 (Free expansion). An ideal gas expands into a vacuum in a thermally insulated rigid container. There is no external work and no heat transfer, so $\Delta U = 0$ and $T_f = T_i$. Nevertheless $\Delta S = Nk_B \ln(V_f / V_i) > 0$. The reversible adiabatic relation $pV^\gamma = \text{constant}$ does not apply to this irreversible process.

For an ideal gas, $H = U + pV = U + Nk_B T$, so differentiation at fixed particle number gives $C_p = C_V + Nk_B$. To derive the reversible adiabatic law, set $\delta Q = 0$ in the first law:

$$C_V dT = -pdV = -\frac{Nk_B T}{V} dV, \quad (16.8)$$

$$\frac{dT}{T} = -\frac{Nk_B}{C_V} \frac{dV}{V} = -(\gamma - 1) \frac{dV}{V}, \quad (16.9)$$

$$\ln T + (\gamma - 1) \ln V = \text{constant}. \quad (16.10)$$

Exponentiation gives $TV^{\gamma-1} = \text{constant}$; using $T = pV / (Nk_B)$ gives $pV^\gamma = \text{constant}$. The use of gas pressure in the work term requires the reversible mechanical path assumed here. Setting heat to zero alone would not justify this derivation.

16.4 Engines and refrigerators

Track one complete cycle so that the working substance returns to its initial state. Its net changes of internal energy and entropy are then zero. The reservoirs, however, have changed. Energy conservation determines the work; their total entropy change limits how large that work can be.

A cyclic engine absorbs Q_h from a hot reservoir and rejects magnitude Q_c to a cold reservoir. Its efficiency is $\eta = W/Q_h = 1 - Q_c/Q_h$. The entropy inequality gives

$$\eta \leq 1 - \frac{T_c}{T_h}, \quad (16.11)$$

with equality for a reversible Carnot engine. A refrigerator's coefficient of performance is $\text{COP} = Q_c/W \leq T_c/(T_h - T_c)$. These ratios use absolute temperatures.

The hot reservoir loses entropy Q_h/T_h and the cold reservoir gains Q_c/T_c . Consequently

$$\Delta S_{\text{total}} = -\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0 \implies \frac{Q_c}{Q_h} \geq \frac{T_c}{T_h}. \quad (16.12)$$

Insert this into $\eta = 1 - Q_c/Q_h$ to obtain the Carnot bound. The result does not depend on whether the working substance is a gas or something else. It follows from the reservoir temperatures and the second law, with equality only when the total entropy production vanishes.

16.5 Thermodynamic potentials and Maxwell relations

Choose a thermodynamic potential by asking what the surroundings control. A bath fixes temperature more naturally than entropy; an external load may fix pressure more naturally than volume. A Legendre transform changes the natural variables so that the potential matches those controls. The extra terms have a purpose: they cancel the differentials we no longer want to use as independent inputs.

Legendre transforms exchange natural variables:

$$H = U + pV, \quad dH = TdS + Vdp + \mu dN, \quad (16.13)$$

$$F = U - TS, \quad dF = -SdT - pdV + \mu dN, \quad (16.14)$$

$$G = U - TS + pV, \quad dG = -SdT + Vdp + \mu dN. \quad (16.15)$$

At fixed T, V, N , equilibrium minimizes Helmholtz free energy F ; at fixed T, p, N , it minimizes Gibbs free energy G , subject to the available states and other constraints. Equality of mixed partial derivatives gives, for example,

$$\left(\frac{\partial S}{\partial V}\right)_{T,N} = \left(\frac{\partial p}{\partial T}\right)_{V,N}. \quad (16.16)$$

Such Maxwell relations turn hard-to-measure derivatives into accessible ones.

At fixed N , start with $dU = TdS - pdV$ and define $F = U - TS$. The product rule gives

$$dF = dU - TdS - SdT = -SdT - pdV. \quad (16.17)$$

Thus $S = -(\partial F/\partial T)_V$ and $p = -(\partial F/\partial V)_T$. Differentiate each with respect to the other variable. Equality of mixed derivatives of a smooth $F(T, V)$ yields

$$\left(\frac{\partial S}{\partial V}\right)_T = -\frac{\partial^2 F}{\partial V \partial T} = -\frac{\partial^2 F}{\partial T \partial V} = \left(\frac{\partial p}{\partial T}\right)_V. \quad (16.18)$$

Rather than memorizing a disconnected list of Maxwell relations, derive the needed one from the differential of the appropriate potential.

16.6 Phase equilibrium and stability

When liquid and vapor coexist, neither phase can lower the total Gibbs energy by taking particles from the other. Their chemical potentials must match. Following that equality as temperature and pressure change gives the slope of the coexistence curve.

Coexisting phases share T, p , and chemical potential for each exchanged species. For a first-order transition, the coexistence curve obeys the Clapeyron equation

$$\frac{dp}{dT} = \frac{\Delta s}{\Delta v} = \frac{\ell}{T\Delta v}, \quad (16.19)$$

where s, v, ℓ are entropy, volume, and latent heat per particle (or all per mole). Stability requires positive appropriate heat capacities and positive isothermal compressibility, $\kappa_T = -V^{-1}(\partial V/\partial p)_T$.

A critical point ends a first-order coexistence line and can exhibit large fluctuations and diverging correlation length. The third law, in its usual perfect-crystal formulation, gives entropy approaching zero for a unique ground state as $T \rightarrow 0$; ground-state degeneracy can produce residual entropy.

For a pure phase, the chemical potential obeys $d\mu = -sdT + vdp$, with s and v taken per particle. Along coexistence, $d\mu_1 = d\mu_2$, so

$$(v_2 - v_1)dp = (s_2 - s_1)dT. \quad (16.20)$$

A reversible phase conversion at temperature T transfers latent heat per particle $\ell = T(s_2 - s_1)$. Combining the two statements produces the Clapeyron equation. The sign of the volume change matters: it determines whether increasing pressure raises or lowers the transition temperature for a positive latent heat.

Try it yourself

A gas expands while being heated. State how you would account for heat, work, and internal-energy change, including your sign convention.

Chapter 17

Statistical mechanics

Thermodynamics gave us macroscopic rules without asking what each molecule was doing. Statistical mechanics now opens that box carefully. Probability, counting, and the Gaussian ideas from the foundations explain why smooth thermodynamic quantities emerge from fluctuating microscopic states. Thermodynamics told us which macroscopic changes are possible. Statistical mechanics asks why those rules emerge from microscopic states. The strategy is to assign probabilities consistently with the system's surroundings, count the states, and average the quantities we can measure. The partition function will become a compact way to perform many of those calculations at once.

17.1 Microstates, macrostates, and ensembles

A gas can have the same pressure and temperature while its molecules occupy enormously many different positions and momenta. A macrostate records the coarse information; a microstate specifies the microscopic configuration. An ensemble is a probability model for such microstates, chosen to reflect what the system can exchange with its surroundings. i labels a microstate, E_i its energy, p_i its probability, and Z the partition function that normalizes the weights.

A macrostate groups many microscopic configurations. Boltzmann's entropy is $S = k_B \ln \Omega$ for equally likely accessible microstates. For general probabilities,

$$S = -k_B \sum_i p_i \ln p_i. \quad (17.1)$$

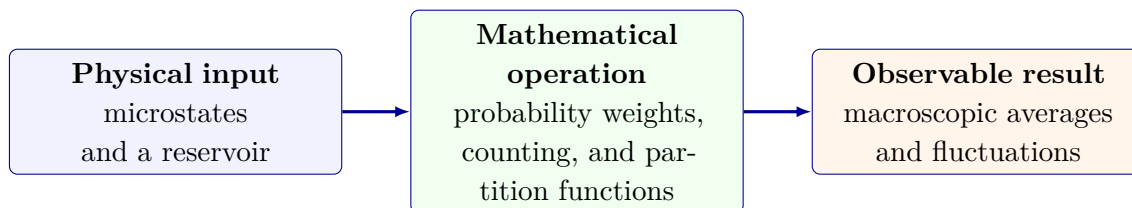


Figure 17.1: Visual model for statistical mechanics: the physical question becomes mathematics and returns as an observable prediction.

The microcanonical ensemble fixes E, V, N . The canonical ensemble fixes T, V, N by exchanging energy with a reservoir. The grand canonical ensemble fixes T, V, μ by exchanging energy and particles. Ensemble equivalence typically emerges for large ordinary systems away from qualifications such as long-range interactions.

If there are Ω equally probable states, set $p_i = 1/\Omega$ in the general entropy expression. Then

$$S = -k_B \sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_B \ln \Omega. \quad (17.2)$$

The logarithm also makes entropy additive for independent systems: their numbers of states multiply, $\Omega_{AB} = \Omega_A \Omega_B$, so $S_{AB} = S_A + S_B$. This is a useful bridge between microscopic counting and macroscopic thermodynamics.

17.2 The canonical partition function

Consider a small system exchanging energy with a much larger reservoir. If the system occupies a state of energy E_i , the reservoir has that much less energy available. The probability of the state is proportional to the number of reservoir microstates compatible with the remaining energy. This is the physical source of the Boltzmann factor. The parameter $\beta = 1/(k_B T)$ increases as temperature falls. Averages use the probabilities p_i ; they are not automatically the value of a single microstate.

A reservoir's multiplicity gives probability proportional to $e^{-\beta E_i}$, where $\beta = 1/(k_B T)$. Define

$$Z = \sum_i e^{-\beta E_i}, \quad p_i = \frac{e^{-\beta E_i}}{Z}, \quad F = -k_B T \ln Z, \quad U = -\frac{\partial \ln Z}{\partial \beta}. \quad (17.3)$$

Further derivatives yield

$$p = k_B T \left(\frac{\partial \ln Z}{\partial V} \right)_{T,N}, \quad \langle (\Delta E)^2 \rangle = k_B T^2 C_V. \quad (17.4)$$

The fluctuation relation links microscopic variability to a macroscopic response.

Worked example 17.1 (Two-level system). For one particle with nondegenerate energies 0 and $\Delta > 0$, $Z = 1 + e^{-\beta \Delta}$ and

$$U = \frac{\Delta}{e^{\beta \Delta} + 1}, \quad C_V = k_B (\beta \Delta)^2 \frac{e^{\beta \Delta}}{(1 + e^{\beta \Delta})^2}. \quad (17.5)$$

At low temperature the excited state is rarely occupied. At high temperature both states become nearly equally populated, so $U \rightarrow \Delta/2$ and the heat capacity again tends to zero.

Write the reservoir entropy as $S_R(E_{\text{tot}} - E_i)$ and expand about E_{tot} :

$$S_R(E_{\text{tot}} - E_i) \simeq S_R(E_{\text{tot}}) - \left(\frac{\partial S_R}{\partial E_R} \right) E_i = S_R(E_{\text{tot}}) - \frac{E_i}{T}. \quad (17.6)$$

Because reservoir multiplicity is proportional to e^{S_R/k_B} , the relative weight is $e^{-\beta E_i}$. Summing all weights gives Z , which normalizes the probabilities. Now differentiate:

$$-\frac{\partial \ln Z}{\partial \beta} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{\sum_i E_i e^{-\beta E_i}}{Z} = \langle E \rangle. \quad (17.7)$$

For a temperature-independent spectrum, a second derivative gives $\partial_\beta^2 \ln Z = \langle E^2 \rangle - \langle E \rangle^2$. Since $d\beta/dT = -1/(k_B T^2)$, this becomes $\langle (\Delta E)^2 \rangle = k_B T^2 C_V$. Differentiating a normalization sum has revealed both the mean energy and its fluctuations.

17.3 Classical ideal gas and equipartition

For a dilute noninteracting gas, position and momentum can be integrated separately. Every location in the container contributes equally, while large momenta are suppressed by their kinetic-energy Boltzmann factor. The Gaussian integral from calculus therefore becomes the basic counting tool for the gas.

For N identical dilute particles with negligible interactions,

$$Z_N = \frac{1}{N!} \left(\frac{V}{\lambda_T^3} \right)^N, \quad \lambda_T = \frac{h}{\sqrt{2\pi m k_B T}}. \quad (17.8)$$

The $N!$ factor removes classical overcounting of permutations of identical particles. The Maxwell–Boltzmann regime requires $n\lambda_T^3 \ll 1$, where $n = N/V$. Differentiating $\ln Z$ yields $pV = Nk_B T$ and $U = 3Nk_B T/2$.

Each independent quadratic term contributes $k_B T/2$ under classical equilibrium conditions. The speed density is

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/(2k_B T)}, \quad v \geq 0. \quad (17.9)$$

It gives $v_{\text{rms}} = \sqrt{3k_B T/m}$, distinct from the mean and most probable speeds.

For one spinless particle, the classical phase-space partition integral is

$$Z_1 = \frac{1}{h^3} \int_V d^3 r \int d^3 p e^{-\beta(p_x^2 + p_y^2 + p_z^2)/(2m)} \quad (17.10)$$

$$= \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} = \frac{V}{\lambda_T^3}. \quad (17.11)$$

For N identical particles, use $Z_N = Z_1^N/N!$. Then $\ln Z_N$ contains $N \ln V - (3N/2) \ln \beta$ plus terms independent of V and β . Its derivatives immediately give $p = Nk_B T/V$ and $U = 3N/(2\beta)$. Each of the three quadratic momentum components contributes equally to that energy.

17.4 Quantum statistics

The new ingredient is the allowed occupation of one quantum state. A fermionic state can contain zero or one particle; a bosonic state can contain any nonnegative integer. Counting those possibilities with the same energy and particle-exchange weight produces the two distribution functions. Their different denominators encode a physical difference in allowed states.

For noninteracting particles occupying single-particle levels ϵ_j , mean occupations are

$$\bar{n}_j = \frac{1}{e^{\beta(\epsilon_j - \mu)} - 1} \quad (\text{bosons}), \quad \bar{n}_j = \frac{1}{e^{\beta(\epsilon_j - \mu)} + 1} \quad (\text{fermions}). \quad (17.12)$$

Bosons allow multiple occupancy of a single state; fermions obey Pauli exclusion. Both approach Maxwell–Boltzmann statistics when occupations are small. For fermions at $T = 0$, levels below the Fermi energy are filled. For a uniform three-dimensional ideal Bose gas with one internal state,

$$T_c = \frac{2\pi\hbar^2}{mk_B} \left[\frac{n}{\zeta(3/2)} \right]^{2/3}, \quad \frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2} \quad (T < T_c). \quad (17.13)$$

This thermodynamic-limit result depends on dimension, dispersion, and confinement; it is not a universal formula for all traps or interacting systems.

For one level of energy ϵ , put $x = e^{-\beta(\epsilon - \mu)}$. Its weight at occupation n is x^n . Fermions give $Z_{\text{level}} = 1 + x$, while bosons give the geometric series $Z_{\text{level}} = \sum_{n=0}^{\infty} x^n = 1/(1 - x)$ for $x < 1$. In either case,

$$\bar{n} = x \frac{d}{dx} \ln Z_{\text{level}}. \quad (17.14)$$

Thus $\bar{n}_F = x/(1 + x)$ and $\bar{n}_B = x/(1 - x)$, which are the distributions above. When $x \ll 1$, both reduce to x : exclusion and enhanced multiple occupancy matter little when a typical state is almost always empty.

17.5 Blackbody radiation

An equilibrium cavity contains electromagnetic standing-wave modes. Count how many modes lie in a small frequency interval, then multiply by the average thermal energy of each mode. This separates the problem into geometry, which determines the density of modes, and statistics, which determines their occupation.

Photons have two transverse polarizations and equilibrium chemical potential zero. Counting modes and using Bose statistics gives spectral energy density per angular frequency,

$$u_{\omega}(T) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\hbar\omega/(k_B T)} - 1}. \quad (17.15)$$

Integration yields $u = aT^4$ and isotropic pressure $p = u/3$. A black surface emits flux $j^{\star} = \sigma T^4$, where

$$a = \frac{\pi^2 k_B^4}{15\hbar^3 c^3}, \quad \sigma = \frac{ac}{4}. \quad (17.16)$$

The spectral peak shifts inversely with temperature, but its numerical location depends on whether the spectrum is expressed per wavelength or per frequency. These peaks cannot be converted merely by $\lambda = c/\nu$ because the density's Jacobian changes.

Including two polarizations, the number of modes per volume in a spherical shell of wave-vector space is

$$\frac{dN_{\text{modes}}}{V} = 2 \frac{4\pi k^2 dk}{(2\pi)^3} = \frac{\omega^2 d\omega}{\pi^2 c^3}. \quad (17.17)$$

Multiply by photon energy $\hbar\omega$ and mean Bose occupation $1/(e^{\beta\hbar\omega} - 1)$ to obtain u_ω . For the total thermal energy density, set $x = \hbar\omega/(k_B T)$:

$$u = \frac{(k_B T)^4}{\pi^2 \hbar^3 c^3} \int_0^\infty \frac{x^3}{e^x - 1} dx. \quad (17.18)$$

The integral is a pure number, so the T^4 law is already established before evaluating it. Its value $\pi^4/15$ fixes the coefficient. Excluding vacuum zero-point energy here is appropriate because we are calculating the thermal radiation energy.

17.6 Interactions, order, and critical behavior

Independent-particle counting cannot describe every material: neighbors can favor one another's alignment. A free-energy model expresses this competition between order and thermal disorder in terms of a collective variable. To find the predicted equilibrium, locate stationary points and then check which are minima.

The Ising model $H = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i$, with $s_i = \pm 1$, illustrates collective order. An order parameter such as magnetization distinguishes phases. A schematic Landau free energy is

$$F(m) = F_0 + a(T - T_c)m^2 + bm^4, \quad a, b > 0. \quad (17.19)$$

Minimization predicts $m = 0$ above T_c and $m^2 = a(T_c - T)/(2b)$ below it. Fluctuations can invalidate these mean-field exponents, especially in low dimensions. Universality groups very different systems by shared symmetry, dimension, and interaction range.

For the stated Landau model,

$$\frac{dF}{dm} = 2a(T - T_c)m + 4bm^3 = 2m[a(T - T_c) + 2bm^2]. \quad (17.20)$$

Above T_c , only $m = 0$ is a minimum. Below T_c , the curvature there becomes negative and two minima appear at $m = \pm \sqrt{a(T_c - T)/(2b)}$. The model respects $m \rightarrow -m$, but either ordered equilibrium chooses one sign. This is the elementary picture of spontaneous symmetry breaking; the quantitative role of fluctuations requires more than minimizing a uniform polynomial.

Try it yourself

A two-level system has a low-energy state and a high-energy state. Describe the likely occupations at low and high temperature and relate them to the mean energy.

Part VI

Electricity and Magnetism

Chapter 18

Electrostatics, potentials, and matter

The continuum chapters taught us to think in fields and fluxes; thermodynamics taught us to respect boundaries and state variables. Electromagnetism combines both habits. We begin with the simplest field, electrostatics, before allowing charge to move and electric and magnetic fields to evolve together. The electric field describes what a test charge would experience at each location. Potential describes the same electrostatic situation through energy per charge. We will move between these descriptions according to the geometry: symmetry often makes fields simple, while superposition and boundary-value problems often favor potential.

18.1 Electric fields and Gauss's law

Start with one charge and a sphere centered on it. The field is radial and equally strong everywhere on that sphere, so its flux is the field magnitude times the area. This is the simplest example of using symmetry before integration. For less symmetric sources, Gauss's law still constrains the field, but it may not determine it by itself. $\mathbf{E}$ is electric field, ϕ electric potential, ρ charge density, Q charge, and $\epsilon_0 = \epsilon_0$ vacuum permittivity. The relation $\mathbf{E} = -\nabla\phi$ means the field points toward decreasing potential. In Gauss's law, Q_{enc} is the charge enclosed by the chosen surface; the surface orientation determines the flux sign.

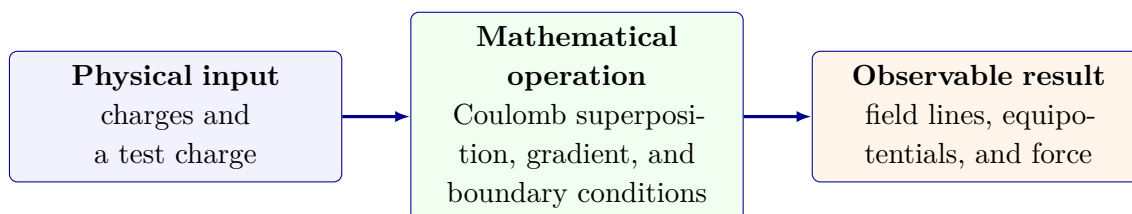


Figure 18.1: Visual model for electrostatics: the physical question becomes mathematics and returns as an observable prediction.

For a point charge,

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, \quad \mathbf{F} = q_{\text{test}} \mathbf{E}. \quad (18.1)$$

Fields superpose linearly in vacuum. Gauss's law is

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (18.2)$$

It is always valid, but determines $\mathbf{E}$ directly from a simple Gaussian surface only when sufficient symmetry is present. Electrostatics also has $\nabla \times \mathbf{E} = 0$, so

$$\mathbf{E} = -\nabla\phi, \quad \nabla^2\phi = -\rho/\epsilon_0. \quad (18.3)$$

In a charge-free region the potential obeys Laplace's equation. Specified boundary values and appropriate regularity yield uniqueness; this justifies methods such as image charges in the physical region.

For a centered charge, the r^{-2} field and r^2 area cancel:

$$\oint \mathbf{E} \cdot d\mathbf{S} = 4\pi r^2 \frac{q}{4\pi\epsilon_0 r^2} = \frac{q}{\epsilon_0}. \quad (18.4)$$

For a uniformly charged solid sphere of radius R , an interior Gaussian sphere encloses only $Q_{\text{enc}} = \rho(4\pi r^3/3)$. Therefore $E(r) = \rho r/(3\epsilon_0)$ inside, while outside the field falls as $1/r^2$. The linear increase inside arises because enclosed charge grows faster than surface area. To obtain the potential equation, substitute $\mathbf{E} = -\nabla\phi$ into $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, giving $\nabla^2\phi = -\rho/\epsilon_0$.

18.2 Conductors and boundary conditions

If an electrostatic field persisted inside an ideal conductor, mobile charges would continue rearranging. Equilibrium is reached when their rearrangement cancels the interior field. At the surface, the field can change abruptly in its normal component because a surface charge is present. Thin integration regions let us calculate that jump without resolving microscopic detail.

In electrostatic equilibrium an ideal conductor has zero internal electric field and constant potential. Excess charge lies on its surface. Across an interface,

$$E_{2\perp} - E_{1\perp} = \sigma/\epsilon_0, \quad \mathbf{E}_{2\parallel} = \mathbf{E}_{1\parallel} \quad (18.5)$$

when σ denotes total surface charge in vacuum-form equations. The normal points from region 1 to 2. Boundary conditions follow by applying Gauss's law to a thin pillbox and the zero-curl law to a narrow loop.

A pillbox of face area A straddling a charged surface has flux $(E_{2\perp} - E_{1\perp})A$ as its height shrinks. The enclosed charge is σA , so Gauss's law gives the normal jump. A thin rectangular loop across the same interface has circulation $(E_{2\parallel} - E_{1\parallel})\ell$ in the corresponding limit. Electrostatic zero circulation forces tangential continuity. At a conductor bordering vacuum, $E_1 = 0$ therefore gives $E_{2\perp} = \sigma/\epsilon_0$ and zero tangential field.

18.3 Multipoles and dipoles

From far away, a compact charge distribution is hard to distinguish from a point charge. If its total charge vanishes, however, that first approximation says nothing. Expand in source size divided by observation distance and retain the first term that survives the charge cancellations.

At distances large compared with a localized charge distribution, expand its potential:

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} + \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} + \cdots \right], \quad \mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d^3r'. \quad (18.6)$$

The leading nonzero multipole dominates far away. An ideal permanent dipole in a slowly varying external field has $U = -\mathbf{p} \cdot \mathbf{E}$, torque $\mathbf{p} \times \mathbf{E}$, and force $\nabla(\mathbf{p} \cdot \mathbf{E})$ when its moment is held fixed.

For $r' \ll r$, expand the inverse distance:

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} \left(1 - \frac{2\hat{\mathbf{r}} \cdot \mathbf{r}'}{r} + \frac{r'^2}{r^2} \right)^{-1/2} \simeq \frac{1}{r} + \frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{r^2} + \cdots. \quad (18.7)$$

Insert this into the source integral for ϕ . The first integral is total charge Q ; the second is the dipole moment projected along $\hat{\mathbf{r}}$. For charges $+q$ and $-q$ separated by vector $\mathbf{d}$ pointing from negative to positive, $\mathbf{p} = q\mathbf{d}$. Their leading distant potential falls as $1/r^2$, faster than a monopole potential, because the net charge contribution cancels.

18.4 Capacitance and field energy

Charging a capacitor becomes progressively harder: each added piece of charge must move through the voltage already established by earlier charge. The energy is therefore the integral of a changing voltage, rather than final voltage times total charge. Begin with a nearly uncharged capacitor and assemble it incrementally.

For a two-conductor capacitor, $C = Q/\Delta\phi$. Its stored energy is

$$U = \frac{Q^2}{2C} = \frac{1}{2}C(\Delta\phi)^2, \quad U_E = \frac{\epsilon_0}{2} \int E^2 dV. \quad (18.8)$$

Parallel plates of area A , spacing $d \ll \sqrt{A}$, and vacuum between them have $C \simeq \epsilon_0 A/d$, neglecting fringing. Capacitors add in parallel, while inverse capacitances add in series.

Worked example 18.1 (Energy and the battery constraint). Insert a dielectric with relative permittivity κ into an ideal capacitor. If disconnected, Q stays fixed, $C \rightarrow \kappa C$, and stored energy falls by $1/\kappa$. If connected to a voltage source, voltage stays fixed and stored energy rises by κ ; battery work must be included in the total energy accounting.

At intermediate charge q , the voltage is q/C . The required work is

$$U = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}. \quad (18.9)$$

For wide parallel plates, $E = Q/(\varepsilon_0 A)$ and $\Delta\phi = Ed$, so $C = Q/\Delta\phi = \varepsilon_0 A/d$. Substituting into $U = Q\Delta\phi/2$ gives $U = (\varepsilon_0 E^2/2)Ad$. The plate volume Ad identifies $\varepsilon_0 E^2/2$ as the vacuum field-energy density. This route connects the circuit element to the field filling its interior.

18.5 Polarization and dielectric response

A dielectric need not transport free charge across the whole sample to respond to an electric field. Positive and negative charge can shift slightly within its microscopic constituents, producing many small dipoles. Polarization averages those dipoles into a field. The distinction between free and bound charge is a bookkeeping choice that makes material response easier to organize.

The polarization $\mathbf{P}$ is electric dipole moment per volume. Bound charges obey $\rho_b = -\nabla \cdot \mathbf{P}$ and $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$. Define

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad \nabla \cdot \mathbf{D} = \rho_{\text{free}}. \quad (18.10)$$

For a linear isotropic dielectric, $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$ and $\mathbf{D} = \varepsilon \mathbf{E}$. Real materials can be anisotropic, nonlinear, dispersive, or lossy. The energy density $\frac{1}{2} \mathbf{E} \cdot \mathbf{D}$ applies to the usual linear nondispersive electrostatic case, not every material response.

Insert $\rho_{\text{total}} = \rho_{\text{free}} + \rho_b$ and $\rho_b = -\nabla \cdot \mathbf{P}$ into Gauss's law:

$$\varepsilon_0 \nabla \cdot \mathbf{E} = \rho_{\text{free}} - \nabla \cdot \mathbf{P} \implies \nabla \cdot (\varepsilon_0 \mathbf{E} + \mathbf{P}) = \rho_{\text{free}}. \quad (18.11)$$

This motivates $\mathbf{D}$. For a homogeneous linear dielectric filling a parallel-plate capacitor, it gives $D = Q_{\text{free}}/A$ and $E = D/\varepsilon$, hence $C = \varepsilon A/d$. The constitutive relation is the extra material information needed to convert a free-charge constraint into an electric field.

Try it yourself

Sketch the electric field of a positive point charge. Explain how the field and potential differ in the information they provide.

Chapter 19

Currents, magnetism, and circuits

Electrostatics gave us fields created by charge at rest. The next step is motion: charge conservation becomes a continuity equation, magnetic force appears, and the field description can be reduced to circuits when the geometry permits. Allow charges to move and the description acquires currents and magnetic fields. Circuit models compress the field behavior into a few variables, but their laws still express charge conservation, induction, and energy transfer. We will use those connections to explain both steady response and the time-dependent approach to it.

19.1 Charge transport and resistance

Current counts charge crossing a surface per unit time. For carriers of number density n , charge q , and mean drift velocity $\mathbf{v}_d$, the current density is $\mathbf{J} = nq\mathbf{v}_d$. Negative carriers can drift opposite to the conventional current direction. In a uniform wire, relating this flow to the electric field yields the familiar resistance law. $\mathbf{J}$ is current density, I total current, $\mathbf{B}$ magnetic field, q particle charge, and $\mathbf{v}$ particle velocity.

Current is $I = dQ/dt = \int \mathbf{J} \cdot d\mathbf{S}$. Local charge conservation is

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \quad (19.1)$$

Ohmic response is $\mathbf{J} = \sigma \mathbf{E}$. A uniform wire has $R = L/(\sigma A)$ and voltage drop $V = IR$. Joule

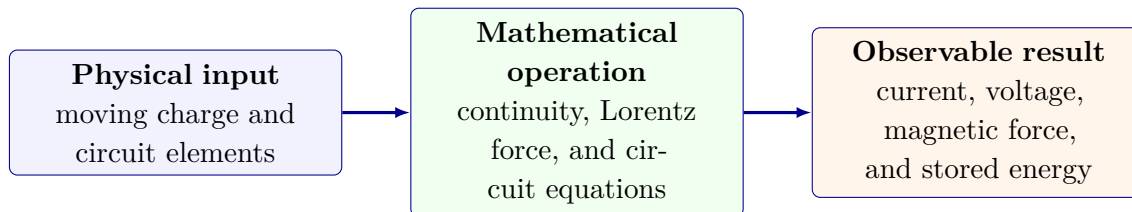


Figure 19.1: Visual model for currents and magnetism: the physical question becomes mathematics and returns as an observable prediction.

heating is $P = IV = I^2 R$. Ohm's law is a constitutive approximation; it is not a universal law of all components.

For uniform axial fields and current, $I = JA$ and $V = EL$. With $J = \sigma E$,

$$I = \sigma A \frac{V}{L} \implies R = \frac{V}{I} = \frac{L}{\sigma A}. \quad (19.2)$$

A charge dQ loses potential energy VdQ crossing the resistor, so the rate is $P = VdQ/dt = VI$. Locally this energy transfer is $\mathbf{J} \cdot \mathbf{E}$ per volume. Integrating that local expression over the wire gives the same circuit power.

19.2 Magnetic forces and steady fields

The cross product in the magnetic force immediately tells us its direction and its limitation: the force is perpendicular to velocity. A uniform magnetic field can therefore bend a trajectory without changing the particle's kinetic energy. Resolve velocity into parallel and perpendicular components to understand the resulting helix. In $q\mathbf{v} \times \mathbf{B}$, the cross product fixes the force direction by the right-hand rule.

The Lorentz force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (19.3)$$

A magnetic field alone changes direction but not the speed of a point charge, since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$. In a uniform field a nonrelativistic charge has gyroradius $r_L = mv_\perp/(|q|B)$ and cyclotron frequency $\omega_c = |q|B/m$.

For steady currents in vacuum,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} d^3r', \quad \oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}. \quad (19.4)$$

A long straight wire has $B = \mu_0 I/(2\pi r)$; a long ideal solenoid has $B \simeq \mu_0 nI$ internally. A small current loop has magnetic moment $\mathbf{m} = IA\hat{\mathbf{n}}$, torque $\mathbf{m} \times \mathbf{B}$, and energy $-\mathbf{m} \cdot \mathbf{B}$ in an external field.

The parallel component feels no magnetic force. In the perpendicular plane, equate force magnitude to the centripetal requirement:

$$|q|v_\perp B = \frac{mv_\perp^2}{r_L}, \quad r_L = \frac{mv_\perp}{|q|B}, \quad \omega_c = \frac{v_\perp}{r_L} = \frac{|q|B}{m}. \quad (19.5)$$

For the field of a long straight wire, cylindrical symmetry makes B constant and tangent around a centered circle. Ampere's law then reads $B(2\pi r) = \mu_0 I$, giving the $1/r$ result. In both examples, geometry does much of the work before the algebra begins.

19.3 Magnetized matter

A magnetized sample contains microscopic magnetic moments. Small current loops provide a useful picture of how neighboring contributions cancel in the interior while leaving an effective surface current. The magnetization field summarizes these moments per volume; it should not be confused with the magnetic field they help produce.

Magnetization $\mathbf{M}$ produces bound currents $\mathbf{J}_b = \nabla \times \mathbf{M}$ and $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$. Define $\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$. A linear isotropic material can satisfy $\mathbf{B} = \mu\mathbf{H}$. Diamagnetism, paramagnetism, and ferromagnetism have different microscopic origins; ferromagnetic hysteresis requires more than a single constant permeability.

In magnetostatics, substitute $\mathbf{J}_{\text{total}} = \mathbf{J}_{\text{free}} + \nabla \times \mathbf{M}$ into $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_{\text{total}}$. Rearranging gives

$$\nabla \times \left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M} \right) = \mathbf{J}_{\text{free}}, \quad \nabla \times \mathbf{H} = \mathbf{J}_{\text{free}}. \quad (19.6)$$

This parallels the role of $\mathbf{D}$ in electrostatics: the auxiliary field places the free sources on one side, while a constitutive relation supplies the material response. A history-dependent magnet cannot be captured by one constant proportionality between $\mathbf{B}$ and $\mathbf{H}$.

19.4 Induction and inductance

A changing magnetic flux drives circulation of electric field, and moving a conductor through a magnetic field can also drive charge. The general loop law accounts for both. Its negative sign says that the induced response opposes the flux change; the energy needed to sustain that change must be supplied from somewhere.

For a stationary loop, Faraday's law is $\mathcal{E} = \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -d\Phi_B/dt$. More generally a moving material loop obeys

$$\mathcal{E} = \oint (\mathbf{E} + \mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_{S(t)} \mathbf{B} \cdot d\mathbf{S}. \quad (19.7)$$

The minus sign expresses Lenz's law. A linear inductor has flux linkage LI , induced emf $-L\dot{I}$, and stored energy $LI^2/2$. Coupled circuits have mutual inductance terms. The vacuum magnetic energy density is $B^2/(2\mu_0)$.

For constant inductance, building the current requires an applied voltage $V = L\dot{I}$ to balance the induced emf. The supplied power is $P = VI = LI\dot{I}$, so

$$U(I) - U(0) = \int_0^I LI' dI' = \frac{1}{2} LI^2. \quad (19.8)$$

The mathematical structure resembles capacitor charging, but the stored variable is now current. When current falls, the magnetic field returns energy to the circuit. An inductor therefore resists rapid changes of current rather than opposing current merely because it is nonzero.

19.5 Lumped circuits and transients

For each circuit, choose a charge or current variable and write a voltage balance with consistent signs. A resistor dissipates energy, a capacitor stores it electrically, and an inductor stores it magnetically. Those roles predict whether a response relaxes monotonically or oscillates before you solve the equation. In circuits, V is voltage, R resistance, C capacitance, and L inductance; their units and signs follow the selected voltage and current directions.

Kirchhoff's current law expresses charge conservation at nodes when charge accumulation is negligible. The usual voltage law assumes a consistent lumped description, with changing magnetic flux represented by inductive elements.

An RC charging circuit obeys $R\dot{Q} + Q/C = V_0$, giving $Q = CV_0(1 - e^{-t/(RC)})$ for an initially uncharged capacitor. An RL circuit has time constant L/R . A series RLC circuit satisfies

$$L\ddot{Q} + R\dot{Q} + Q/C = V(t), \quad (19.9)$$

mathematically identical to a driven damped oscillator.

With phasors proportional to $e^{i\omega t}$,

$$Z_R = R, \quad Z_L = i\omega L, \quad Z_C = \frac{1}{i\omega C}. \quad (19.10)$$

Series impedances add; parallel admittances add. Average power is $V_{\text{rms}}I_{\text{rms}}\cos\phi$. The lumped approximation becomes inadequate when circuit dimensions approach relevant signal wavelengths or propagation delays matter.

For RC charging, define the departure from final charge as $q = Q - CV_0$. Then $R\dot{q} + q/C = 0$, so $q = q(0)e^{-t/(RC)}$. With $Q(0) = 0$, this reconstructs $Q = CV_0(1 - e^{-t/(RC)})$. The same method solved linear drag: subtract the equilibrium value and solve for the decaying departure.

For a sinusoidal current or voltage, a derivative becomes multiplication by $i\omega$. Thus $V_L = L\dot{I}$ gives $Z_L = i\omega L$, while $I = C\dot{V}_C$ gives $Z_C = 1/(i\omega C)$. Dividing the series RLC equation by L identifies

$$\omega_0^2 = \frac{1}{LC}, \quad 2\beta = \frac{R}{L}. \quad (19.11)$$

All the oscillator methods now apply, with charge taking the role of displacement and inductance taking the role of inertia.

Try it yourself

In a simple resistor-capacitor discharge, describe how the voltage and current change with time. Identify the quantity that sets the time scale.

Chapter 20

Maxwell's equations and electromagnetic waves

The previous two chapters treated electric and magnetic phenomena in separate settings. Maxwell's synthesis shows that the separation was a matter of approximation: changing fields sustain one another and carry energy through space. The wave equation from the continuum unit returns here in a new physical form. The separate electrical and magnetic laws fit into one dynamical system. Electric and magnetic fields can sustain one another through time-dependent coupling, even far from their sources. The wave speed will emerge from the field equations themselves, and an energy balance will tell us how those waves carry power.

20.1 The unified field equations

Read the divergence equations as statements about sources and the curl equations as statements about circulation and time evolution. Before solving them, check whether they agree with charge conservation. This is a powerful example of using one established physical requirement to constrain the form of another law. $\mathbf{E}$ and $\mathbf{B}$ are fields, ρ charge density, and $\mathbf{J}$ current density.

In vacuum with total charge and current densities,

$$\nabla \cdot \mathbf{E} = \rho/\varepsilon_0, \qquad \nabla \cdot \mathbf{B} = 0, \qquad (20.1)$$

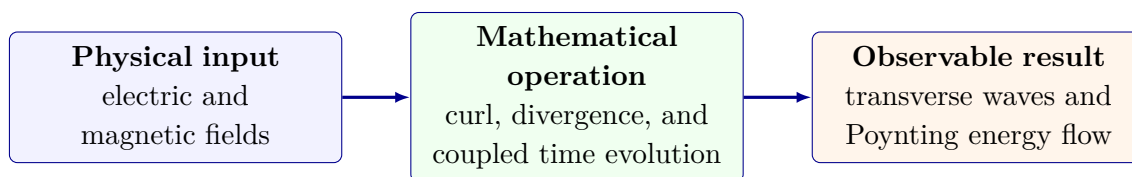


Figure 20.1: Visual model for Maxwell's equations: the physical question becomes mathematics and returns as an observable prediction.

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial_t \mathbf{E}. \quad (20.2)$$

Taking the divergence of the last equation and using Gauss's law yields charge continuity. The displacement-current term is necessary for this consistency, including in a charging capacitor.

Take the divergence of the Ampere–Maxwell equation. Since the divergence of a curl is zero,

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \partial_t (\nabla \cdot \mathbf{E}) = \mu_0 (\nabla \cdot \mathbf{J} + \partial_t \rho). \quad (20.3)$$

Without the displacement-current term, we would instead require $\nabla \cdot \mathbf{J} = 0$ everywhere, excluding local charge accumulation. In a charging capacitor, current reaches the plates while the electric field between them changes. Maxwell's added term lets those two descriptions belong to one consistent field law.

20.2 Potentials and gauge freedom

Potentials help solve the field equations because some constraints become identities. Writing $\mathbf{B}$ as a curl automatically makes its divergence zero. Once that is done, adding $-\partial_t \mathbf{A}$ to the electric field automatically accounts for Faraday induction. The price is that more than one pair of potentials describes the same measurable fields.

Write

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \phi - \partial_t \mathbf{A}. \quad (20.4)$$

The fields are unchanged by $\mathbf{A} \rightarrow \mathbf{A} + \nabla \chi$, $\phi \rightarrow \phi - \partial_t \chi$. In Lorenz gauge, $\nabla \cdot \mathbf{A} + c^{-2} \partial_t \phi = 0$, the potentials satisfy sourced wave equations. The causal vacuum solutions are

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}', t - R/c)}{R} d^3r', \quad (20.5)$$

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t - R/c)}{R} d^3r', \quad R = |\mathbf{r} - \mathbf{r}'|. \quad (20.6)$$

Retarded time expresses finite propagation speed.

Check a gauge transformation directly:

$$\mathbf{B}' = \nabla \times (\mathbf{A} + \nabla \chi) = \mathbf{B}, \quad (20.7)$$

$$\mathbf{E}' = -\nabla(\phi - \partial_t \chi) - \partial_t(\mathbf{A} + \nabla \chi) = \mathbf{E}. \quad (20.8)$$

The cancellation uses commuting space and time derivatives for smooth χ . A gauge condition selects a convenient representative, not a new physical field. In the retarded solution, source distance R means a travel time R/c , so the field here and now depends on the source at $t - R/c$. This is how causality enters the integral explicitly.

20.3 Wave derivation and polarization

Set charge and current to zero in the region of interest. The fields may still be nonzero because disturbances can arrive from elsewhere. To obtain an equation for the electric field alone, take one more curl of Faraday's law and eliminate the magnetic field using the other curl equation. The constants ε_0 and μ_0 describe vacuum response, with $c = 1/\sqrt{\mu_0\varepsilon_0}$.

In a source-free vacuum, take the curl of Faraday's law and use the curl-curl identity:

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \partial_t^2 \mathbf{E} = 0, \quad c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}. \quad (20.9)$$

The magnetic field obeys the same wave equation. For a plane wave, $\mathbf{E} \perp \mathbf{B} \perp \mathbf{k}$, $B = E/c$, and $\mathbf{B} = \hat{\mathbf{k}} \times \mathbf{E}/c$. Light is an electromagnetic wave.

Carry out that elimination without skipping the signs:

$$\nabla \times (\nabla \times \mathbf{E}) = -\partial_t(\nabla \times \mathbf{B}) = -\mu_0\varepsilon_0 \partial_t^2 \mathbf{E}, \quad (20.10)$$

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0\varepsilon_0 \partial_t^2 \mathbf{E}. \quad (20.11)$$

Since $\nabla \cdot \mathbf{E} = 0$ here, both minus signs cancel. Comparing with $\partial_t^2 u = v^2 \nabla^2 u$ identifies $c^2 = 1/(\mu_0\varepsilon_0)$. For a plane wave $\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$, Gauss's law gives $\mathbf{k} \cdot \mathbf{E}_0 = 0$ and Faraday's law gives $\mathbf{B}_0 = (\mathbf{k} \times \mathbf{E}_0)/\omega$. Together with $\omega = ck$, these establish transversality and the field-amplitude ratio.

20.4 Energy, momentum, and radiation

A wave that heats an absorber or pushes a mirror must carry energy and momentum. To derive the energy balance, form the rate of change of squared field strengths. Dot the electric field into the Ampere–Maxwell equation and the magnetic field into Faraday's equation; their combination turns curls into a divergence. The Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ points in the direction of electromagnetic energy transport; its sign is set by the field orientation.

Poynting's theorem is

$$\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}, \quad u = \frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}, \quad \mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (20.12)$$

It states that decreasing field energy supplies outgoing flux and work on charges. Vacuum momentum density is $\mathbf{g} = \mathbf{S}/c^2$. Normal radiation pressure is I/c for full absorption and $2I/c$ for ideal reflection.

Accelerating charges radiate. For a nonrelativistic point charge in vacuum, the Larmor power is

$$P = \frac{q^2 a^2}{6\pi\varepsilon_0 c^3}. \quad (20.13)$$

Near fields and radiation fields have different distance dependence: the radiative electric field scales as $1/r$, so its intensity scales as $1/r^2$. Radiation reaction requires care beyond inserting this power loss into every motion without checking consistency.

The needed vector identity is

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}). \quad (20.14)$$

Substituting Maxwell's equations gives

$$\frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) = -\frac{1}{\mu_0} \mathbf{B} \cdot \partial_t \mathbf{B} - \varepsilon_0 \mathbf{E} \cdot \partial_t \mathbf{E} - \mathbf{J} \cdot \mathbf{E}. \quad (20.15)$$

Use $\mathbf{E} \cdot \partial_t \mathbf{E} = \partial_t (E^2/2)$ and the analogous magnetic identity. Rearrangement yields Poynting's theorem. The right side is negative when fields do positive work on matter. For a vacuum plane wave, $S = \varepsilon_0 c E^2$; averaging a sinusoid gives $I = \varepsilon_0 c E_0^2/2$. Energy flux divided by c is incident momentum flux, which explains the pressure on an absorber.

20.5 Plasmas and electromagnetic response

If electrons move slightly relative to a nearly fixed positive background, charge separation creates a restoring electric field. The resulting collective oscillation resembles a spring, but its stiffness comes from electrostatics. This provides a simple route to the plasma frequency before considering more complicated kinetic or magnetic effects.

A plasma is a many-particle charged medium with collective behavior. With fixed ions and a cold electron fluid, small electron-density oscillations have frequency

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \varepsilon_0}}. \quad (20.16)$$

Thermal electrostatic screening has electron Debye length $\lambda_D = \sqrt{\varepsilon_0 k_B T_e / (n_e e^2)}$ when other species' screening can be neglected. Multiple mobile species contribute to the inverse squared screening length. In an unmagnetized collisionless cold plasma, $\omega^2 = \omega_p^2 + c^2 k^2$ for transverse waves. Below ω_p , these waves are evanescent in the ideal model.

Magnetohydrodynamics treats a sufficiently conducting plasma as a fluid coupled to magnetic fields. Ideal MHD uses $\mathbf{E} + \mathbf{v} \times \mathbf{B} = 0$ and predicts magnetic-flux transport with the fluid. Finite resistivity, kinetic effects, and reconnection can invalidate this ideal description.

In a simple uniform slab displacement model, displace the electron fluid by x . Surface charge separation gives a restoring-field magnitude $n_e e |x| / \varepsilon_0$, with the sign such that

$$m_e \ddot{x} = -\frac{n_e e^2}{\varepsilon_0} x \implies \ddot{x} + \omega_p^2 x = 0. \quad (20.17)$$

The ions are treated as fixed and thermal pressure is neglected. The density dependence $\omega_p \propto \sqrt{n_e}$ reflects a stronger restoring field when more charge separates for the same displacement. In the stated transverse-wave dispersion relation, $k^2 = (\omega^2 - \omega_p^2)/c^2$ becomes negative below cutoff, producing exponential spatial decay rather than propagation in this ideal model.

Try it yourself

Name the pair of changing fields in an electromagnetic wave. Explain why the wave can travel through a vacuum and state one measurable property of it.

Part VII

Relativity and Gravitation

Chapter 21

Special relativity

Classical mechanics and electromagnetism used different ideas about space and time. The electromagnetic wave speed gives us a reason to revisit that foundation. We will preserve the successful conservation laws while replacing absolute time with an invariant spacetime interval. Newtonian mechanics assumes that different inertial observers share a common time. Electromagnetic waves force us to reconsider that assumption: all inertial observers measure the same vacuum light speed. The solution is to transform space and time together. We will derive consequences by comparing events and invariants rather than relying on pictures of moving objects alone.

21.1 Events, intervals, and Lorentz transformations

An event is a location and a time, such as a flash at a detector. Two observers can assign different coordinates to that event without disagreeing about what happened. Begin with transformations that are linear for uniform relative motion, and demand that light trajectories and the spacetime interval retain their form. x, t are coordinates assigned by one observer, primes denote another observer's coordinates, v relative frame speed, and c invariant light speed. The Lorentz factor γ is always at least one for $|v| < c$. The interval ds^2 is invariant even though dx and dt separately change.

The laws of physics have the same form in inertial frames, and vacuum light speed is invariant. For a boost along x ,

$$x' = \gamma(x - vt), \quad t' = \gamma(t - vx/c^2), \quad \gamma = (1 - v^2/c^2)^{-1/2}. \quad (21.1)$$

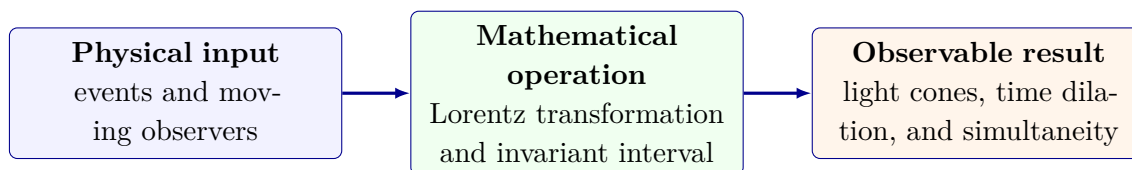


Figure 21.1: Visual model for special relativity: the physical question becomes mathematics and returns as an observable prediction.

The invariant interval is $ds^2 = -c^2 dt^2 + d\mathbf{r}^2$. For a timelike worldline, proper time satisfies

$$d\tau = \sqrt{-ds^2}/c = dt\sqrt{1 - v^2/c^2}. \quad (21.2)$$

Time dilation gives $\Delta t = \gamma\Delta\tau$ for a uniformly moving clock. Length contraction gives $L = L_0/\gamma$ when the endpoints are measured simultaneously in the observing frame. Simultaneity itself is frame dependent.

For a boost along x , consider $x' = A(x - vt)$ and $t' = A(t - vx/c^2)$. These send both $x = ct$ and $x = -ct$ into light trajectories of the same speed. Their interval satisfies

$$x'^2 - c^2 t'^2 = A^2(1 - v^2/c^2)(x^2 - c^2 t^2). \quad (21.3)$$

Invariance and continuity with the identity at $v = 0$ require $A = \gamma$. For a clock at rest in the primed frame, $dx' = 0$ and $dt' = d\tau$, so invariance gives $d\tau = dt/\gamma$. For a moving rod, instead compare its endpoints at the same unprimed time: $\Delta t = 0$ gives $L_0 = \Delta x' = \gamma\Delta x = \gamma L$. Clock comparison and length measurement use different pairs of events; this is why specifying simultaneity matters.

21.2 Velocity addition, light cones, and causality

Velocity is a ratio of coordinate differences. If time transforms as well as distance, subtracting frame velocities is no longer sufficient. Derive the new rule by dividing the transformed displacement by the transformed time interval along the particle's path.

Collinear velocities transform as

$$u'_x = \frac{u_x - v}{1 - u_x v/c^2}. \quad (21.4)$$

A light signal stays at c . Timelike and null separated events can have a causal connection; spacelike separated events cannot exchange a signal at or below light speed. Spacelike event order may change under a boost without causing a causal contradiction.

For light from a source receding directly along the line of sight at speed v ,

$$\frac{f_{\text{obs}}}{f_{\text{src}}} = \sqrt{\frac{1 - \beta}{1 + \beta}}, \quad \beta = v/c. \quad (21.5)$$

This combines time dilation and changing propagation distance.

From the Lorentz transformation,

$$u'_x = \frac{dx'}{dt'} = \frac{\gamma(dx - vdt)}{\gamma(dt - vdx/c^2)} = \frac{u_x - v}{1 - u_x v/c^2}. \quad (21.6)$$

Putting $u_x = c$ makes numerator and denominator cancel to c . At low speeds the denominator approaches one, recovering $u'_x \simeq u_x - v$. For the receding light source, successive emissions separated by source proper time T_0 are separated in the observer frame by γT_0 . The source moves farther away during that interval, adding travel delay $v\gamma T_0/c$. Therefore $T_{\text{obs}} = \gamma T_0(1 + \beta)$, whose reciprocal gives the stated Doppler factor.

21.3 Four-momentum and relativistic energy

The proper-time derivative provides a velocity that transforms consistently as a four-vector. Multiplying by invariant mass gives four-momentum. Its temporal component contains energy and its spatial components contain ordinary three-momentum. A single invariant then links quantities that different observers measure differently.

Four-velocity is $U^\mu = dx^\mu/d\tau$, with $U^\mu U_\mu = -c^2$. Four-momentum for a massive particle is

$$p^\mu = mU^\mu = (E/c, \mathbf{p}), \quad E = \gamma mc^2, \quad \mathbf{p} = \gamma m\mathbf{v}, \quad (21.7)$$

so

$$\boxed{E^2 = p^2 c^2 + m^2 c^4.} \quad (21.8)$$

Rest energy is mc^2 and kinetic energy is $(\gamma - 1)mc^2 \rightarrow mv^2/2$ for $v \ll c$. For a photon, $m = 0$ and $E = pc = h\nu$. The invariant mass of a multiparticle system includes kinetic and binding energy in its center-of-momentum frame.

Worked example 21.1 (Two photons can have nonzero system mass). Two photons each have energy E and travel in opposite directions. Total momentum is zero and total energy is $2E$, so the system has invariant mass $M = 2E/c^2$, even though each photon is individually massless.

For a massive particle, substitute $E = \gamma mc^2$ and $p = \gamma mv$:

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 (1 - v^2/c^2) = m^2 c^4. \quad (21.9)$$

The low-speed expansion makes the Newtonian limit explicit:

$$E = mc^2 + \frac{1}{2}mv^2 + \frac{3mv^4}{8c^2} + \dots \quad (21.10)$$

Rest energy is the leading term, and Newtonian kinetic energy is the first correction. For several particles, add their energies and momenta first, then evaluate $M^2 c^4 = E_{\text{total}}^2 - c^2 |\mathbf{p}_{\text{total}}|^2$. Adding individual rest masses alone would miss internal motion and binding.

21.4 Collisions and thresholds

At a production threshold, the final particles have the least possible energy compatible with their total momentum. This is easiest to describe in the center-of-momentum frame, where threshold means no relative motion among the products. Use an invariant to connect that simple condition to the laboratory setup.

Conserve total four-momentum, not Newtonian kinetic energy. For a projectile with total energy E_a and rest mass m_a striking a stationary target m_b , define the squared center-of-momentum energy

$$s = E_{\text{CM}}^2 = m_a^2 c^4 + m_b^2 c^4 + 2m_b c^2 E_a. \quad (21.11)$$

At threshold, final particles have no relative motion in that frame, so $E_{\text{CM}} = \sum_f m_f c^2$. A fixed-target experiment spends much of its energy on center-of-momentum motion; colliders can use beam energy more efficiently for creating invariant mass.

For a stationary target, initial energy is $E_a + m_b c^2$ and initial momentum is $\mathbf{p}_a$. Hence

$$s = (E_a + m_b c^2)^2 - p_a^2 c^2 \quad (21.12)$$

$$= (E_a^2 - p_a^2 c^2) + 2m_b c^2 E_a + m_b^2 c^4. \quad (21.13)$$

Use the projectile mass-shell relation to obtain the expression above. Setting $s = (\sum_f m_f c^2)^2$ then gives

$$E_{a,\text{th}} = \frac{(\sum_f m_f)^2 - m_a^2 - m_b^2}{2m_b} c^2. \quad (21.14)$$

This is total projectile energy; subtract $m_a c^2$ for kinetic energy. Apply it to a kinematically and quantum-number-allowed channel, remembering that incident total energy cannot fall below the projectile's rest energy.

21.5 Electromagnetic covariance

An observer who moves relative to a charge distribution sees a different combination of charge density and current. It is therefore natural that the same physical electromagnetic situation splits into electric and magnetic parts differently for different observers. The transformation formulas below quantify that split; the invariant combinations provide a way to check it.

Electric and magnetic fields are components of a single antisymmetric field tensor. Their division depends on the observer. For a frame moving at $\mathbf{v}$ relative to the original frame,

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad \mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}, \quad (21.15)$$

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}), \quad \mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \mathbf{v} \times \mathbf{E}/c^2). \quad (21.16)$$

The scalars $E^2 - c^2 B^2$ and $\mathbf{E} \cdot \mathbf{B}$ are invariant. A field that is purely electric for one observer can include magnetism for another.

For a field with $\mathbf{B} = 0$ and $\mathbf{E} \perp \mathbf{v}$, the transformed fields satisfy $E' = \gamma E$ and $B' = \gamma v E / c^2$. Then

$$E'^2 - c^2 B'^2 = \gamma^2 E^2 (1 - v^2 / c^2) = E^2. \quad (21.17)$$

A magnetic component has appeared, but the invariant is unchanged. For a vacuum plane wave, both $E^2 - c^2 B^2$ and $\mathbf{E} \cdot \mathbf{B}$ vanish. No inertial boost can turn a nonzero such wave into a purely electric static field.

Try it yourself

Two observers disagree about elapsed time between events. Identify what each must know about the events and their relative motion before comparing measurements.

Chapter 22

Differential geometry and introductory topology

Special relativity taught us that coordinates are not measurements by themselves; the interval is what all inertial observers agree on. To let that lesson survive when gravity and coordinates vary from place to place, we need a language for curved spaces, tangent directions, transport, and curvature. Classical vectors are usually introduced in a fixed Euclidean basis, but relativity compares measurements made in curved spacetime and in many coordinate systems. Geometry supplies the language for doing this without confusing coordinates with physical objects. Topology then records which global features cannot be removed by smooth stretching.

22.1 Manifolds, tangent spaces, and coordinate changes

A smooth manifold is a space that looks like $\mathbb{R}^n$ in a sufficiently small neighborhood, even if no single coordinate chart covers it without distortion. A point p has a tangent space T_pM containing the possible infinitesimal directions through that point. A tangent vector can be defined by how it acts on smooth functions:

$$v[f] = \left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} \quad (22.1)$$

for any curve γ with $\gamma(0) = p$ and tangent v . In coordinates, $v = v^\mu \partial_\mu$, but the components change when coordinates change. M is a manifold, T_pM the tangent space at point p , $g_{\mu\nu}$ the metric

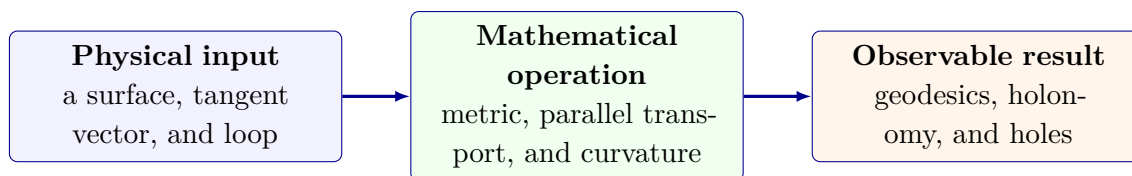


Figure 22.1: Visual model for differential geometry and topology: the physical question becomes mathematics and returns as an observable prediction.

components, and $\Gamma^\rho_{\mu\nu}$ connection coefficients. Greek indices label coordinates and are summed when repeated.

The dual space T_p^*M contains covectors, which act linearly on tangent vectors. The differential df is the covector satisfying $df(v) = v[f]$. A metric is a smoothly varying inner product g_p on each tangent space:

$$\|v\|^2 = g_{\mu\nu}v^\mu v^\nu. \quad (22.2)$$

It determines lengths, angles, and causal character in relativity, but it need not be positive definite. The spacetime metric has one timelike and three spacelike directions.

22.2 Differential forms and integration

A differential one-form is a covector field such as $A = A_\mu dx^\mu$. Its exterior derivative is the two-form

$$dA = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)dx^\mu \wedge dx^\nu. \quad (22.3)$$

The exterior derivative obeys $d^2 = 0$, so an exact form is automatically closed. In electromagnetism, writing the field strength as $F = dA$ makes one pair of Maxwell equations identities; the remaining pair contains the sources.

The generalized Stokes theorem is

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega. \quad (22.4)$$

The familiar fundamental theorem of calculus, Green's theorem, Stokes' theorem, and the divergence theorem are different-dimensional cases of this statement. Orientation is part of the integration data: reversing it changes the sign of the integral.

22.3 Connections, parallel transport, and curvature

Vectors at different points belong to different tangent spaces, so there is no automatic way to subtract them. A connection defines such a comparison. Along a curve with tangent u^μ , parallel transport of V^μ means

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\nu\rho}u^\nu V^\rho = 0. \quad (22.5)$$

The connection coefficients depend on coordinates, but the statement that a vector is parallel transported is geometric. For the unique torsion-free connection compatible with a metric, the coefficients are the Levi-Civita symbols used in the GR chapter. A coordinate component may change under relabeling even when the geometric vector does not; curvature is the coordinate-independent content measured by transport around loops.

Transport around a small closed loop need not return a vector to its original direction. The curvature tensor measures this failure:

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma. \quad (22.6)$$

Flat polar coordinates have nonzero connection coefficients but zero curvature. This is the clean distinction between coordinate effects and genuine gravitational tidal geometry.

22.4 Geodesics and topology

A geodesic is locally straight according to the connection. Equivalently, it makes the metric length stationary under endpoint-fixed variations. In a curved space, geodesics can converge, diverge, or fail to remain globally minimizing. This is why local free fall and global orbital structure are separate questions.

Topology ignores distances and angles and keeps only properties preserved by continuous deformation. A space is connected if it cannot be separated into two disjoint open pieces. A loop may or may not be continuously shrinkable to a point; the distinction detects holes. The circle has a noncontractible loop, while a disk does not. Orientability asks whether a consistent choice of local orientation exists across the whole space; a Mobius strip fails this test.

These ideas are deliberately introductory. General relativity primarily needs smooth manifolds, metrics, connections, and curvature; topology becomes especially important when discussing horizons, defects, gauge fields, and topological phases. The purpose here is to give those later claims a precise conceptual home before the formulas become dense.

Try it yourself

Give an example of a surface that can be curved even though a small patch looks nearly flat. Explain what a metric lets us calculate on that surface.

Chapter 23

General relativity

The geometry chapter gave names to the structures that special relativity only needed in flat form. We can now return to gravity with a precise question: how does matter determine the geometry, and how does that geometry guide free motion? The Newtonian limit will be our bridge back to familiar mechanics. Special relativity describes measurements in flat spacetime. General relativity makes spacetime geometry dynamical and uses it to describe gravity. This chapter introduces the sequence of ideas needed to read the equations: the metric defines intervals, the connection defines parallel transport, curvature measures the failure of transport to be path independent, and stress-energy helps determine curvature. Some full solutions require techniques beyond this introduction; we will develop the accessible steps and identify where a deeper calculation enters.

23.1 Equivalence principle and geometry

Inside a small freely falling laboratory, nearby loose objects appear almost weightless. Over a larger region their accelerations differ, revealing tidal effects. This suggests distinguishing the acceleration removable by choosing a freely falling frame from the relative acceleration that remains. The metric is the mathematical object that lets us make that distinction without privileging one coordinate system. $g_{\mu\nu}$ is the spacetime metric, $\Gamma^\rho_{\mu\nu}$ the Levi-Civita connection, $R^\rho_{\sigma\mu\nu}$ curvature, and $T_{\mu\nu}$ stress-energy. The indices are spacetime labels, not powers.

Locally, a freely falling laboratory removes the leading effect of a uniform gravitational field. Tidal

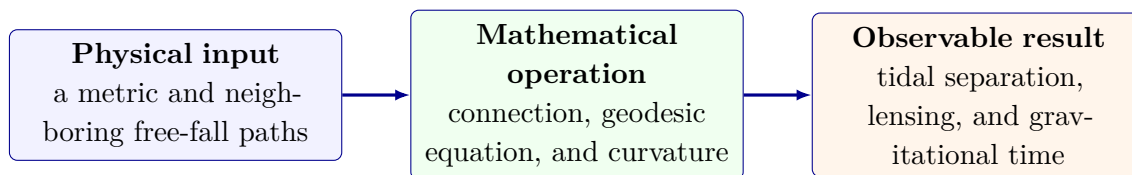


Figure 23.1: Visual model for general relativity: the physical question becomes mathematics and returns as an observable prediction.

effects remain and measure curvature. General relativity describes gravity with a spacetime metric,

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (23.1)$$

The equivalence principle is local: curvature cannot generally be removed throughout an extended region by changing coordinates.

Even flat space can have position-dependent metric components. In polar coordinates, a short displacement has squared length $d\ell^2 = dr^2 + r^2 d\phi^2$. The factor r^2 reflects the arc length associated with an angular change; it does not imply curvature. Likewise, nonzero connection coefficients alone do not establish a gravitational tidal field. Curvature, built from derivatives and products of those coefficients, is the stronger test.

23.2 Covariant derivatives and geodesics

A vector's components may change because the vector changes or because the coordinate basis changes. The ordinary derivative mixes those effects. A covariant derivative includes the basis change so that the result transforms as a tensor. A geodesic is a trajectory whose tangent is parallel transported along itself. In the geodesic equation, τ is proper time for massive particles; in Einstein's equation, curvature responds to energy, momentum, pressure, and stress.

The Levi-Civita connection is

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}). \quad (23.2)$$

For a vector, $\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\rho}^\nu V^\rho$. A freely falling massive particle follows

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (23.3)$$

Light follows null geodesics with an affine parameter in place of proper time. Curvature can be defined by

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda. \quad (23.4)$$

We use $R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}$ and $R = g^{\mu\nu} R_{\mu\nu}$. Sign conventions differ among texts, so tensor definitions and field equations must be used consistently.

To derive the geodesic equation, use an affine parameter λ and the path Lagrangian $L = \frac{1}{2} g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta$, where dots in this paragraph mean $d/d\lambda$. Its derivatives give

$$g_{\mu\nu} \ddot{x}^\nu + (\partial_\alpha g_{\mu\nu}) \dot{x}^\alpha \dot{x}^\nu - \frac{1}{2} (\partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0. \quad (23.5)$$

The product $\dot{x}^\alpha \dot{x}^\beta$ is symmetric, so symmetrize the middle coefficient and multiply by the inverse metric. The combination of metric derivatives is precisely $\Gamma_{\alpha\beta}^\rho$, leaving $\ddot{x}^\rho + \Gamma_{\alpha\beta}^\rho \dot{x}^\alpha \dot{x}^\beta = 0$. For massive particles choose proper time; for light use a different affine parameter because proper time along a null path vanishes.

23.3 Einstein's field equations

The metric cannot be chosen arbitrarily once matter is specified. Einstein's equation is the gravitational field law relating a particular curvature tensor to stress-energy. It is not a consequence of the geodesic equation alone: one law determines geometry, while the other describes free motion within it. Its coefficient is fixed by requiring the weak-field limit to recover Newtonian gravity.

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.} \quad (23.6)$$

Stress-energy includes energy density, momentum density, pressure, and stress. The geometric identity $\nabla_\mu G^{\mu\nu} = 0$ implies $\nabla_\mu T^{\mu\nu} = 0$. This is local covariant conservation; a globally conserved total energy requires additional structure and does not exist in every spacetime.

For weak, slowly varying gravity, $g_{00} \simeq -(1 + 2\Phi/c^2)$ and the geodesic equation gives $\ddot{\mathbf{r}} = -\nabla\Phi$. Einstein's equations reduce to $\nabla^2\Phi = 4\pi G\rho$ when pressure and relativistic corrections are negligible and Λ is neglected.

We can check the force-law limit directly. Use $x^0 = ct$, a static weak field, $g_{00} \simeq -(1 + 2\Phi/c^2)$, and nearly Cartesian spatial coordinates. Then

$$\Gamma_{00}^i \simeq -\frac{1}{2} \partial_i g_{00} = \frac{\partial_i \Phi}{c^2}. \quad (23.7)$$

For slow motion, $dt/d\tau \simeq 1$ and the leading spatial geodesic equation is $\ddot{x}^i \simeq -c^2 \Gamma_{00}^i = -\partial_i \Phi$. This recovers Newton's acceleration law. Deriving the source equation as well requires inserting the weak-field metric into the curvature tensors; the result is the Poisson equation stated above.

23.4 Schwarzschild geometry and tests

Spherical symmetry removes much of the complexity of the vacuum field equations. The Schwarzschild solution describes the exterior geometry under the stated assumptions. Rather than treating its metric as one intimidating formula, first read off what it says about a clock held at a fixed radius, then compare clocks at different radii.

Outside a spherical nonrotating mass in vacuum with $\Lambda = 0$,

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad r_s = \frac{2GM}{c^2}. \quad (23.8)$$

Static clocks satisfy $d\tau = \sqrt{1 - r_s/r} dt$. For static emission and reception outside the horizon,

$$\frac{\nu_r}{\nu_e} = \sqrt{\frac{1 - r_s/r_e}{1 - r_s/r_r}}. \quad (23.9)$$

Light climbing outward redshifts. Leading weak-field predictions include deflection $\alpha \simeq 4GM/(bc^2)$ and periapsis advance per orbit $\Delta\phi \simeq 6\pi GM/[a(1 - e^2)c^2]$.

The surface $r = r_s$ is an event horizon for a Schwarzschild black hole, not a material surface. The metric's singular form there is coordinate dependent; curvature diverges at $r = 0$ in the classical solution. Static observers cannot remain at the horizon.

For a static clock, $dr = d\Omega = 0$ and $ds^2 = -c^2 d\tau^2$, giving $d\tau = \sqrt{1 - r_s/r} dt$. A fixed coordinate-time interval is therefore less proper time for a clock closer to the mass. In this stationary geometry, successive wave crests have the same coordinate-time separation at emission and reception, so their locally measured frequencies have the ratio shown above.

Far from the source, expand the clock factor:

$$\sqrt{1 - \frac{2GM}{rc^2}} \simeq 1 - \frac{GM}{rc^2} = 1 + \frac{\Phi}{c^2}. \quad (23.10)$$

This connects the relativistic clock effect to the Newtonian potential. Light deflection and periastris advance require solving geodesics, not merely applying this clock factor; their formulas summarize that further calculation.

23.5 Gravitational waves and black-hole thermodynamics

A changing gravitational field can transport energy away from a source. For an orbiting binary, losing energy means the orbit tightens rather than simply slowing down at a fixed radius. Black-hole thermodynamics adds another layer: quantum fields in a black-hole background connect horizon area with entropy. These are distinct extensions of the geometry we have introduced.

Weak perturbations of flat spacetime have two transverse tensor polarizations propagating at c . Gravitational radiation is sourced at leading order by a changing mass quadrupole; isolated monopole and dipole radiation are excluded by conservation laws. For a circular weak-field binary of separation a ,

$$P = \frac{32}{5} \frac{G^4 \mu^2 M^3}{c^5 a^5}, \quad M = m_1 + m_2. \quad (23.11)$$

Energy loss shrinks the orbit and increases its frequency. Strong-field merger physics generally requires numerical relativity.

Semiclassical theory assigns a Schwarzschild black hole temperature and entropy

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}, \quad S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar}. \quad (23.12)$$

These connect gravity, quantum theory, and thermodynamics; a complete microscopic account of general quantum spacetime remains beyond established theory.

The leading binary power law can be turned into an inspiral law using Newtonian orbital energy $E = -G\mu M/(2a)$:

$$\frac{dE}{dt} = \frac{G\mu M}{2a^2} \dot{a} = -P \implies \dot{a} = -\frac{64}{5} \frac{G^3 \mu M^2}{c^5 a^3}. \quad (23.13)$$

The negative sign shows shrinking separation. Since orbital frequency scales as $a^{-3/2}$, the signal grows faster as the system loses energy. This argument assumes a slowly evolving circular weak-field orbit. The Hawking temperature and entropy, by contrast, are semiclassical results quoted here; they cannot be obtained by Newtonian energy balance alone.

Try it yourself

Describe in words how a freely falling object moves in curved spacetime. Give one observation that distinguishes the theory from Newtonian gravity.

Part VIII

Quantum Physics

Chapter 24

Quantum principles and mathematical structure

Classical waves taught us to track amplitudes and phase, while statistical physics taught us to distinguish microscopic possibilities from measured averages. Quantum mechanics combines those lessons in a new state space: amplitudes evolve linearly, but measurements return probabilities. Quantum mechanics changes the object we predict. Instead of assigning a definite trajectory to every particle, we evolve a state and calculate probabilities for specified measurements. The algebra of vectors, operators, and complex amplitudes is the machinery for doing that. We will distinguish the theory's postulates from consequences derived from them, so the starting assumptions stay visible.

24.1 Why classical physics is incomplete

A useful theory must survive contact with experiment. Discrete atomic spectra and the frequency dependence of photoelectron energies cannot be explained by simply refining a classical trajectory. They motivate a new framework. The photon energy relation is an empirical starting point for that framework, not something derived from the classical wave equation.

Classical equipartition predicts an ultraviolet divergence for thermal radiation. Classical electrodynamics does not yield stable atoms with discrete spectral lines. The photoelectric effect associates

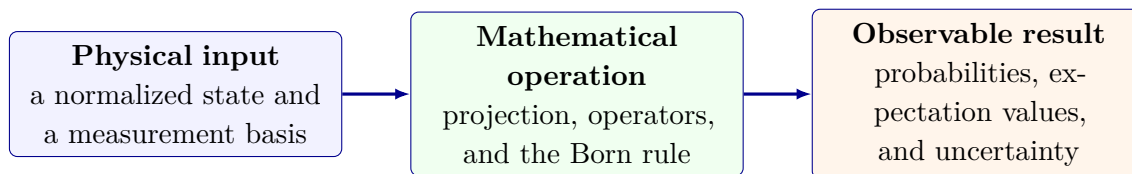


Figure 24.1: Visual model for quantum principles: the physical question becomes mathematics and returns as an observable prediction.

the emitted electron's maximum kinetic energy with frequency,

$$K_{\max} = h\nu - W, \quad (24.1)$$

where W is the material's work function in the ideal single-photon description. Increasing intensity changes the photon flux, not the energy $h\nu$ of each photon. Matter also has wave character, with de Broglie wavelength $\lambda = h/p$.

Quantum theory replaces definite classical trajectories with states and probability amplitudes. Classical behavior is recovered under appropriate scale, state, and decoherence conditions, rather than by simply declaring quantum effects absent for every large object.

In the ideal photoelectric model, one photon supplies $h\nu$. Removing an electron costs at least the work function W , leaving maximum kinetic energy $h\nu - W$. The threshold is therefore $\nu_0 = W/h$. Below it, increasing intensity cannot supply the missing energy to a single electron within the single-photon model. Above it, increasing frequency increases the maximum kinetic energy, while intensity mainly changes how many photons arrive. This separates energy per event from event rate.

24.2 States, observables, and the Born rule

Choose the measurement you want to predict and expand the state in that observable's eigenbasis. The expansion coefficients are amplitudes; their squared magnitudes are probabilities. This is the Born rule, one of the physical postulates. Linear algebra then lets us calculate the resulting averages and changes of basis. $|\psi\rangle$ is a state vector, α and β amplitudes, $|\alpha|^2$ and $|\beta|^2$ probabilities in a chosen basis, and $\hat{H}$ the Hamiltonian operator. A hat or operator symbol means the object acts on a state rather than being an ordinary number. The phase of an amplitude can be unobservable alone but becomes observable through interference with another amplitude.

A pure state is a normalized vector $|\psi\rangle$ in a complex Hilbert space, with physically irrelevant overall phase. Observables are represented by self-adjoint operators. For a discrete nondegenerate orthonormal eigenbasis $A|a_n\rangle = a_n|a_n\rangle$,

$$|\psi\rangle = \sum_n c_n |a_n\rangle, \quad P(a_n) = |c_n|^2, \quad \langle A \rangle = \langle \psi | A | \psi \rangle. \quad (24.2)$$

Degenerate outcomes use projectors onto eigenspaces. Continuous spectra replace sums with integrals and require appropriate normalization conventions. A position wavefunction satisfies

$$\int |\psi(\mathbf{r})|^2 d^3r = 1, \quad P(\mathbf{r} \in V) = \int_V |\psi|^2 d^3r. \quad (24.3)$$

An ideal projective measurement conditioned on outcome a changes the state to $P_a|\psi\rangle/\sqrt{\langle\psi|P_a|\psi\rangle}$. More general measurements can be represented by measurement operators.

Orthonormality gives $c_n = \langle a_n | \psi \rangle$ and normalization gives $\sum_n |c_n|^2 = 1$. Insert the expansion on both sides of A :

$$\langle \psi | A | \psi \rangle = \sum_{m,n} c_m^* c_n \langle a_m | A | a_n \rangle = \sum_n |c_n|^2 a_n. \quad (24.4)$$

Thus the operator expression is exactly the probability-weighted mean of measurement outcomes. An average need not be an allowed individual outcome. Multiplying all coefficients by the same phase changes no probability, while changing relative phases can change interference in another measurement basis.

24.3 Operators and uncertainty

The order of two operator actions can matter. Apply the position and momentum operators to a smooth test wavefunction and compare their two orders: the product rule leaves a nonzero remainder. The resulting commutator controls a lower bound on the spreads of the corresponding measurement distributions.

In the position representation, $\hat{\mathbf{p}} = -i\hbar\nabla$, and $[x_i, p_j] = i\hbar\delta_{ij}$. For suitable states and operator domains,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|, \quad \Delta x \Delta p_x \geq \hbar/2. \quad (24.5)$$

The spreads refer to repeated measurements on identically prepared states. This is not merely a statement about disturbing a particle with an instrument. Time usually enters nonrelativistic quantum mechanics as a parameter, so energy-time relations need interpretation different from the position-momentum commutator.

In one dimension,

$$(xp - px)\psi = -i\hbar x\psi' + i\hbar(x\psi)' = i\hbar\psi. \quad (24.6)$$

For the uncertainty bound, define $\tilde{A} = A - \langle A \rangle$ and $\tilde{B} = B - \langle B \rangle$. Cauchy-Schwarz applied to $\tilde{A}|\psi\rangle$ and $\tilde{B}|\psi\rangle$ gives

$$(\Delta A)^2 (\Delta B)^2 \geq |\langle \tilde{A} \tilde{B} \rangle|^2 \geq \left| \frac{\langle [A, B] \rangle}{2i} \right|^2. \quad (24.7)$$

The second step retains the imaginary part of the complex expectation value. Taking square roots gives the stated inequality. A state with very small position spread must therefore have a sufficiently broad momentum distribution, even before specifying a measurement apparatus.

24.4 Schrodinger evolution and continuity

The time-dependent Schrodinger equation is the dynamical postulate for this nonrelativistic model. Once the Hamiltonian and initial state are given, it predicts the evolving amplitudes. For a time-independent potential, try separating spatial shape from time dependence; this reveals why energy eigenstates acquire a simple phase.

For a spinless nonrelativistic particle,

$$i\hbar\partial_t\psi = \hat{H}\psi, \quad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t). \quad (24.8)$$

A time-independent Hamiltonian has stationary states $\psi_n = \phi_n e^{-iE_n t/\hbar}$ with $\hat{H}\phi_n = E_n \phi_n$. Their probability densities are stationary, although the phase evolves. Superpositions can have time-dependent interference.

For real scalar V , subtracting the conjugate Schrodinger equation gives

$$\partial_t |\psi|^2 + \nabla \cdot \mathbf{j} = 0, \quad \mathbf{j} = \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi). \quad (24.9)$$

With electromagnetic coupling, $\hat{H} = [-i\hbar \nabla - q\mathbf{A}]^2/(2m) + q\phi$ and the current includes $-q\mathbf{A}|\psi|^2/m$. Gauge transformations must transform the wavefunction phase as well as the potentials.

Insert $\psi(\mathbf{r}, t) = \phi(\mathbf{r})T(t)$ into the equation and separate variables:

$$i\hbar \frac{\dot{T}}{T} = \frac{\hat{H}\phi}{\phi} = E, \quad T(t) = T(0)e^{-iEt/\hbar}. \quad (24.10)$$

The spatial problem is the energy eigenvalue equation. To check probability conservation, differentiate $|\psi|^2$ and use both the equation and its conjugate:

$$\partial_t |\psi|^2 = \frac{i\hbar}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = -\nabla \cdot \left[\frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right]. \quad (24.11)$$

The real potential terms cancel. The bracket is the probability current, so the wave equation preserves total probability when no net probability escapes the boundary.

24.5 Ehrenfest's theorem and pictures of dynamics

A bridge to classical mechanics should compare measurable averages. Differentiate $\langle \psi | A | \psi \rangle$ using the product rule: one term comes from the bra, one from the operator if it depends on time, and one from the ket. Substituting Schrodinger evolution into the state terms produces a commutator.

$$\frac{d}{dt} \langle A \rangle = \frac{i}{\hbar} \langle [H, A] \rangle + \langle \partial_t A \rangle. \quad (24.12)$$

Thus $d\langle x \rangle/dt = \langle p \rangle/m$ and $d\langle p \rangle/dt = -\langle V'(x) \rangle$. These resemble Newton's equations, but generally $\langle V'(x) \rangle \neq V'(\langle x \rangle)$; a sufficiently narrow packet can make the classical approximation work.

In the Schrodinger picture states evolve; in the Heisenberg picture operators evolve. For time-independent H , $U(t) = e^{-iHt/\hbar}$ is unitary, preserving probabilities and inner products.

For $H = p^2/(2m) + V(x)$, the commutators are $[H, x] = -i\hbar p/m$ and $[H, p] = i\hbar V'(x)$. Ehrenfest's formula then yields the stated mean position and momentum equations. Expand the force around $\bar{x} = \langle x \rangle$ to see the approximation involved:

$$\langle V'(x) \rangle = V'(\bar{x}) + \frac{1}{2} V^{(3)}(\bar{x}) (\Delta x)^2 + \dots \quad (24.13)$$

The linear fluctuation term averages to zero. A narrow packet in a smoothly varying force can follow approximately classical mean motion; a quadratic potential makes the displayed correction vanish exactly.

Try it yourself

A state is a sum of two alternatives. Explain why amplitudes are added before probabilities are calculated. What does normalization guarantee?

Chapter 25

Exactly solvable quantum systems

The previous chapter introduced states and measurements. We now let the Hamiltonian evolve those states in concrete systems. Boundary conditions, just as in waves, select allowed patterns; the difference is that the patterns now describe probability amplitudes. Simple quantum models are valuable because each exposes a different mechanism: boundary conditions quantize a box, forbidden regions produce exponential tails, and operator algebra organizes an oscillator. In every case, begin with the potential, solve the appropriate differential or algebraic equation, and impose the conditions that make a state physically admissible.

25.1 Free particles and wave packets

A plane wave has one exact momentum and extends across all space. To describe a localized particle, superpose momenta. Their different propagation speeds then reshape the packet even without a force. Free spreading is therefore compatible with constant average momentum.

A momentum eigenfunction is proportional to $e^{i(kx-\omega t)}$ with $p = \hbar k$ and $E = \hbar\omega = \hbar^2 k^2 / (2m)$. It is not square-integrable on an infinite line; physical localized states are superpositions. For an initially minimum-uncertainty Gaussian with position variance σ_0^2 and no initial position-momentum covariance,

$$\sigma_x^2(t) = \sigma_0^2 + \frac{\hbar^2 t^2}{4m^2 \sigma_0^2}. \quad (25.1)$$

Localization produces momentum spread, which causes dispersive broadening.

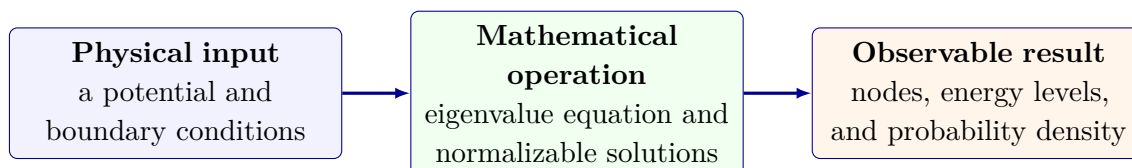


Figure 25.1: Visual model for solvable quantum systems: the physical question becomes mathematics and returns as an observable prediction.

In the Heisenberg description, $x(t) = x(0) + p(0)t/m$. The variance consequently satisfies

$$\sigma_x^2(t) = \sigma_x^2(0) + \frac{2t}{m} \text{Cov}(x, p) + \frac{t^2}{m^2} \sigma_p^2(0), \quad (25.2)$$

where the covariance uses the symmetrized product. For the initially uncorrelated minimum-uncertainty Gaussian, covariance is zero and $\sigma_p = \hbar/(2\sigma_0)$. Substitution produces the spreading law above. A smaller initial width gives a larger momentum spread and hence faster eventual broadening.

25.2 Infinite and finite wells

For the infinite well, begin inside the region where the potential is zero. Solve there first, then demand that the wavefunction vanish at both walls. Those two boundary conditions eliminate most mathematical solutions and leave only a discrete set of energies. $V(x)$ is potential energy as a function of position, ϕ_n a stationary spatial amplitude, E_n its energy, and n an integer level label. The wavefunction must be normalized, so $|\phi|^2$ can be interpreted as a probability density. Nodes are points where the amplitude is zero, not places where the particle is classically stuck.

For $V = 0$ on $0 < x < L$ and infinite walls,

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, \dots \quad (25.3)$$

Boundary conditions quantize the wave number. The ground state has nonzero kinetic energy because confinement forbids a zero-momentum constant wavefunction satisfying both walls.

For finite steps with equal mass and finite potential, ψ and ψ' are continuous. Bound states penetrate classically forbidden regions with decay constant $\kappa = \sqrt{2m(V - E)}/\hbar$. Singular delta potentials instead produce a derivative jump fixed by integrating the Schrodinger equation across the singularity.

Worked example 25.1 (A superposition in a box). For $\psi(x, 0) = (\phi_1 + \phi_2)/\sqrt{2}$, the mean energy is $(E_1 + E_2)/2$, while an energy measurement yields E_1 or E_2 with equal probability. The density contains $\phi_1 \phi_2 \cos[(E_2 - E_1)t/\hbar]$, so it oscillates despite the time-independent Hamiltonian.

The stationary equation inside the well is

$$-\frac{\hbar^2}{2m} \phi'' = E\phi \implies \phi = A \sin kx + B \cos kx, \quad k^2 = \frac{2mE}{\hbar^2}. \quad (25.4)$$

At $x = 0$, $\phi = 0$ sets $B = 0$. At $x = L$, it requires $kL = n\pi$. The $n = 0$ case is the zero function, which cannot be normalized; negative n duplicates the same states up to a sign. Finally,

$$1 = |A|^2 \int_0^L \sin^2 \frac{n\pi x}{L} dx = |A|^2 \frac{L}{2} \quad (25.5)$$

fixes $A = \sqrt{2/L}$ up to an irrelevant phase. For a finite barrier, the equation becomes $\phi'' = \kappa^2 \phi$ in a forbidden region. This yields exponential tails instead of an abrupt zero at the interface.

25.3 Tunneling through a barrier

Divide the line into the left free region, the barrier, and the right free region. Solve the stationary equation in each, then match both the wavefunction and its derivative at each finite interface. Tunneling emerges from that matching; it does not require inserting a separate rule that permits a particle to cross.

For a rectangular barrier of height V_0 and width a , with zero potential on both sides and $0 < E < V_0$,

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)} \right]^{-1}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \quad (25.6)$$

For a thick barrier, the dominant suppression is $e^{-2\kappa a}$. Transmission probability is a ratio of probability currents; when the asymptotic speeds differ, it is not just a squared amplitude ratio. Tunneling does not mean the particle temporarily violates energy conservation.

With unit incident amplitude, choose

$$\psi_I = e^{ikx} + re^{-ikx}, \quad \psi_{II} = Ce^{\kappa x} + De^{-\kappa x}, \quad \psi_{III} = te^{ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (25.7)$$

There is no incoming wave from the right. The four matching conditions are

$$1 + r = C + D, \quad ik(1 - r) = \kappa(C - D), \quad (25.8)$$

$$te^{ika} = Ce^{\kappa a} + De^{-\kappa a}, \quad ikte^{ika} = \kappa(Ce^{\kappa a} - De^{-\kappa a}). \quad (25.9)$$

Eliminating C, D, r gives

$$\frac{1}{T} = 1 + \frac{(k^2 + \kappa^2)^2}{4k^2\kappa^2} \sinh^2(\kappa a). \quad (25.10)$$

Use $k^2 + \kappa^2 = 2mV_0/\hbar^2$ to recover the quoted formula. Equal asymptotic wave numbers make $T = |t|^2$ here. For $\kappa a \gg 1$, $\sinh^2 \kappa a \simeq e^{2\kappa a}/4$, exposing the exponential suppression.

25.4 The harmonic oscillator

The oscillator potential is quadratic, so position and momentum enter symmetrically after suitable rescaling. Ladder operators exploit that symmetry. Instead of solving a new differential equation for every level, we will construct the spectrum by identifying an operator that raises or lowers energy by a fixed amount.

For $H = p^2/(2m) + m\omega^2 x^2/2$, define

$$a = \sqrt{\frac{m\omega}{2\hbar}} x + \frac{ip}{\sqrt{2m\hbar\omega}}, \quad [a, a^\dagger] = 1. \quad (25.11)$$

Then $H = \hbar\omega(a^\dagger a + 1/2)$, with

$$E_n = \hbar\omega(n + 1/2), \quad a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (25.12)$$

The ground state is Gaussian and has $\Delta x \Delta p = \hbar/2$. A coherent state satisfies $a|\alpha\rangle = \alpha|\alpha\rangle$; its expectation values follow the classical oscillator motion while retaining quantum fluctuations.

Multiply the definition of a by its adjoint and retain the order of x and p :

$$a^\dagger a = \frac{m\omega x^2}{2\hbar} + \frac{p^2}{2m\hbar\omega} + \frac{i[x, p]}{2\hbar} = \frac{H}{\hbar\omega} - \frac{1}{2}. \quad (25.13)$$

The last term comes from the commutator and produces the zero-point energy. Positivity of $\|a|\psi\rangle\|^2$ prevents repeated lowering from continuing to negative occupation. The lowest state obeys $a|0\rangle = 0$. In position space,

$$\left(x + \frac{\hbar}{m\omega} \frac{d}{dx}\right) \phi_0 = 0 \implies \phi_0(x) \propto e^{-m\omega x^2/(2\hbar)}. \quad (25.14)$$

Repeated raising then generates excited states, with normalization fixing the square-root coefficients in the ladder formulas.

25.5 Angular momentum and spin

Quantum angular momentum describes how a state responds to rotations. Its components do not all commute, so we choose a compatible pair, conventionally J^2 and J_z . Raising and lowering the z component while keeping total angular momentum fixed produces a finite chain of allowed outcomes.

Angular momentum obeys $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$. Simultaneous eigenstates satisfy

$$J^2|jm\rangle = \hbar^2 j(j+1)|jm\rangle, \quad J_z|jm\rangle = \hbar m|jm\rangle, \quad m = -j, \dots, j. \quad (25.15)$$

Orbital angular momentum has integer ℓ and angular wavefunctions $Y_{\ell m}$. Intrinsic spin may be integer or half-integer and is not literally a little sphere spinning in ordinary space.

For spin one-half, $\mathbf{S} = \hbar\boldsymbol{\sigma}/2$, where

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (25.16)$$

A spin state aligned along a unit vector $\hat{\mathbf{n}}$ has probability $\cos^2(\theta/2)$ of being measured spin-up along an axis making angle θ with $\hat{\mathbf{n}}$. Adding angular momenta gives $j = |j_1 - j_2|, \dots, j_1 + j_2$.

Define $J_\pm = J_x \pm iJ_y$. The commutators imply $[J_z, J_\pm] = \pm\hbar J_\pm$ and

$$\|J_\pm|jm\rangle\|^2 = \hbar^2[j(j+1) - m(m \pm 1)]. \quad (25.17)$$

A norm cannot be negative, so the ladder stops at $m = j$ above and $m = -j$ below. For spin one-half, a state pointing at polar angle θ and azimuth ϕ can be written

$$|+\hat{\mathbf{n}}\rangle = \cos \frac{\theta}{2} |+\rangle + e^{i\phi} \sin \frac{\theta}{2} |-\rangle. \quad (25.18)$$

Projection onto $|+\rangle$ and the Born rule give the $\cos^2(\theta/2)$ probability. The half-angle is a property of spinor rotations, not the classical projection of a rotating little ball.

25.6 Hydrogenic atoms

The Coulomb potential depends only on radius, so separate angular dependence using angular-momentum eigenfunctions. The remaining radial equation contains both attraction and an angular-momentum barrier. Bound-state energies are selected by regularity at the origin and decay at infinity, much as box energies were selected by the walls.

For a point nucleus of charge $+Ze$ and a single electron, use reduced mass μ and potential $-Ze^2/(4\pi\epsilon_0 r)$. Separation gives $\psi_{n\ell m} = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$ and

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}, \quad a_\mu = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2}. \quad (25.19)$$

For ordinary hydrogen, the ground-state binding energy is approximately 13.6 eV and the characteristic radius is about $a_0 = 5.29 \times 10^{-11}$ m. Allowed labels are $n \geq 1$, $0 \leq \ell \leq n-1$, and $-\ell \leq m \leq \ell$. Spin, relativistic effects, nuclear structure, and radiative corrections refine this Coulomb spectrum.

Write $u(r) = rR(r)$ and $\kappa = Ze^2/(4\pi\epsilon_0)$. Separation gives

$$-\frac{\hbar^2}{2\mu}u'' + \left[\frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} - \frac{\kappa}{r} \right] u = Eu. \quad (25.20)$$

For the ground state, try $R(r) = Ae^{-r/a}$ with $\ell = 0$. Its radial Laplacian is $(a^{-2} - 2/(ar))R$, so the Schrodinger equation becomes

$$\left[-\frac{\hbar^2}{2\mu a^2} + \frac{1}{r} \left(\frac{\hbar^2}{\mu a} - \kappa \right) \right] R = ER. \quad (25.21)$$

The $1/r$ coefficient must vanish: $a = \hbar^2/(\mu\kappa) = a_\mu/Z$. The remaining constant gives $E_1 = -\mu\kappa^2/(2\hbar^2)$. Higher levels require solving the radial polynomial problem with the same boundary conditions; that further step yields the $1/n^2$ spectrum.

Try it yourself

For a particle in an infinite well, sketch the lowest stationary wavefunction. Identify where it must vanish and why its energy is not zero.

Chapter 26

Quantum approximation methods and scattering

Exact models are our training ground, not the usual destination. The same approximation habits learned from Taylor series, small oscillations, and geometric optics now return in quantum form: identify a small parameter, keep the leading correction, and test where the approximation breaks. Exact solutions are landmarks, not the typical case. When a problem differs slightly from a solvable one, we can organize corrections. When no nearby exact solution is available, we may seek a bound or a semiclassical approximation. The important question is not just how to apply a method, but which small parameter, gap, or length scale makes it trustworthy.

26.1 Time-independent perturbation theory

Start with a Hamiltonian whose eigenstates you know, and describe the remaining interaction as a perturbation. Expand both the energy and the state in the same bookkeeping parameter. Matching equal powers then turns one hard eigenproblem into a sequence of simpler equations involving the known states. H_0 is the solvable reference Hamiltonian, V the added perturbation, and λ a bookkeeping parameter that is set to one only after the expansion is organized.

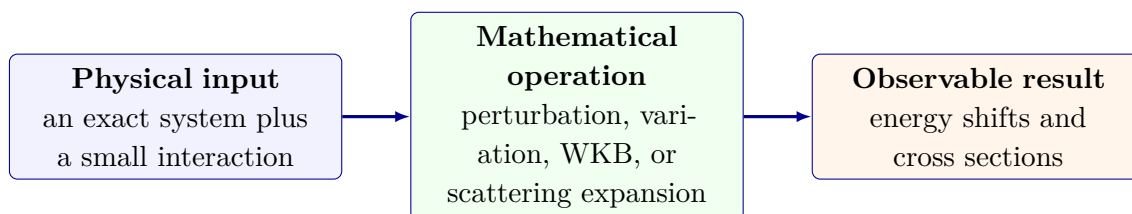


Figure 26.1: Visual model for quantum approximations: the physical question becomes mathematics and returns as an observable prediction.

Let $H = H_0 + \lambda V$. For a nondegenerate unperturbed level,

$$E_n = E_n^{(0)} + \lambda \langle n|V|n \rangle + \lambda^2 \sum_{m \neq n} \frac{|\langle m|V|n \rangle|^2}{E_n^{(0)} - E_m^{(0)}} + \cdots, \quad (26.1)$$

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m|V|n \rangle}{E_n^{(0)} - E_m^{(0)}} |m\rangle. \quad (26.2)$$

The states on the right are unperturbed. The expansion fails near degeneracy unless the perturbation is first diagonalized within the degenerate or nearly degenerate subspace.

Worked example 26.1 (Weak quartic anharmonicity). For $V = \lambda x^4$ added to a harmonic oscillator, the ground-state Gaussian has $\langle x^4 \rangle = 3[\hbar/(2m\omega)]^2$. The first-order shift is therefore $\Delta E_0 = 3\lambda\hbar^2/(4m^2\omega^2)$. Its smallness relative to $\hbar\omega$ provides a practical perturbative criterion.

Use $E_n = E_n^{(0)} + \lambda E_n^{(1)} + \cdots$ and $|\psi_n\rangle = |n\rangle + \lambda|n^{(1)}\rangle + \cdots$. At first order,

$$(H_0 - E_n^{(0)})|n^{(1)}\rangle = -(V - E_n^{(1)})|n\rangle. \quad (26.3)$$

Project onto $\langle n|$ to obtain $E_n^{(1)} = \langle n|V|n\rangle$. Project onto $\langle m|$ for $m \neq n$ to obtain

$$\langle m|n^{(1)}\rangle = \frac{\langle m|V|n\rangle}{E_n^{(0)} - E_m^{(0)}}. \quad (26.4)$$

With $\langle n|n^{(1)}\rangle = 0$ as a normalization convention, these projections construct the first-order state. At second order, projection onto $\langle n|$ gives $E_n^{(2)} = \langle n|V|n^{(1)}\rangle$, producing the stated sum. Small denominators are a warning: even a modest matrix element can strongly mix nearly degenerate states.

26.2 Variational principle and WKB

The variational method asks how low an energy you can obtain within a chosen family of trial states. WKB asks how a wave evolves when its local wavelength changes slowly. These approaches use different information and answer different questions; neither requires an exact solution of the full eigenvalue problem.

For any normalized trial state in the Hamiltonian's domain,

$$E_0 \leq \langle \psi_{\text{trial}} | H | \psi_{\text{trial}} \rangle. \quad (26.5)$$

Expanding the trial state in energy eigenstates proves the bound: its expected energy is a weighted average no lower than the ground energy. Optimizing trial parameters improves an upper bound, not necessarily the exact wavefunction everywhere.

In one dimension where the local momentum varies slowly,

$$\psi(x) \approx \frac{1}{\sqrt{p(x)}} \left[A e^{\frac{i}{\hbar} \int^x p dx'} + B e^{-\frac{i}{\hbar} \int^x p dx'} \right], \quad p = \sqrt{2m(E - V)}. \quad (26.6)$$

WKB fails near turning points and requires matching. For two smooth simple turning points,

$$\int_{x_1}^{x_2} p dx = (n + 1/2)\pi\hbar. \quad (26.7)$$

A smooth thick barrier gives leading transmission

$$T \sim \exp \left[-\frac{2}{\hbar} \int \sqrt{2m(V - E)} dx \right], \quad (26.8)$$

where the integral extends over the forbidden interval.

For a normalized trial state $|\psi\rangle = \sum_n c_n |n\rangle$, the energy difference is

$$\langle\psi|H|\psi\rangle - E_0 = \sum_n |c_n|^2 (E_n - E_0) \geq 0. \quad (26.9)$$

This is the entire upper-bound argument. A concrete trial width b for an oscillator, $\psi \propto e^{-x^2/(2b^2)}$, gives

$$E(b) = \frac{\hbar^2}{4mb^2} + \frac{m\omega^2 b^2}{4}. \quad (26.10)$$

Narrowing lowers potential energy but raises kinetic energy. Setting $dE/db = 0$ yields $b^2 = \hbar/(m\omega)$ and $E = \hbar\omega/2$; the result is exact because this trial family contains the true ground state.

For WKB, insert $\psi = A(x)e^{iS(x)/\hbar}$ into the stationary equation. At leading order, $S'^2 = 2m(E - V) = p^2$. At next order, $2A'S' + AS'' = 0$, or $(A^2 S')' = 0$, so $A \propto p^{-1/2}$. This explains both the phase integral and the amplitude factor. At a turning point $p = 0$, that factor diverges and the approximation must be replaced locally and matched.

26.3 Time-dependent transitions

A time-dependent perturbation can transfer amplitude between energy eigenstates. Separate the phase each state would acquire on its own from the changing expansion coefficients. The resulting equation shows that transitions depend on both the coupling matrix element and the relative phase accumulated during the interaction.

To first order for a perturbation $V(t)$ switched on at $t = 0$,

$$c_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t \langle f|V(t')|i\rangle e^{i(E_f - E_i)t'/\hbar} dt'. \quad (26.11)$$

For weak coupling to a dense set of final states and a suitable long-time regime, Fermi's golden rule is

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f), \quad (26.12)$$

with energy conservation selecting the resonant final energy, including any energy supplied by a periodic drive. It is a rate approximation, not a universal short-time law.

An isolated resonantly driven two-level system instead exhibits coherent Rabi oscillations, $P_e(t) = \sin^2(\Omega_R t/2)$ under the rotating-wave approximation and standard definition of Ω_R .

Write $|\psi\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle$ using the unperturbed eigenbasis. Substitution into Schrodinger's equation cancels the unperturbed evolution and leaves

$$i\hbar \dot{c}_f = \sum_n V_{fn}(t) c_n(t) e^{i(E_f - E_n)t/\hbar}. \quad (26.13)$$

For a weak perturbation and initial state i , replace c_i by one and the other coefficients by zero on the right. Integration gives the first-order amplitude above. A rapidly oscillating phase tends to cancel contributions from different times; a resonant drive compensates that phase and lets amplitude accumulate. A dense continuum can support a rate description, while an isolated pair of levels can coherently exchange population back and forth.

26.4 Adiabatic evolution and geometric phase

Slow change is meaningful only relative to the system's internal time scales. If two levels approach each other, their small energy gap makes transitions easier and the needed change may have to be much slower. Even when the system remains in one instantaneous eigenstate, its phase records more than the integral of its energy.

A system follows an instantaneous nondegenerate eigenstate approximately when changes are slow compared with the relevant inverse energy gaps and transition matrix elements are small. Closing gaps can defeat this approximation. In a cyclic parameter change, the state can acquire both a dynamical phase and Berry phase

$$\gamma_n = i \oint \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle \cdot d\mathbf{R}. \quad (26.14)$$

The accumulated geometric phase can have observable interference consequences.

For parameters changing in time, try $|\psi\rangle = e^{-\frac{i}{\hbar} \int E_n dt} e^{i\gamma_n(t)} |n(t)\rangle$ and neglect transitions out of level n . Projecting Schrodinger's equation onto $\langle n(t) |$ gives

$$\dot{\gamma}_n = i \langle n | \dot{n} \rangle. \quad (26.15)$$

Integrating along a closed parameter path produces the Berry phase. The eigenvector's phase convention can change the local integrand, but the closed-loop phase modulo 2π , with consistent endpoint identification, has observable interference consequences. This is another reason to distinguish relative phase from an arbitrary overall phase at one instant.

26.5 Scattering and cross sections

A scattering experiment prepares incoming flux and counts particles emerging into different directions. Solve for the outgoing amplitude far from the target, where its radial spreading is simple. The cross section translates that amplitude into a count rate normalized by the incident beam. In

scattering, $f(\theta, \phi)$ is an amplitude while $d\sigma/d\Omega$ is a measurable cross section; amplitude is squared only after all alternatives have been added.

For an incoming plane wave and outgoing spherical wave in three dimensions,

$$\psi \sim e^{ikz} + f(\theta, \phi) \frac{e^{ikr}}{r}, \quad \frac{d\sigma}{d\Omega} = |f|^2 \quad (26.16)$$

for elastic scattering with equal incident and outgoing wave numbers. The differential cross section is scattered rate per solid angle divided by incident flux; it has units of area per solid angle.

For a short-range central potential,

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta), \quad \sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{\ell} (2\ell+1) \sin^2 \delta_{\ell}. \quad (26.17)$$

At sufficiently low energy, the s wave often dominates. Long-range Coulomb scattering requires special treatment of the asymptotic form. The first Born approximation treats a weak scattering potential perturbatively; it is not justified merely because an integral is easy to evaluate.

For incident amplitude one, $j_{\text{in}} = \hbar k/m$. Far away, the outgoing spherical term gives radial current $j_r \simeq (\hbar k/m) |f|^2 / r^2$. A detector subtending solid angle $d\Omega$ intercepts area $r^2 d\Omega$, so

$$\frac{j_r r^2 d\Omega}{j_{\text{in}}} = |f|^2 d\Omega. \quad (26.18)$$

The distance cancels, explaining $d\sigma/d\Omega = |f|^2$. For a central potential, partial waves separate the angular-momentum channels; phase shifts measure how much each radial wave differs from free motion. The complicated interaction region is encoded in those shifts before the outgoing channels are recombined.

Try it yourself

Before using perturbation theory, what should you compare to judge whether the added interaction is small? Name one result that the approximation can predict.

Chapter 27

Identical particles, entanglement, and quantum information

A single quantum state was already a vector, but several quantum systems require a larger space built from tensor products. This is where the linear algebra introduced earlier becomes physically consequential: correlations can belong to the joint state even when no subsystem has a pure state of its own. With several quantum systems, the state space grows by tensor products. This introduces two ideas with no direct classical-particle counterpart: exchanging identical particles constrains the allowed states, and a joint state need not assign a pure state to each part. Density operators let us describe both incomplete preparation knowledge and subsystems of entangled states.

27.1 Exchange symmetry

Labels such as “particle 1” and “particle 2” are bookkeeping devices for identical particles. Swapping them must not produce a physically distinguishable labeling of the same configuration. Bosonic and fermionic exchange symmetry impose different amplitude relations, with important consequences even when the Hamiltonian contains no explicit exchange-force term.

For identical bosons the total state is symmetric under exchange; for identical fermions it is antisymmetric. Two fermions cannot occupy the same complete one-particle state. A Slater determinant implements antisymmetry for independent-particle orbitals.

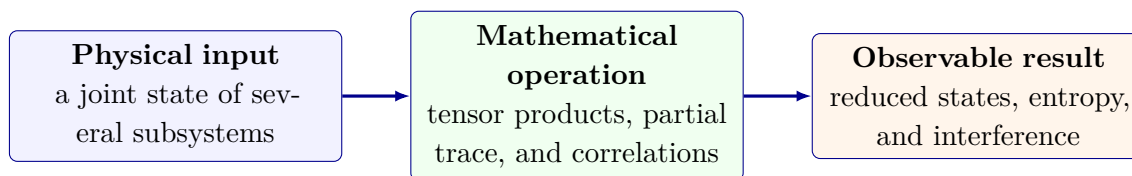


Figure 27.1: Visual model for entanglement and quantum information: the physical question becomes mathematics and returns as an observable prediction.

For two electrons, a spin singlet is antisymmetric in spin and must be paired with a symmetric spatial state; spin triplets require antisymmetric spatial states. Exchange effects do not arise from an extra classical force, although they can strongly modify energies and correlations.

For orthonormal orbitals a, b , form

$$\psi_{\pm}(1, 2) = \frac{\phi_a(1)\phi_b(2) \pm \phi_b(1)\phi_a(2)}{\sqrt{2}}. \quad (27.1)$$

Exchange preserves the plus state and reverses the sign of the minus state. If $a = b$, the antisymmetric numerator vanishes identically, so no normalized state exists: this is Pauli exclusion. For electrons, apply antisymmetry to the complete spatial-and-spin state. A symmetric spatial factor is allowed when the spin factor is antisymmetric.

27.2 Density operators and subsystems

Suppose a preparation produces state $|\psi_i\rangle$ with probability p_i , and you do not know which occurred. Average the measurement predictions, not the amplitudes. The density operator collects that average into one object and also handles the state available to an observer who can access only part of a larger system. ρ is a density operator, ρ_{AB} the joint state, and $\rho_A = \text{Tr}_B \rho_{AB}$ the state available when subsystem B is ignored. Purity $\text{Tr} \rho^2$ distinguishes pure from mixed states.

A statistical mixture is represented by

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad \text{Tr} \rho = 1, \quad \rho \geq 0. \quad (27.2)$$

Expectations are $\langle A \rangle = \text{Tr}(\rho A)$. A pure state has $\rho^2 = \rho$; a mixed state has $\text{Tr} \rho^2 < 1$. Closed-system evolution obeys $i\hbar\dot{\rho} = [H, \rho]$. The reduced state of subsystem A is $\rho_A = \text{Tr}_B \rho_{AB}$.

For an observable A ,

$$\langle A \rangle = \sum_i p_i \langle \psi_i | A | \psi_i \rangle = \text{Tr} \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| A \right). \quad (27.3)$$

This motivates ρ . A partial trace is defined by requiring all local predictions to agree: $\text{Tr}_{AB}[\rho_{AB}(A \otimes I_B)] = \text{Tr}_A(\rho_A A)$. In an orthonormal basis $|b\rangle$ of the inaccessible subsystem,

$$\rho_A = \sum_b \langle b | \rho_{AB} | b \rangle, \quad (27.4)$$

where the contraction is only over subsystem B . We are discarding access to its outcomes, not asserting that the global state ceased to exist.

27.3 Entanglement and correlations

A joint pure state is entangled when it cannot be written as one state for each subsystem multiplied together. The Bell state provides a small example whose entire algebra can be inspected. Its local

uncertainty coexists with perfect correlations between selected joint outcomes. Entanglement is a property of factorization, not merely of a probability distribution.

The Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (27.5)$$

cannot be factored into a product of states of its two parts. Each subsystem has $\rho_A = I/2$, although the combined state is pure. Entanglement entropy for a pure bipartite state is $S_A = -k_B \text{Tr}(\rho_A \ln \rho_A)$.

Bell inequalities constrain local hidden-variable models under their stated assumptions; suitable quantum correlations violate them. Such correlations do not permit controllable faster-than-light signaling. Measurement outcomes are correlated, while each observer's unconditioned local statistics remain unaffected by the distant measurement choice.

Expand its density operator:

$$\rho_{AB} = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|). \quad (27.6)$$

Tracing over B kills the cross terms because $\langle 0|1\rangle = 0$, leaving $\rho_A = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2$. The eigenvalues are both $1/2$, so purity is $1/2$ and entropy is $k_B \ln 2$. The global state's purity remains one. Local mixedness alone does not prove entanglement for an arbitrary mixed joint state, but for a pure joint state it signals entanglement.

27.4 Qubits, interference, and decoherence

A qubit's relative phase matters because future operations can convert it into an observable probability difference. Losing access to that phase information changes what interference can reveal. We can see the distinction with two Hadamard operations and with a simple model of environmental entanglement.

A qubit is $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. A Hadamard operation maps $|0\rangle$ to $(|0\rangle + |1\rangle)/\sqrt{2}$; a controlled-NOT flips a target when its control is $|1\rangle$. Together they can create the Bell state from $|00\rangle$.

A quantum computation manipulates amplitudes so useful outcomes interfere constructively. It does not make every amplitude individually readable. Unknown arbitrary quantum states cannot be perfectly cloned by a universal operation. Quantum error correction encodes information in a larger space and extracts error syndromes without directly measuring the encoded logical amplitudes.

Environmental entanglement suppresses interference in a subsystem's reduced state: this is decoherence. It explains the practical emergence of preferred stable observables, but should not be silently identified with every interpretive claim about definite measurement outcomes.

Since $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, applying H twice returns $|0\rangle$. The unwanted output amplitudes cancel. By contrast, an equal incoherent mixture remains $I/2$ under a Hadamard and gives equal output probabilities.

If an environment becomes correlated with the qubit, the joint state may be $\alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle$. After tracing out the environment, the qubit's off-diagonal element is

$$(\rho_{\text{qubit}})_{01} = \alpha\beta^*\langle e_1|e_0\rangle. \quad (27.7)$$

Nearly orthogonal environmental records suppress this coherence. This calculation shows how interference can become inaccessible locally while the full system still undergoes quantum evolution.

Try it yourself

A pair of qubits is in an entangled pure state. Explain how one qubit can have a mixed reduced state even though the pair has a pure state.

Part IX

Quantum Computing

Chapter 28

Quantum computing from bits to algorithms

The preceding quantum chapters developed states, operators, tensor products, measurement, and entanglement. Quantum computing uses those ideas deliberately as an information-processing language. We will move slowly from the familiar classical bit to interference that can be arranged into an algorithm. Quantum computing is not simply faster computation with smaller hardware. It is a theory of information in which amplitudes evolve linearly, measurement produces probabilities, and carefully arranged interference changes which answers are likely to be observed. We will begin with ordinary bits, then add one new idea at a time: a qubit, a gate, a circuit, an algorithm, and finally the noise that makes real devices difficult.

28.1 Bits, states, and reversible operations

A classical bit is either 0 or 1. A register of n bits has one definite string at a time, although an algorithm may manipulate many possible inputs over many runs. A reversible classical operation is a permutation of those strings; it has an inverse and loses no information.

For two bits, a controlled-NOT maps (a, b) to $(a, a \text{ xor } b)$. Applying it twice returns the original pair. Reversibility matters because a quantum gate must be represented by a unitary transformation, and unitary transformations preserve inner products and total probability.

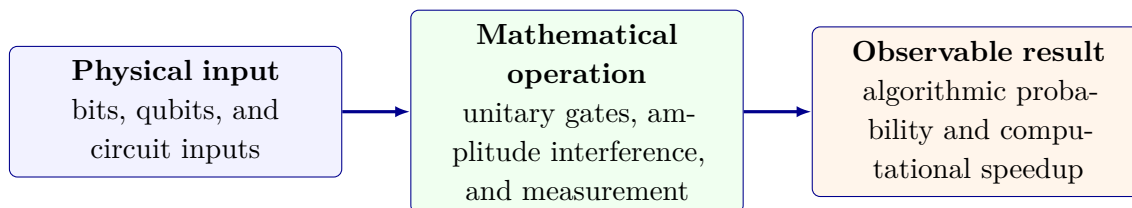


Figure 28.1: Visual model for quantum computing: the physical question becomes mathematics and returns as an observable prediction.

28.2 One qubit and the Bloch sphere

A qubit is a normalized complex vector

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (28.1)$$

The coefficients are amplitudes, not probabilities. Measuring in the computational basis returns 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$. The relative phase between α and β can later change an interference pattern. α, β are qubit amplitudes, H is the Hadamard gate, and N is the number of search items in amplitude amplification.

Ignoring an overall phase, every pure qubit can be written

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (28.2)$$

The pair (θ, ϕ) places the state on the Bloch sphere. The sphere is a picture of a two-complex-dimensional vector after normalization and removal of one physically irrelevant phase; it is not an ordinary position space.

28.3 Gates as rotations and interference

Quantum gates are unitary matrices. The Pauli matrices generate rotations, while the Hadamard gate changes between computational and interference-friendly bases:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (28.3)$$

Applying H twice returns the input. The second application adds amplitudes with signs, so cancellation and reinforcement occur before measurement. This is the central visual idea of quantum algorithms. Gates are unitary matrices, so they preserve $|\alpha|^2 + |\beta|^2 = 1$.

For a general state, a phase gate changes β without changing computational-basis probabilities immediately. A later Hadamard converts that hidden phase into a measurable probability difference. Therefore an experiment must display both amplitudes and the final measurement histogram; either one alone hides part of the calculation.

28.4 Registers, tensor products, and circuits

Two qubits occupy the tensor-product space with basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. A product state has amplitudes that factor, while an entangled state does not. A circuit is a time-ordered product of gates, with wires showing subsystems and measurement symbols showing where probabilities are extracted.

The controlled-NOT acts as

$$\text{CNOT}|a, b\rangle = |a, a \text{ xor } b\rangle. \quad (28.4)$$

Starting from $|00\rangle$, apply H to the first qubit and then CNOT:

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (28.5)$$

The final state is entangled. Measuring one qubit predicts the corresponding result of the other, but neither result is individually determined before measurement.

28.5 A first algorithm: amplitude amplification

Suppose one item among N possibilities is marked by an oracle. Classical search needs order N queries in the worst case. Grover's algorithm prepares a uniform superposition, flips the marked amplitude, and reflects amplitudes about their average. Each round rotates the state toward the marked direction. In Grover's formula, θ is the angle toward the marked state and k counts iterations, not time steps.

In the two-dimensional space spanned by the marked state $|w\rangle$ and the normalized unmarked state $|r\rangle$, write

$$|\psi\rangle = \sin \theta |w\rangle + \cos \theta |r\rangle, \quad \sin \theta = \frac{1}{\sqrt{N}}. \quad (28.6)$$

The oracle reflection and diffusion reflection together rotate by approximately 2θ . After k iterations,

$$P_k = \sin^2[(2k+1)\theta]. \quad (28.7)$$

The quadratic speedup comes from rotation in amplitude space, not from reading all amplitudes individually. Measuring too late rotates past the marked direction, so the number of iterations is part of the algorithm.

28.6 Noise, error correction, and what the picture must show

Real qubits lose phase coherence, suffer gate errors, and can be measured imperfectly. A density matrix describes the resulting mixed state. Error correction encodes one logical qubit into several physical qubits so that error syndromes can be measured without directly revealing the logical state.

For a simple phase-damping model, the off-diagonal density-matrix element evolves as

$$\rho_{01}(t) = \rho_{01}(0)e^{-t/T_2}, \quad (28.8)$$

while the diagonal probabilities remain fixed. A useful visual therefore shows the Bloch vector shrinking toward the measurement axis, alongside the circuit and the changing probability histogram. The picture must distinguish a changing amplitude from a changing classical probability and must report the noise model used.

Try it yourself

Apply a Hadamard gate twice to the state $|0\rangle$. What state results, and what does this show about interference?

Part X

Atoms, Materials, Nuclei, and Particles

Chapter 29

Atomic and molecular physics

Quantum mechanics first gave us ideal particles and ideal potentials. Atoms ask what happens when those states interact with fields, nuclei, and other electrons; molecules then ask how the same quantum structure produces collective bonds and vibrations. The hydrogen atom gave us a starting model. Real atoms add electron-electron interactions, spin couplings, and corrections at smaller energy scales. Molecules add moving nuclei and bonding. We will build a hierarchy of approximations, using each simpler model to identify which additional terms are likely to matter next.

29.1 Atomic structure beyond hydrogen

Write the interactions before choosing orbitals. For a fixed nucleus, each electron is attracted to the nucleus and repelled by every other electron. The repulsion depends on pairs of coordinates, so it prevents the Hamiltonian from separating into independent one-electron hydrogen problems. Z is nuclear charge number, n principal quantum number, ℓ orbital angular-momentum label, and m its projection label.

In many-electron atoms, electron-electron repulsion prevents exact separation into independent hydrogenic problems. Useful approximations include a central mean field, Hartree–Fock orbitals with exchange, and correlation corrections. Shell structure, spin, and Pauli exclusion organize the periodic table.

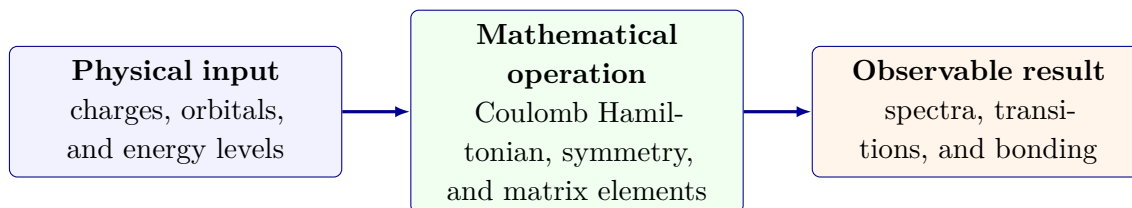


Figure 29.1: Visual model for atomic and molecular physics: the physical question becomes mathematics and returns as an observable prediction.

Fine structure combines relativistic kinetic corrections, spin-orbit coupling, and the Darwin term. Hyperfine structure couples nuclear and electronic angular momentum. Radiative corrections and finite nuclear size produce additional shifts, including the Lamb shift. These corrections are smaller than gross Coulomb energies but essential for precision spectra.

Ignoring relativistic effects and nuclear motion, the starting Hamiltonian is

$$H = \sum_i \left[-\frac{\hbar^2}{2m_e} \nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i} \right] + \sum_{i<j} \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|}. \quad (29.1)$$

The pair sum counts each electron pair once. A mean-field approximation replaces the detailed influence of other electrons by an effective one-electron potential. Because that potential depends on the occupied orbitals, solve for orbitals, rebuild the field, and repeat toward self-consistency. Antisymmetrization adds exchange; correlation describes physics beyond a single independent-particle determinant. This hierarchy explains why shell models are useful without making them exact.

29.2 Light-matter interaction and selection rules

A transition matrix element can vanish because of symmetry even when the photon has the right energy. Before integrating a complicated wavefunction, inspect how the initial state, final state, and interaction transform. Selection rules identify these zeros and save work while explaining which spectral lines are strong or absent in a given approximation. The symbols $\langle f | \mathbf{d} | i \rangle$ describe a transition matrix element between final and initial states. A selection rule says a matrix element vanishes by symmetry; it is not merely an energy mismatch.

When the optical wavelength is large compared with an atom, the electric dipole interaction is approximately $H' = -\mathbf{d} \cdot \mathbf{E}(t)$. Transition strength depends on $|\langle f | \mathbf{d} | i \rangle|^2$. For single-electron orbital states, electric dipole selection rules include

$$\Delta\ell = \pm 1, \quad \Delta m = 0, \pm 1, \quad (29.2)$$

with parity changing. For total angular momentum, $\Delta J = 0, \pm 1$ except $J = 0 \rightarrow 0$; spin is conserved in the ideal spin-independent dipole approximation. Higher multipoles and state mixing can enable weaker transitions.

A finite excited-state lifetime τ produces a characteristic natural energy width of order $\hbar/\tau$. Doppler motion, collisions, and instrumentation broaden spectral lines further.

For a dipole transition, one component contains an integral of the form $\int \psi_f^* x \psi_i d^3r$. Under inversion, x changes sign. If the two states have the same parity, the integrand is odd and its integral vanishes. Opposite parity is therefore necessary. The dipole's vector character supplies the angular-momentum rules quoted above. Energy matching and a nonzero matrix element are separate requirements: satisfying the first does not guarantee the second.

29.3 Magnetic fields and resonance

A magnetic field distinguishes states with different projections of magnetic moment along it. First calculate their energy separation; then compare it with the energy of the applied oscillatory field. Resonance occurs when the drive can efficiently connect those levels, subject to the coupling matrix element.

A magnetic moment contributes $H_Z = -\boldsymbol{\mu} \cdot \mathbf{B}$. In weak fields where angular momentum coupling remains valid, atomic shifts are often written $\Delta E = g_J \mu_B m_J B$, with $\mu_B = e\hbar/(2m_e)$. Resonant oscillatory fields drive transitions at the level splitting divided by $\hbar$.

Electron spin resonance and nuclear magnetic resonance probe different magnetic moments and environments. Nuclear chemical shifts and spin couplings reflect electronic structure; they are not described fully by the isolated-nucleus gyromagnetic ratio alone.

For adjacent allowed values of m_J in the stated linear Zeeman regime, the splitting magnitude is $|\Delta E| = |g_J| \mu_B B$. Matching to $\hbar\omega$ gives

$$\omega_{\text{res}} = \frac{|g_J| \mu_B B}{\hbar}. \quad (29.3)$$

The static field sets the energy separation, while a suitable transverse oscillating field induces transitions. As the magnetic interaction becomes comparable to internal angular-momentum couplings, the weak-field labels and effective g_J description must be reconsidered.

29.4 Molecules and the Born–Oppenheimer approximation

Start by holding the nuclei fixed and solving the electronic problem for that arrangement. Repeat for neighboring arrangements to obtain an electronic energy surface. The nuclei then move approximately on that surface. This approach turns the complicated joint problem into a fast electronic calculation followed by slower nuclear dynamics, provided nonadiabatic couplings are small.

Nuclei are much heavier than electrons. Solving electronic motion at approximately fixed nuclear positions produces potential energy surfaces for slower nuclear motion. This separation can fail near electronic degeneracies or during rapid nonadiabatic processes.

A diatomic molecule near equilibrium has approximate vibrational energies $E_v = \hbar\omega(v + 1/2)$ and rigid-rotor energies

$$E_J = \frac{\hbar^2 J(J+1)}{2I}, \quad I = \mu R_e^2. \quad (29.4)$$

Electronic, vibrational, and rotational energy scales are typically successively smaller. Covalent bonding involves shared electronic states; ionic bonding involves charge transfer and electrostatic attraction; dispersion forces arise from correlated charge fluctuations.

For a diatomic molecule, let $U(R)$ include electronic energy and nuclear repulsion. Expand around its minimum R_e :

$$U(R) \simeq U(R_e) + \frac{1}{2}U''(R_e)(R - R_e)^2, \quad \omega = \sqrt{\frac{U''(R_e)}{\mu}}. \quad (29.5)$$

This is the oscillator again, now for relative nuclear motion. Rotation at nearly fixed separation has classical energy $L^2/(2I)$; replacing L^2 by its eigenvalue $\hbar^2 J(J+1)$ yields the rotor spectrum. The approximations neglect bond stretching during rotation and anharmonicity at large vibration amplitude, giving clear directions for later corrections.

Try it yourself

Explain how a photon can cause a transition between two atomic energy levels. Why does matching the energy gap alone not guarantee that the transition occurs?

Chapter 30

Condensed matter and collective phenomena

Molecular vibrations showed how many-body motion can still be organized around modes. A crystal repeats that idea across an enormous lattice: translation symmetry creates phonons and bands, while interactions and collective order determine which simple pictures survive. A solid contains too many interacting particles to solve by listing all their coordinates. Its regularities suggest better variables: lattice waves, electron bands, collective order, and quasiparticles. The purpose of these descriptions is to retain the information needed for a particular scale and measurement. Their success comes from symmetry and controlled approximations, not from pretending the microscopic interactions disappear.

30.1 Crystals, reciprocal space, and phonons

A periodic arrangement makes waves emitted or scattered from different cells acquire related phases. Reciprocal space records the phase conditions for constructive addition. For vibrations, the same periodicity allows traveling patterns that diagonalize the coupling between lattice sites. a is lattice spacing, $\mathbf{R}$ a lattice translation, $\mathbf{k}$ crystal wave vector, $\mathbf{G}$ a reciprocal-lattice vector, and ω mode frequency.

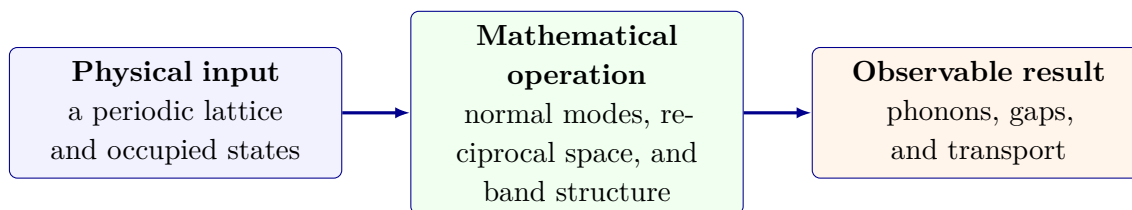


Figure 30.1: Visual model for condensed matter: the physical question becomes mathematics and returns as an observable prediction.

A crystal is periodic under lattice translations. Reciprocal vectors satisfy $\mathbf{G} \cdot \mathbf{R} = 2\pi n$, organizing diffraction and wave states. Bragg's law is $2d \sin \theta = n\lambda$ when θ is measured from the lattice planes.

Small lattice vibrations become coupled harmonic modes. Quantization produces phonons with energies $\hbar\omega_{\mathbf{k},s}(n + 1/2)$. Acoustic branches have $\omega \rightarrow 0$ as $k \rightarrow 0$; multi-atom unit cells can also have optical branches. The Debye model predicts low-temperature lattice heat capacity $C_V \propto T^3$ in three dimensions, while the high-temperature classical limit is approximately $3Nk_B$.

For a one-dimensional chain with spacing a , mass m , and nearest-neighbor spring constant K , the displacement satisfies

$$m\ddot{u}_j = K(u_{j+1} - 2u_j + u_{j-1}). \quad (30.1)$$

Try $u_j = u_0 e^{i(kja - \omega t)}$. Cancel the common phase and use $e^{ika} + e^{-ika} = 2 \cos ka$:

$$\omega^2 = \frac{2K}{m}(1 - \cos ka) = \frac{4K}{m} \sin^2 \frac{ka}{2}. \quad (30.2)$$

For $|ka| \ll 1$, $\omega \simeq a\sqrt{K/m}|k|$, an acoustic branch. Quantizing each normal mode supplies phonons. In three dimensions, acoustic mode counting gives a low-frequency density proportional to ω^2 ; thermally populating modes up to scale $k_B T / \hbar$ leads to energy proportional to T^4 and heat capacity proportional to T^3 .

30.2 Bloch's theorem and electronic bands

A periodic Hamiltonian commutes with lattice translations. We can therefore choose states that acquire only a phase under each lattice translation. Extracting that phase as a plane-wave factor leaves a function with the periodicity of the lattice: this is the content of Bloch's theorem. The band label n distinguishes solutions with the same $\mathbf{k}$; it is not particle number. A gap is an energy interval with no allowed band state in the ideal model.

For a periodic single-particle potential,

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}), \quad u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r}). \quad (30.3)$$

Allowed energies form bands $E_n(\mathbf{k})$ separated in some cases by gaps. A partially filled band supports metallic behavior in the simple band picture; filled bands separated from empty bands by a gap give an insulator at zero temperature. Correlations and disorder can alter this classification.

A semiclassical wave packet in an isolated ordinary band has

$$\mathbf{v} = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n, \quad \hbar \dot{\mathbf{k}} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (30.4)$$

Berry curvature can add an anomalous velocity term. Near an isotropic parabolic extremum, $E \simeq E_0 + \hbar^2 k^2 / (2m^*)$ defines effective mass. Crystal momentum is not generally the free-space mechanical momentum.

If $\psi(\mathbf{r} + \mathbf{R}) = e^{i\mathbf{k} \cdot \mathbf{R}} \psi(\mathbf{r})$, define $u(\mathbf{r}) = e^{-i\mathbf{k} \cdot \mathbf{r}} \psi(\mathbf{r})$. Direct substitution gives $u(\mathbf{r} + \mathbf{R}) = u(\mathbf{r})$. Distinct solutions within a cell supply the band label n . Near a one-dimensional band minimum,

$$E(k) \simeq E_0 + \frac{1}{2} E''(k_0) (k - k_0)^2, \quad \frac{1}{m^*} = \frac{E''(k_0)}{\hbar^2}. \quad (30.5)$$

Using $v = (1/\hbar)dE/dk$ and $\hbar\dot{k} = F$ gives $\dot{v} = F/m^*$ locally. The effective mass measures band curvature and hence response to force within this approximation.

30.3 Fermi gases and electron transport

At zero temperature, fermions fill available one-particle states from the bottom up. The Fermi wave number is determined by counting enough states to hold all electrons. Transport then asks how fields and scattering slightly change that occupation, rather than treating every electron as a freely accelerating isolated charge.

For a three-dimensional noninteracting spin-one-half electron gas,

$$k_F = (3\pi^2 n)^{1/3}, \quad E_F = \frac{\hbar^2 k_F^2}{2m}. \quad (30.6)$$

At low temperature only states near the Fermi surface can change occupation, yielding electronic heat capacity proportional to T rather than the classical constant.

The Drude model with relaxation time τ gives $\sigma = ne^2\tau/m$; a band effective mass can replace m in a suitable simple band model. The single-carrier Hall coefficient is $R_H = -1/(ne)$ for electrons. Multiband transport, scattering anisotropy, and strong correlations require more complete models.

With two spin states per wave vector, counting inside a sphere of radius k_F gives

$$N = 2 \frac{V}{(2\pi)^3} \frac{4\pi k_F^3}{3} \implies n = \frac{k_F^3}{3\pi^2}. \quad (30.7)$$

For Drude transport, model momentum relaxation by $m\dot{\mathbf{v}}_d = -e\mathbf{E} - m\mathbf{v}_d/\tau$. In steady state $\mathbf{v}_d = -e\tau\mathbf{E}/m$. Since $\mathbf{J} = -ne\mathbf{v}_d$, we obtain $\sigma = ne^2\tau/m$. The model separates carrier density, inertia, and scattering time, while leaving the microscopic value of τ to additional physics.

30.4 Semiconductors and junctions

A filled valence band and an empty conduction band carry little ordinary direct current at zero temperature in the simple band model. Excitation or doping creates mobile carriers. Bringing differently doped regions together initially drives diffusion, which leaves fixed ionized dopants behind and builds an opposing electric field.

A semiconductor has a band gap small enough that thermal excitation and doping can control carrier density. Donors tend to supply electrons; acceptors tend to supply holes. A hole is a useful positive-charge quasiparticle describing an unfilled valence-band state.

A p - n junction develops a depletion region and built-in electric field. Under ideal diffusion-dominated conditions, its current-voltage relation is

$$I = I_s \left(e^{eV/(n_i k_B T)} - 1 \right), \quad (30.8)$$

with ideality factor $n_{\text{id}} = 1$ in the simplest model. Recombination, series resistance, and breakdown change this relation. Photons above the gap can create electron-hole pairs, enabling photodetectors and solar cells.

At equilibrium, drift from the built-in field balances diffusion. A forward applied bias lowers the barrier for carrier injection. In the ideal diffusion model, the injected minority-carrier population grows by a Boltzmann factor $e^{eV/(k_B T)}$. Subtracting the equilibrium contribution explains the form $I_s(e^{eV/(k_B T)} - 1)$. A full derivation of I_s requires solving minority-carrier diffusion and recombination with junction boundary conditions. Thus the exponential form and its prefactor carry different pieces of the physics.

30.5 Magnetism, superconductivity, and superfluidity

Collective order means the low-energy state is organized across many constituents. In a magnet, spin correlations organize orientation. In a superconductor or superfluid, a coherent phase becomes a central variable. These descriptions let us predict long-distance behavior without solving every microscopic degree of freedom at once.

Exchange interactions can favor aligned or alternating spins, producing ferromagnetic or antiferromagnetic order. A Heisenberg model uses $H = -\sum_{ij} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$ with units of J_{ij} chosen for the spin convention. Collective spin excitations are magnons.

A superconductor exhibits dissipationless direct-current transport under suitable limits and the Meissner effect, expelling weak magnetic fields from its bulk. London theory gives magnetic penetration over a length λ_L ; the coherence length ξ characterizes spatial variation of the order parameter. In conventional BCS superconductors, paired electrons form a coherent many-body state, with weak-coupling result

$$\Delta(0) \simeq 1.764 k_B T_c, \quad \Phi_0 = \frac{h}{2e}. \quad (30.9)$$

The first relation is not universal across all superconductors. The second is the usual flux quantum for charge- $2e$ condensates. Josephson junctions satisfy $I = I_c \sin \varphi$ and $\dot{\varphi} = 2eV/\hbar$ in the ideal relations.

Superfluidity involves coherent flow with restricted dissipation channels. A useful Landau criterion is $v_c = \min_p [\epsilon(p)/p]$ for elementary excitation production in the ideal setting; vortices and boundaries can lower observed critical velocities.

One useful derivation is the ideal Landau criterion. In the rest frame of an obstacle, creating an excitation with momentum $\mathbf{p}$ in a fluid moving at $\mathbf{v}$ can cost energy $\epsilon(p) - \mathbf{v} \cdot \mathbf{p}$. If this quantity becomes negative, creating the excitation can lower the energy. Requiring it to remain nonnegative for all momenta gives

$$v \leq \min_p \frac{\epsilon(p)}{p}. \quad (30.10)$$

This identifies an energetic threshold for that excitation mechanism. It does not include every way real flow can dissipate, such as vortex nucleation at a boundary. Likewise, the quoted BCS gap relation needs a microscopic pairing calculation; it is not implied by phase coherence alone.

30.6 Topology and emergence

Some distinctions between phases survive smooth changes in material parameters. An integer invariant cannot vary continuously: changing it requires leaving the conditions under which it was defined, often through a gap closing. This offers a different kind of classification from a local order parameter such as magnetization.

Macroscopic phases can differ through global properties of their quantum states as well as local order parameters. In an ideal integer quantum Hall plateau,

$$\sigma_{xy} = \nu \frac{e^2}{h}, \quad \nu \in \mathbb{Z}. \quad (30.11)$$

The integer relates to a topological invariant of occupied states. Edge transport and bulk topology are connected, subject to the system's symmetry and gap conditions. Fractional quantum Hall states require interactions and support excitations with fractionalized properties.

Quasiparticles, order parameters, and effective theories illustrate emergence: the most useful degrees of freedom at long distances need not resemble the microscopic ingredients directly.

Read the Hall formula as two separate statements. The combination e^2/h supplies the conductance scale; the integer ν contains the topological information. Dimensional analysis can identify a possible scale but cannot prove integer quantization or determine which integer a sample realizes. Those conclusions require the quantum states and their response. This is a useful place to remember the boundary between an instructive scaling argument and a full microscopic derivation.

Try it yourself

Describe how a repeating crystal structure changes the allowed electron states. What physical difference would you expect between a filled band and a partly filled band?

Chapter 31

Nuclear physics

Atomic physics taught us to separate energy scales and to distinguish an allowed transition from a likely one. Nuclear physics applies the same discipline at a smaller length scale, where binding, tunneling, statistics, and relativistic reaction kinematics meet. Nuclear physics brings together binding energy, quantum statistics, tunneling, and reaction kinematics at much shorter distances. Start by accounting for mass and charge consistently, then distinguish whether a process is energetically allowed from whether it happens rapidly. Favorable energy release alone does not determine a decay or reaction rate.

31.1 Nuclear scales and binding

A bound nucleus has less mass than its separated constituents because energy was released when the bound state formed. The binding energy measures the energy required to separate it again. Before inserting tabulated masses, decide whether they refer to bare nuclei or neutral atoms so that electron contributions are handled consistently. $A = Z + N$ is mass number, Z proton number, and N neutron number. B is binding energy and Q reaction energy release; positive Q does not guarantee a rapid reaction.

A nucleus contains Z protons and N neutrons, with mass number $A = Z + N$. A useful empirical radius scale is $R \simeq r_0 A^{1/3}$ with $r_0 \simeq 1.2$ fm. Nuclear binding energy is

$$B = [Zm_p + Nm_n - M_{\text{nucleus}}]c^2. \quad (31.1)$$

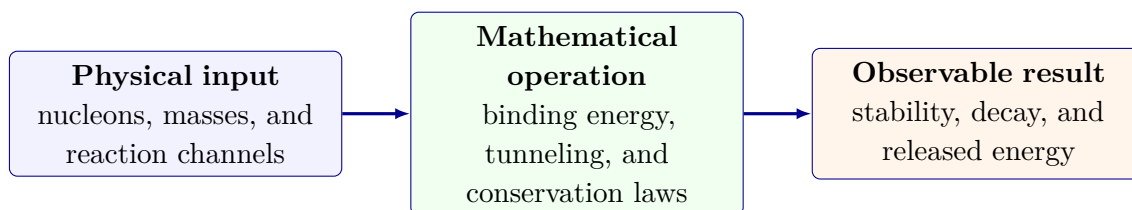


Figure 31.1: Visual model for nuclear physics: the physical question becomes mathematics and returns as an observable prediction.

If atomic masses are used, electron masses and binding must be accounted for consistently. Binding reflects residual strong interactions, opposed in part by proton Coulomb repulsion.

A schematic semi-empirical mass formula is

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + \delta(A, Z). \quad (31.2)$$

Its terms represent volume attraction, surface correction, Coulomb energy, neutron-proton asymmetry, and pairing. Shell effects cause additional structure and explain enhanced stability at magic nucleon numbers.

The empirical relation $R \propto A^{1/3}$ implies volume proportional to nucleon number, suggesting roughly constant interior density. If each nucleon interacts attractively with a limited number of nearby neighbors, the bulk binding scales as A . Surface nucleons have fewer neighbors, producing a correction proportional to area, $A^{2/3}$. Electrostatic repulsion scales roughly as Z^2/R , hence $Z^2/A^{1/3}$. These arguments motivate the first three mass-formula terms; their coefficients and the remaining refinements require nuclear data and more detailed theory.

31.2 Radioactive decay

A constant decay rate means that each surviving nucleus has the same small probability λdt of decaying in the next interval dt . The expected number lost is proportional to the number still present. Exponential decay follows from this local probability statement, without requiring every nucleus to carry a predetermined decay time. In decay, λ is a probability rate and $t_{1/2} = \ln 2/\lambda$ the half-life.

For independent nuclei with constant decay rate λ ,

$$\dot{N} = -\lambda N, \quad N = N_0 e^{-\lambda t}, \quad t_{1/2} = \frac{\ln 2}{\lambda}, \quad \mathcal{A} = \lambda N. \quad (31.3)$$

The exponential law is an excellent intermediate-time description; exact quantum survival amplitudes need not be exponential at arbitrarily short or long times. Competing decay modes have total rate $\lambda = \sum_i \lambda_i$ and branching fractions λ_i/λ .

Alpha decay emits a helium nucleus and is enabled by tunneling through a Coulomb barrier. Beta-minus decay changes a neutron into a proton, electron, and electron antineutrino; beta-plus decay changes a proton into a neutron, positron, and electron neutrino when energetically allowed. Gamma decay changes nuclear excitation without changing A or Z .

Separate $dN/dt = -\lambda N$ and integrate:

$$\int_{N_0}^{N(t)} \frac{dN'}{N'} = -\lambda \int_0^t dt' \implies N = N_0 e^{-\lambda t}. \quad (31.4)$$

Set $N(t_{1/2}) = N_0/2$ to obtain $t_{1/2} = \ln 2/\lambda$. For several independent channels, their probabilities add over a short interval, giving total rate $\sum_i \lambda_i$. The integrated probability for channel i is $\int_0^\infty \lambda_i e^{-\lambda t} dt = \lambda_i/\lambda$, the branching fraction.

31.3 Reaction energetics and cross sections

List the initial and final particles and compare their total rest energies. This determines the energy release, but momentum must also be conserved. For an endothermic reaction against a stationary target, some incident energy necessarily remains as collective motion, so the beam energy threshold is generally larger than the rest-energy deficit alone.

A reaction's energy release is

$$Q = (M_{\text{initial}} - M_{\text{final}})c^2. \quad (31.5)$$

Positive Q is exothermic. Negative Q requires sufficient incident energy, with recoil making the laboratory threshold exceed $|Q|$ in general. A beam with flux Φ incident on N_t thin-target nuclei gives event rate approximately $R = \Phi N_t \sigma$ when attenuation and multiple interactions are negligible.

Fusion of light nuclei and fission of sufficiently heavy nuclei can release binding energy because products may be more tightly bound per nucleon. Their rates depend on barriers, quantum states, and reaction channels, not simply on a favorable Q value. Stellar fusion occurs through tunneling at energies below the classical Coulomb barrier.

The cross section is an effective interaction area, not necessarily the geometric area of a nucleus. A beam flux Φ has units of particles per area per time, so one target produces rate $\Phi\sigma$. For N_t independently exposed targets, rates add to $R = \Phi N_t \sigma$. If a target is thick enough to attenuate the beam, its flux changes with depth and this simple multiplication must be replaced by a transport calculation. This connects a microscopic interaction probability to an experimentally measured event count.

31.4 Nuclear models and dense matter

Choose the model according to the feature you want to explain. Smooth binding trends suggest a liquid-drop picture; unusually stable nucleon numbers suggest shells; low-lying rotations suggest collective deformation. Different descriptions can be useful for the same nucleus because they organize different observables.

The liquid-drop model explains bulk trends; the shell model treats nucleons in an effective mean field with strong spin-orbit effects. Collective rotational and vibrational models describe other spectra. No single simple model captures all nuclear structure.

At extreme density, neutron-rich matter is governed by degeneracy, nuclear interactions, and relativistic gravity. Its equation of state relates pressure to energy density and composition. Neutron-star structure therefore links microscopic many-body physics to macroscopic gravitational observations.

For dense matter, the key output needed by a stellar model is an equation of state. Microscopic calculations determine how pressure changes with energy density and composition; gravitational balance then determines the density profile. This is a chain of models with distinct jobs. Agreement with one nuclear binding trend does not by itself establish the high-density equation of state needed inside a neutron star.

Try it yourself

Two nuclei have different binding energies. Explain what additional information you need to decide whether a proposed reaction releases energy.

Chapter 32

Relativistic quantum theory and particle physics

The quantum chapters allowed a fixed number of particles, while relativity allows energy to become new particles. Quantum field theory is the next synthesis: fields are fundamental, particles are excitations, and symmetries organize the interactions. Quantum mechanics and special relativity meet most naturally in quantum field theory, where particle number can change and fields supply the underlying degrees of freedom. A full development requires a substantial course of its own. Here we build the entry points carefully: relativistic dispersion, field equations, quantized modes, and the role of interactions and symmetries. Treat the more advanced results as a map for further study, with their assumptions visible.

32.1 Why quantum fields are needed

At low enough energies, a fixed number of particles can be an excellent approximation. Once enough energy is available to produce additional particles, that approximation cannot describe all possible outcomes. A field description accommodates sectors with different occupation numbers while preserving a common dynamical framework. ϕ is a scalar field, ψ a spinor field, m mass, and ∂_μ a spacetime derivative.

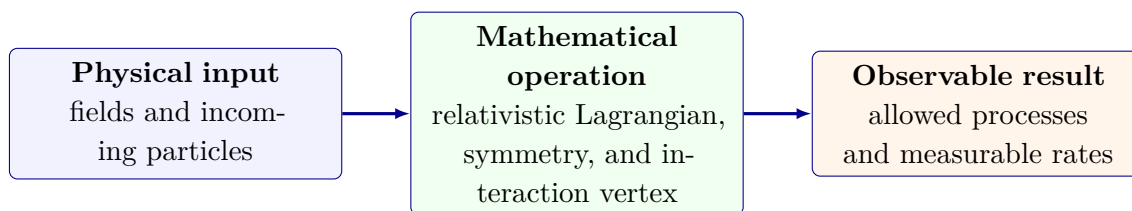


Figure 32.1: Visual model for particle physics: the physical question becomes mathematics and returns as an observable prediction.

Relativity permits energy to create particle-antiparticle pairs. A fixed-particle-number wavefunction is therefore insufficient for general high-energy processes. Quantum field theory assigns operators to fields; particles are excitations of those fields, and interactions can change particle number.

For the remainder of this chapter use $\hbar = c = 1$ unless restored explicitly. Keep the metric signature $(-, +, +, +)$ and define $\square = \partial_\mu \partial^\mu = -\partial_t^2 + \nabla^2$. A free scalar field obeys

$$(\square - m^2)\phi = 0, \quad (32.1)$$

which gives $E^2 = \mathbf{p}^2 + m^2$. The Lagrangian density $\mathcal{L} = -\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2}m^2 \phi^2$ yields this equation with our signature.

Check the scalar equation by inserting a plane wave $\phi \propto e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}$. With this chapter's signature,

$$\square\phi = (E^2 - \mathbf{p}^2)\phi, \quad (\square - m^2)\phi = 0 \implies E^2 = \mathbf{p}^2 + m^2. \quad (32.2)$$

The equation can also be obtained from the displayed Lagrangian density: $\partial\mathcal{L}/\partial(\partial_\mu\phi) = -\partial^\mu\phi$ and $\partial\mathcal{L}/\partial\phi = -m^2\phi$. The field Euler–Lagrange equation then gives $-\square\phi + m^2\phi = 0$. This checks the signs against a physical dispersion relation before more complicated calculations are attempted.

32.2 Spinors, antiparticles, and quantization

A relativistic equation first order in time and space requires coefficients that cannot all be ordinary numbers. Matrices can cancel the unwanted cross terms when the operator is squared. The Dirac algebra is precisely the condition that makes this construction recover the relativistic energy relation while introducing spinor components. The metric convention fixes signs in $\square$ and the gamma-matrix algebra; changing signature requires changing those definitions consistently.

A Dirac equation can be written

$$(i\gamma^\mu \partial_\mu - m)\psi = 0, \quad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}I, \quad (32.3)$$

using this explicitly chosen gamma-matrix convention with the mostly-plus metric. Squaring the operator reproduces the stated Klein–Gordon dispersion. Other texts often use the opposite metric and corresponding Clifford-algebra convention.

A free field decomposes into harmonic modes, promoted to creation and annihilation operators. Bosonic operators obey commutators and fermionic ones anticommutators. In local relativistic quantum theories satisfying the usual assumptions, the spin-statistics connection links integer spin with bosons and half-integer spin with fermions. Antiparticles have the same mass and spin as their partners and opposite additive internal charges.

Multiply the Dirac operator by its opposite-mass partner:

$$(i\gamma^\mu \partial_\mu + m)(i\gamma^\nu \partial_\nu - m) = -\frac{1}{2}\{\gamma^\mu, \gamma^\nu\}\partial_\mu \partial_\nu - m^2 = \square - m^2. \quad (32.4)$$

Only the symmetric matrix combination survives because the derivatives commute. With the explicitly chosen anticommutator, each component satisfies the scalar mass-shell equation. Quantization then promotes the field's mode amplitudes to operators; the oscillator algebra familiar

from quantum mechanics becomes the starting point for particle creation and annihilation. The interpretation of antiparticles requires that field framework, rather than treating negative-energy solutions as ordinary classical trajectories.

32.3 Interactions, diagrams, and effective theories

Begin with a free theory you can solve and add interaction terms allowed by the symmetries. Perturbation theory organizes their repeated action in powers of coupling. Diagrams are a way to label those algebraic contributions and their integrals, so their meaning should be traced back to amplitudes rather than imagined particle paths. A Feynman diagram organizes an amplitude, not a literal classical trajectory.

An interaction term couples fields. Perturbation theory expands amplitudes in coupling constants; Feynman diagrams organize integrals in that expansion. Internal lines are propagators, not directly observed little particles following classical paths. Observable rates come from amplitudes, interference, phase space, and the appropriate normalization.

Loop integrals require regularization and renormalization. Measured masses and couplings depend on how parameters are defined and, for running couplings, on the energy scale. Effective field theory organizes allowed interactions by symmetry and powers of a small ratio E/Λ , where Λ is a higher scale. It can make precise low-energy predictions without claiming validity at arbitrarily high energy.

If two alternatives contribute amplitudes $\mathcal{M}_1$ and $\mathcal{M}_2$, their probability factor is

$$|\mathcal{M}_1 + \mathcal{M}_2|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 + 2\text{Re}(\mathcal{M}_1\mathcal{M}_2^*). \quad (32.5)$$

The interference term means separate diagrams are generally not independent events with separately addable probabilities. In an effective theory, omitted contributions can often be ordered as powers of E/Λ . Identifying that ratio and the relevant symmetries tells you what accuracy to expect and when new degrees of freedom may need to be included.

32.4 The Standard Model's structure

Read the particle table as a classification of fields and their transformation properties. The gauge groups indicate different internal symmetries and constrain which interactions can occur. Mass, electric charge, color, and flavor are distinct labels; belonging to one generation does not mean the particles have similar masses or identical forces.

The Standard Model is a gauge theory with group $SU(3)_C \times SU(2)_L \times U(1)_Y$. Its matter fields form three generations of quarks and leptons:

Generation	Quarks	Leptons
1	up, down	electron, electron neutrino
2	charm, strange	muon, muon neutrino
3	top, bottom	tau, tau neutrino

Quarks carry color and fractional electric charges; leptons do not carry color. Gluons mediate the strong interaction, the photon electromagnetism, and $W^\pm, Z$ the weak interaction. The Higgs field's nonzero vacuum value is associated with electroweak symmetry breaking and masses for the weak bosons; Yukawa couplings generate charged-fermion masses. Most of the proton's mass is associated with QCD dynamics, rather than simply the sum of its valence-quark masses.

The minimal original Standard Model has massless neutrinos. Observed flavor oscillations require neutrino mass differences and therefore an extension of that minimal formulation; the detailed mass mechanism is not fixed by oscillation data alone.

A helpful learning order is to start with the familiar electromagnetic coupling, then ask which additional charges and fields are needed for weak and strong processes. The table supplies the matter content; the gauge structure supplies interactions; the Higgs sector and its couplings supply part of the mass mechanism. The classification alone does not calculate all particle masses. Many parameters must be measured, and composite masses also require interaction energy, just as binding changed the mass of a nucleus.

32.5 Strong and weak interactions

The strength and character of an interaction can depend on the scale at which it is probed. Strong interactions organize quarks into hadrons, while weak interactions connect different flavors. Neutrino oscillations give a particularly transparent example of flavor change arising from coherent phase evolution rather than a sequence of classical conversions.

Quantum chromodynamics is the $SU(3)$ gauge theory of color. Its coupling weakens at short distances (asymptotic freedom) and becomes strong at large distances. Isolated colored quarks and gluons are not observed as asymptotic free particles; they form color-neutral hadrons. Baryons include three-valence-quark states such as protons and neutrons; mesons include quark-antiquark states, with sea constituents and gluons also contributing.

Weak interactions can change flavor and violate parity. Quark flavor mixing is described by the CKM matrix; lepton mixing by the PMNS matrix. Charge conjugation and parity together are also violated in some processes. In the two-flavor vacuum approximation, restoring $\hbar, c$,

$$P(\nu_\alpha \rightarrow \nu_\beta) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 c^3 L}{4\hbar E}\right), \quad \alpha \neq \beta. \quad (32.6)$$

This assumes relativistic coherent propagation. Three-flavor effects and matter interactions modify the expression.

For two flavors, use mass eigenstates $|\nu_1\rangle, |\nu_2\rangle$ and write

$$|\nu_\alpha\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle, \quad (32.7)$$

$$|\nu_\beta\rangle = -\sin\theta|\nu_1\rangle + \cos\theta|\nu_2\rangle. \quad (32.8)$$

Their propagation phases differ by $\Delta\varphi \simeq \Delta m^2 c^3 L / (2\hbar E)$ in the relativistic coherent limit. Projection onto the second flavor gives an amplitude proportional to $\sin\theta \cos\theta (e^{-i\varphi_2} - e^{-i\varphi_1})$. Squaring yields

$$P_{\alpha \rightarrow \beta} = \sin^2(2\theta) \sin^2(\Delta\varphi/2). \quad (32.9)$$

This recovers the stated formula. Both mixing and a nonzero relative phase are necessary; equal masses would eliminate this oscillation phase difference.

32.6 Symmetries, conservation, and open questions

Before calculating a proposed reaction, apply the conservation laws relevant to the interaction and regime. They can rule out a channel before any difficult integral is done. After that, distinguish what the established model predicts from what it leaves as measured input or an unresolved question.

Energy-momentum and electric charge conservation strongly constrain reactions. Additional quantum numbers may be exact, approximate, or violated by particular interactions; a conservation rule from one regime should not be assumed universally.

Established quantum field theories do not by themselves provide a complete quantum theory of gravity, identify the microscopic nature of dark matter, or explain every cosmological initial condition. These are open problems. Proposed answers should be distinguished from the tested framework used to calculate spectra and scattering.

Conservation is necessary but not sufficient for an appreciable rate. A channel can conserve energy, momentum, and charge yet be suppressed by a small coupling, a symmetry of the matrix element, or limited final-state phase space. This repeats a lesson from atomic selection rules and nuclear barriers: first ask whether the process is allowed, then ask what amplitude and accessible states determine how often it occurs.

Try it yourself

Why can a relativistic collision require a description in which particle number changes? State what a Feynman diagram represents in a calculation.

Part XI

Astrophysics, Cosmology, and Synthesis

Chapter 33

Stars, compact objects, and the expanding universe

The final physics unit brings earlier scales together. Gravity supplies structure, quantum and nuclear physics supply matter, thermodynamics supplies pressure and entropy, and relativity supplies the geometry of compact objects and the expanding universe. Astrophysics is therefore a synthesis, not a separate beginning. Astrophysics combines the previous modules on a grand scale. Stellar pressure is supplied by matter and radiation; gravity sets the required support; nuclear reactions and transport determine energy flow. Cosmology replaces the internal profile of one object with the evolution of a nearly homogeneous universe. In both settings, state the symmetry assumptions before reducing field equations to a manageable model.

33.1 Stellar structure and energy transport

Build a star out of thin spherical shells. Each shell adds mass, contributes or transports energy, and must be supported against the gravitational pull of the enclosed mass. This construction explains why several coupled differential equations are needed: no single pressure balance determines the temperature, luminosity, and density by itself. In stellar equations, M_r is mass enclosed inside radius r and L_r luminosity crossing that shell. A subscript r means “at radius r ,” not a new physical kind of mass or luminosity.

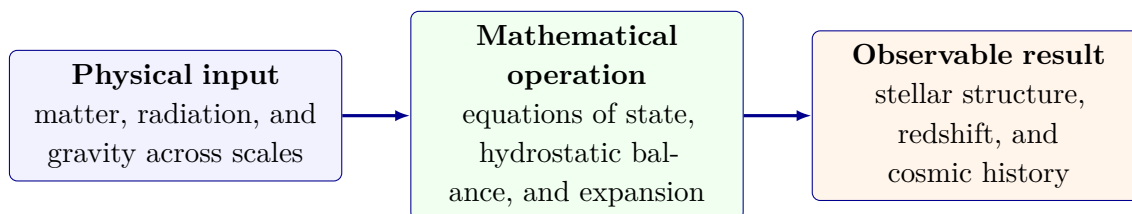


Figure 33.1: Visual model for astrophysics and cosmology: the physical question becomes mathematics and returns as an observable prediction.

A spherical nonrotating star in Newtonian hydrostatic equilibrium satisfies

$$\frac{dM_r}{dr} = 4\pi r^2 \rho, \quad \frac{dp}{dr} = -\frac{GM_r \rho}{r^2}, \quad \frac{dL_r}{dr} = 4\pi r^2 \rho \epsilon, \quad (33.1)$$

where M_r is enclosed mass, L_r is outward luminosity, and ϵ is net local energy generation per mass per time. An equation of state and an energy-transport law complete the model. In an optically thick radiative region,

$$\frac{dT}{dr} = -\frac{3\kappa_R \rho L_r}{16\pi a c r^2 T^3}, \quad (33.2)$$

where κ_R is the Rosseland mean opacity and a the radiation constant. Convection becomes important when the required temperature gradient is sufficiently steep; convection modeling is an additional physical problem.

For a self-gravitating stationary system with negligible surface terms, the virial theorem gives $2\langle K \rangle + \langle U \rangle = 0$ for an inverse-distance potential. A star can heat as it loses total energy through gravitational contraction: its thermal kinetic energy increases while gravitational energy becomes more negative.

A shell of thickness dr has mass $dM_r = 4\pi r^2 \rho dr$, giving the first structure equation. Hydrostatic balance is the fluid relation $dp/dr = -\rho g$ with $g = GM_r/r^2$. Net energy production in the shell is ϵdM_r , giving $dL_r/dr = 4\pi r^2 \rho \epsilon$.

For optically thick radiative diffusion, energy density is $u = aT^4$ and flux is approximately $F = -[c/(3\kappa_R \rho)] du/dr$. Use $du/dr = 4aT^3 dT/dr$ and $L_r = 4\pi r^2 F$ to obtain the temperature gradient above. Central regularity, surface conditions, an equation of state, and material rates complete the boundary-value problem.

33.2 Degeneracy pressure and compact remnants

Fermions cannot all collapse into the same low-momentum state. Increasing their density fills states to larger momenta, producing pressure even at zero temperature. The relation between momentum and energy changes as particles become relativistic, and that change alters how strongly pressure grows during compression.

Electron degeneracy supports white dwarfs. For nonrelativistic degenerate electrons, $p \propto n_e^{5/3}$; in the ultrarelativistic limit, $p \propto n_e^{4/3}$. The change in scaling leads to a limiting white-dwarf mass of order a solar mass, conventionally about $1.4M_\odot$ for common idealized composition assumptions.

Neutron stars require relativistic hydrostatic balance. Writing ρ for total energy density divided by c^2 , the Tolman–Oppenheimer–Volkoff equation is

$$\frac{dp}{dr} = -\frac{G(\rho + p/c^2)(M_r + 4\pi r^3 p/c^2)}{r^2[1 - 2GM_r/(rc^2)]}, \quad \frac{dM_r}{dr} = 4\pi r^2 \rho. \quad (33.3)$$

Pressure contributes to gravitation, and the equation of state controls the possible masses and radii. Sufficiently compact collapse can form a black hole. Rotation, magnetic fields, and accretion complicate these idealized pictures.

State counting gives $p_F \propto n^{1/3}$. For nonrelativistic particles, a characteristic energy is $p_F^2/(2m)$, so energy density and degeneracy pressure scale as $n^{5/3}$. For ultrarelativistic particles, energy is proportional to $p_F c$, giving $n^{4/3}$. To see why the latter produces a mass scale, estimate required gravitational pressure as GM^2/R^4 and relativistic electron pressure as proportional to $(M/R^3)^{4/3}$ at fixed composition. Both scale as R^{-4} , so reducing radius no longer changes their relative strength for a fixed mass. Detailed stellar integration supplies the coefficient and composition dependence of the limiting mass.

33.3 Homogeneity, isotropy, and expansion

Homogeneity means no preferred position in the idealized spatial geometry; isotropy means no preferred direction. These assumptions leave a scale factor to describe how separations between comoving locations change in time. A comoving coordinate stays fixed while its physical separation from another comoving location can grow. ρ is mass-equivalent energy density, p pressure, $a(t)$ the cosmological scale factor, and $H = \dot{a}/a$ the expansion rate.

On sufficiently large scales, the cosmological principle models space as statistically homogeneous and isotropic. The FLRW metric is

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]. \quad (33.4)$$

Here take r dimensionless, a to have dimensions of length, and $k = 0, \pm 1$. The expansion rate is $H = \dot{a}/a$. Comoving wavelength grows with a , so

$$1 + z = \frac{a(t_0)}{a(t_{\text{emission}})}. \quad (33.5)$$

Nearby galaxies approximately satisfy $v \simeq H_0 d$ when peculiar motion is negligible. At large distances, cosmological recession is not simply a special-relativistic velocity in one global inertial frame.

In a spatially flat slice, two nearby comoving points separated by fixed Δr have distance $d = a(t)\Delta r$. Differentiation gives $\dot{d} = (\dot{a}/a)d = Hd$. This is the geometric origin of the local Hubble relation. Photon wavelength stretches with the same scale factor, so

$$\frac{\lambda_0}{\lambda_e} = \frac{a_0}{a_e} = 1 + z, \quad \frac{\nu_0}{\nu_e} = \frac{a_e}{a_0}. \quad (33.6)$$

For gravitationally bound systems, fixed comoving separation is generally not the appropriate description; local dynamics must also be considered.

33.4 Friedmann equations and cosmic contents

The expansion rate depends on both geometry and the matter content. To solve for $a(t)$, first specify an equation of state, then determine how its density changes with expansion. The continuity equation supplies that density law, and the Friedmann equation turns it into a time evolution.

For a perfect fluid with mass-equivalent density ρ ,

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad (33.7)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \quad (33.8)$$

$$\dot{\rho} + 3H\left(\rho + \frac{p}{c^2}\right) = 0. \quad (33.9)$$

Here ρ, p exclude the explicitly displayed cosmological constant. For a separately conserved component with constant $p = w\rho c^2$,

$$\rho \propto a^{-3(1+w)}. \quad (33.10)$$

Nonrelativistic matter has $w \simeq 0$ and dilutes as a^{-3} ; radiation has $w = 1/3$ and scales as a^{-4} because its photons also redshift. Vacuum energy has $w = -1$ and constant density.

In flat universes dominated by a single component and with no separate Λ , $a \propto t^{2/3}$ for matter and $a \propto t^{1/2}$ for radiation. Positive constant vacuum energy instead produces exponential expansion when it dominates. The critical density is $\rho_c = 3H^2/(8\pi G)$. Numerical estimates of cosmological parameters depend on observations and models; no particular current fit is required for these relations.

Worked example 33.1 (Temperature of expanding radiation). For equilibrium freely redshifting radiation, $\rho_{\text{rad}} \propto a^{-4}$ and $u = a_{\text{rad}}T^4$ imply $T \propto a^{-1}$. Thus radiation observed at redshift z had temperature $T(z) = T_0(1+z)$ if no additional entropy was transferred into it during the interval being modeled.

For a comoving volume $V \propto a^3$, local energy balance has the form $d(\rho c^2 V) = -pdV$. Expanding the differential gives

$$d\rho = -\left(\rho + \frac{p}{c^2}\right) \frac{dV}{V} = -3\left(\rho + \frac{p}{c^2}\right) \frac{da}{a}. \quad (33.11)$$

For constant w , integrate to obtain $\rho \propto a^{-3(1+w)}$. In a flat expanding universe containing only this component, $\dot{a}/a \propto a^{-3(1+w)/2}$. For $w > -1$, integration from a vanishing scale factor gives

$$a^{3(1+w)/2} \propto t, \quad a(t) \propto t^{2/[3(1+w)]}. \quad (33.12)$$

Matter and radiation give the quoted powers. The case $w = -1$ must be treated separately: constant density gives constant H and therefore $a \propto e^{Ht}$. The pressure contribution explains why radiation dilutes faster than conserved particle number alone.

33.5 Thermal history and structure formation

Compare two clocks: the time $1/\Gamma$ for a microscopic reaction and the expansion time $1/H$. When reactions are fast, populations can adjust as the universe cools. When expansion is faster, they may retain a nonequilibrium abundance. Structure growth asks a related competition question: can gravitational attraction amplify a density excess despite the expanding background?

The hot expanding-universe model organizes early particle reactions, light-element nucleosynthesis, formation of neutral atoms, and the release of radiation now observed as the cosmic microwave background. Recombination and photon decoupling are related but distinct processes: atomic populations evolve while the photon scattering rate becomes too small to maintain tight coupling.

A reaction remains near equilibrium when its rate Γ greatly exceeds H . Freeze-out occurs roughly when $\Gamma \lesssim H$, although accurate abundances require kinetic equations. For cold nonrelativistic matter on subhorizon scales in the linear pressureless approximation, density contrast obeys

$$\ddot{\delta} + 2H\dot{\delta} - 4\pi G\rho_m\delta = 0. \quad (33.13)$$

Expansion competes with gravitational growth. Pressure, radiation, free streaming, and nonlinear collapse require extensions.

For a flat matter-only background, $H = 2/(3t)$ and $\rho_m = 1/(6\pi Gt^2)$. Try a power law $\delta = t^p$ in the linear growth equation:

$$p(p-1) + \frac{4}{3}p - \frac{2}{3} = 0 \implies p = \frac{2}{3} \text{ or } -1. \quad (33.14)$$

The growing mode has $\delta \propto t^{2/3} \propto a$, and the other mode decays. This provides a concrete prediction within the pressureless linear model. Once $|\delta|$ becomes order one, neglecting nonlinear terms is no longer justified.

33.6 Dark sectors and inflation

Keep the observational role of a component separate from a proposed microscopic explanation. Extra gravitational clustering and accelerated expansion point to different features of cosmological models. A scalar field is one way to produce negative pressure, but deriving its possible behavior does not establish that a particular field drove the early universe.

Dark matter denotes gravitating matter inferred from multiple gravitational effects whose dominant microscopic identity is not established. Dark energy is a description of the component associated with accelerated expansion; a cosmological constant is one model. These terms summarize different phenomena and should not be treated as interchangeable.

Inflation proposes an early accelerated-expansion phase that can address horizon and flatness problems and generate primordial perturbations. Its specific field content and dynamics are model dependent. A homogeneous canonical scalar field in units $c = \hbar = 1$ has $\rho = \dot{\phi}^2/2 + V(\phi)$ and $p = \dot{\phi}^2/2 - V(\phi)$; potential domination yields negative pressure. This mechanism is a theoretical framework, not a complete experimentally selected account of the earliest universe.

In the scalar-field units specified above,

$$\rho + 3p = 2\dot{\phi}^2 - 2V(\phi). \quad (33.15)$$

The acceleration equation with no separate cosmological constant therefore permits $\ddot{a} > 0$ when $V > \dot{\phi}^2$. Potential domination gives $p \simeq -\rho$, analogous to a nearly constant vacuum component.

Establishing a complete inflationary model additionally requires the field's dynamics, an end to the accelerated era, and predictions for perturbations. The inequality identifies the mechanism, not all of those ingredients.

Try it yourself

What balances gravity inside an ordinary star? Explain how a measurement of redshift can give evidence about cosmic expansion.

Chapter 34

Computational physics and model validation

A calculation is not finished when an equation has been written, and it is not validated merely because an animation looks plausible. This chapter gathers the book’s recurring habits—units, limits, conservation, approximation, and comparison—and turns them into a computational workflow. A numerical solution is another approximation, layered on top of the physical model. It can make an unsolvable analytic problem accessible, but smooth output alone does not establish accuracy. The aim of this chapter is to make computational choices explainable: derive the discrete rule, estimate its error, and test it against something whose answer you understand.

34.1 Discretization and numerical error

A computer represents a function at finitely many points. To approximate a derivative from those values, expand nearby samples about the point of interest and combine them so unwanted Taylor terms cancel. The first surviving neglected term tells you how error should change when the grid is refined. h is a spatial or temporal step, Δx and Δt grid spacings, $Q(h)$ a computed quantity at resolution h , and Q_* its limiting value. A numerical index such as j labels a grid cell and n a time level.

Numerical physics approximates a well-posed mathematical model, whose own physical approxima-

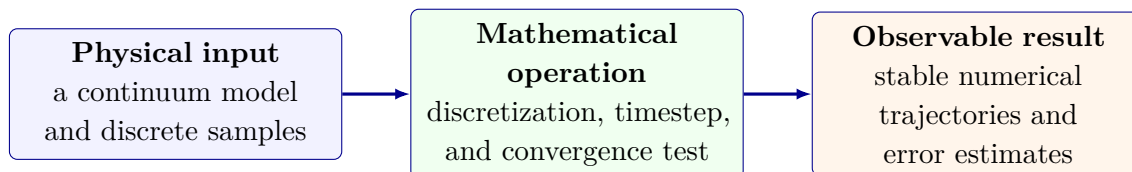


Figure 34.1: Visual model for computational physics: the physical question becomes mathematics and returns as an observable prediction.

tions remain separate from numerical error. A forward difference has error $O(h)$, while

$$f'(x) \simeq \frac{f(x+h) - f(x-h)}{2h} \quad (34.1)$$

has truncation error $O(h^2)$ for a sufficiently smooth function. Taking h extremely small can amplify floating-point cancellation. Numerical integration likewise balances resolution, smoothness, and cost.

Convergence means the computed result approaches a stable limit as resolution improves. Agreement at one step size is insufficient evidence. Compare conserved quantities, analytic limiting cases, and independent formulations whenever feasible.

Subtract the expansions of $f(x+h)$ and $f(x-h)$:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3}f^{(3)}(x) + O(h^5). \quad (34.2)$$

Division by $2h$ gives the centered derivative with leading error $h^2f^{(3)}/6$. In the truncation-dominated regime, halving h should reduce that error by about four. For a numerical quantity $Q(h) = Q_* + Ch^2 + \dots$, the extrapolation

$$Q_* \simeq \frac{4Q(h/2) - Q(h)}{3} \quad (34.3)$$

cancels the leading error. Check that successive refinements actually show the expected trend before relying on the extrapolation; rounding error or unresolved features can break it.

34.2 Time integration and Hamiltonian systems

An integrator advances initial data by a finite time step. Its local update may look accurate while gradually distorting the geometry of a long trajectory. Oscillators are excellent test cases because their exact phase-space motion is bounded, making artificial growth or damping easy to detect.

Forward Euler uses $y_{n+1} = y_n + hf(t_n, y_n)$ and is first-order globally. Fourth-order Runge–Kutta combines several slope estimates for smooth ordinary differential equations. Stiff systems may require implicit or specially designed solvers rather than impractically tiny explicit steps.

For $\ddot{\mathbf{r}} = \mathbf{a}(\mathbf{r})$, velocity Verlet is

$$\mathbf{r}_{n+1} = \mathbf{r}_n + h\mathbf{v}_n + \frac{1}{2}h^2\mathbf{a}_n, \quad (34.4)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{h}{2}(\mathbf{a}_n + \mathbf{a}_{n+1}). \quad (34.5)$$

For separable conservative Hamiltonians it is a second-order symplectic scheme, often maintaining bounded long-term energy error for suitable steps. Symplectic does not mean exact energy conservation at every step.

For $\dot{x} = v$, $\dot{v} = -\omega^2 x$, forward Euler gives $x' = x + hv$ and $v' = v - h\omega^2 x$. Its next-step energy is

$$E' = \frac{1}{2}m(v - h\omega^2 x)^2 + \frac{1}{2}m\omega^2(x + hv)^2 = (1 + h^2\omega^2)E. \quad (34.6)$$

Energy grows on every nonzero step even though the exact system conserves it. Velocity Verlet can instead be organized as a half-step velocity update, full-step position update, and final half-step velocity update:

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{h}{2}\mathbf{a}_n, \quad \mathbf{r}_{n+1} = \mathbf{r}_n + h\mathbf{v}_{n+1/2}, \quad \mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{h}{2}\mathbf{a}_{n+1}. \quad (34.7)$$

Combining these expressions recovers the displayed Verlet formulas. Its structure improves conservative long-time behavior for suitable steps, while convergence and stability still need checking.

34.3 PDE stability and eigenvalue problems

A grid error can be analyzed as a sum of Fourier modes, just like a physical disturbance. Insert one mode into the update rule and calculate how much its amplitude changes in one step. Stability requires that numerical evolution not amplify modes that the underlying diffusive physics should damp. Stability is a property of an update rule; accuracy is how closely its result approaches the continuum solution.

For explicit centered-space diffusion in one dimension,

$$u_j^{n+1} = u_j^n + r(u_{j+1}^n - 2u_j^n + u_{j-1}^n), \quad r = D\Delta t/(\Delta x)^2, \quad (34.8)$$

linear stability requires $r \leq 1/2$. For a common explicit centered wave scheme, the one-dimensional Courant condition is $v\Delta t/\Delta x \leq 1$. Conditions depend on the actual algorithm, geometry, and dimension; these bounds are not universal for every discretization.

Discretizing a Schrodinger or normal-mode operator produces a matrix eigenproblem. Boundary conditions, grid spacing, and domain size affect the spectrum. Verify normalization, orthogonality, and convergence of the lowest states before trusting highly excited numerical modes.

Write $u_j^n = g^n e^{ikj\Delta x}$ and substitute into the diffusion update. Cancel the common factor to obtain

$$g = 1 + r(e^{ik\Delta x} - 2 + e^{-ik\Delta x}) \quad (34.9)$$

$$= 1 - 4r \sin^2(k\Delta x/2). \quad (34.10)$$

For $r \geq 0$, the largest amplification is one and the smallest is $1 - 4r$. Requiring $g \geq -1$ gives $r \leq 1/2$. Halving Δx therefore requires reducing the allowed Δt by four for this method. Spatial refinement without the accompanying time-step change can make a previously stable simulation fail.

34.4 Monte Carlo and statistical sampling

Sampling replaces a difficult sum or integral by an average over representative configurations. Independent random samples make the error analysis simple; a Markov chain trades independence for the ability to reach complicated distributions. Its acceptance rule must preserve the target distribution, and its exploration must be broad enough to represent that distribution in practice.

Monte Carlo integration estimates an expectation using samples, with typical independent-sample error scaling as $N^{-1/2}$ for finite variance. In Markov-chain Monte Carlo, samples are correlated; the effective sample size is smaller. Metropolis sampling accepts a proposed configuration change with probability

$$P_{\text{accept}} = \min(1, e^{-\beta\Delta E}) \quad (34.11)$$

for a symmetric proposal in a canonical ensemble. Asymmetric proposals require an additional proposal-probability ratio. Check equilibration, autocorrelation, and sampling across distinct metastable regions.

For canonical weights $\pi_i \propto e^{-\beta E_i}$ and symmetric proposal probabilities, detailed balance requires

$$\pi_i P_{i \rightarrow j} = \pi_j P_{j \rightarrow i}, \quad \frac{P_{i \rightarrow j}}{P_{j \rightarrow i}} = e^{-\beta(E_j - E_i)}. \quad (34.12)$$

Metropolis acceptance satisfies the required ratio by accepting energy-lowering moves and accepting energy-raising moves with the Boltzmann factor. This ensures the desired stationary weights when combined with the proposal, but does not guarantee rapid exploration. A chain trapped in one region can look quiet while badly underestimating uncertainty.

34.5 A practical verification checklist

Treat verification as part of solving the problem, not a final cosmetic step. For example, a numerical orbit should approach a known circular solution in the appropriate initial state, show predictable refinement behavior, and conserve the relevant quantities to the expected accuracy. Use the following sequence to make those questions routine.

1. Check the continuum equations, units, parameter regime, and boundary conditions.
2. Reproduce an analytic special case or a trusted limiting solution.
3. Refine time step, spatial grid, domain size, or sample count separately.
4. Monitor relevant invariants and calculate uncertainty rather than relying only on plots.
5. Distinguish numerical artifacts from model limitations and measurement disagreement.

When a check fails, change one thing at a time. If halving the time step changes the result but enlarging the spatial domain does not, the evidence points toward time discretization rather than the outer boundary. If the numerical result converges but still disagrees with experiment, revisit model assumptions and parameter uncertainty. Separating these questions prevents you from spending computation to refine the wrong physical model.

Try it yourself

Run a calculation with two time steps. What change in the results would suggest numerical error? Name one conserved quantity or limiting case you could check.

Chapter 35

Connections across the subject

We can now look backward without flattening the differences between subjects. The same mathematical form can describe a spring, a circuit, a field mode, or a quantum oscillator, but the symbols acquire different physical meanings. These connections are the book's invitation to continue learning beyond its last page. By now the same mathematical patterns have appeared in very different settings. Recognizing them lets you reuse an argument without confusing the physical meanings of its symbols. We close the main text by tracing four recurring ideas: local balance, oscillation, symmetry, and scale. Each is a guide to choosing where a new calculation should begin.

35.1 Conservation from local balance

Imagine drawing a boundary around whatever quantity you want to track: mass, charge, energy, or probability. Its amount inside can change because it crosses the boundary or because a local source creates or removes it. Write that statement before selecting a coordinate system or a constitutive law. When the same symbol appears in different chapters, its meaning must be re-established from context. The recurring forms are the important objects: a local balance, an oscillator, a symmetry, or a small parameter. Always translate the symbols back into the physical quantities before claiming that two equations describe the same physics.

Many laws share the structure

$$\partial_t \rho + \nabla \cdot \mathbf{j} = s. \quad (35.1)$$

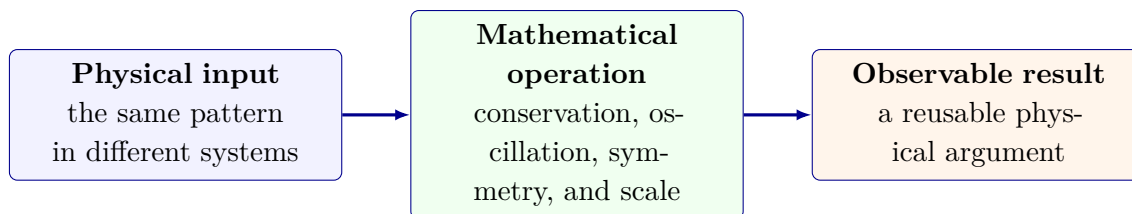


Figure 35.1: Visual model for connections across physics: the physical question becomes mathematics and returns as an observable prediction.

Here ρ is a stored density, $\mathbf{j}$ its flux, and s a source. Integrating over a fixed volume gives

$$\frac{d}{dt} \int_V \rho dV = - \oint_{\partial V} \mathbf{j} \cdot d\mathbf{S} + \int_V s dV. \quad (35.2)$$

Mass continuity, charge conservation, heat flow, and quantum probability conservation are instances. Their different flux laws create different dynamics.

For an insulated boundary with zero normal flux and no sources, the integrated amount remains constant. If the flux is instead $\mathbf{j} = -D\nabla\rho$, the same conservation statement becomes diffusion. If the flux is a velocity times density, it becomes transport. The shared balance law does not make these systems physically identical; the flux relation determines how each one evolves.

35.2 The oscillator as a recurring model

Whenever you find a stable equilibrium, expand the energy around it. The vanishing linear term and positive quadratic term are the invitation to use oscillator mathematics. In several coordinates, diagonalize the quadratic form to discover the independent normal modes.

A spring, an LC circuit, a molecular vibration, an electromagnetic cavity mode, and a free quantum-field mode all reduce to oscillator mathematics in an appropriate regime. Frequencies come from restoring forces and inertia or their generalized analogues. Damping represents coupling to other degrees of freedom; quantization turns normal modes into discrete excitations.

A useful translation table is $m \leftrightarrow L$, $k \leftrightarrow 1/C$, and $x \leftrightarrow Q$ for a spring and an LC circuit. Their energies become $\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ and $\frac{1}{2}L\dot{Q}^2 + Q^2/(2C)$. The variables have different units, but the equations have the same structure. Quantizing a normal mode then adds the common spectrum $\hbar\omega(n + 1/2)$. This is how one solved model reaches from mechanics to circuits, vibrations, and fields.

35.3 Symmetry, gauge freedom, and observables

Ask what a proposed transformation changes physically. Rotating an apparatus can relate distinct physical configurations through a symmetry of the laws. Changing gauge can alter the mathematical representation of the very same fields. Keeping these roles separate helps you identify which quantities should be compared with measurements.

A physical symmetry relates states while preserving the laws. Gauge freedom is a redundancy of description; gauge-related potentials describe the same physical situation when matter fields are transformed consistently. Continuous symmetries generate conservation laws, while symmetry breaking and boundary conditions determine which states occur.

Only observable predictions can be compared directly with experiments. Coordinate components, wavefunction phases, and gauge potentials may be essential calculation tools without being individually observable in every form.

An additive constant in a Newtonian potential changes the displayed energy value but not the force $-\nabla V$ or energy differences. A global wavefunction phase changes amplitudes but not measurement probabilities. These examples teach a practical habit: before assigning physical meaning to a symbol, examine which changes of description leave all predictions invariant. More elaborate gauge theories extend this habit with corresponding transformations of matter fields.

35.4 Scale and effective descriptions

The first decision in a new problem is often the description itself. Estimate the relevant lengths, times, energies, and speeds before choosing equations. A small ratio can justify removing a detail; a ratio approaching one warns that the omitted physics may become important.

The same matter can be modeled as particles, a continuous fluid, a thermodynamic system, or a quantum many-body state. The useful description depends on resolution and the question asked. A hierarchy of small parameters can organize the choice:

Parameter or comparison	Physical significance
v/c	Relativistic kinematic corrections
$GM/(rc^2)$	Strength of relativistic gravitational effects
$\hbar/S_{\text{characteristic}}$	Semiclassical action scale; not a complete criterion for every state
$n\lambda_T^3$	Quantum degeneracy of a dilute particle gas
Mean free path divided by system size	Validity of a local continuum description
Wavelength divided by apparatus size	Wave optics versus a possible ray approximation
Interaction energy divided by a level gap	Suitability of nondegenerate perturbation theory

No one approximation wins everywhere. A good solution identifies the relevant regime and estimates its own neglected effects.

For example, a gas can be dilute enough for classical particle statistics while still requiring a fluid description over distances much larger than its mean free path. A slowly moving object can need quantum mechanics, while a rapidly moving dilute beam can be treated with relativistic particle kinematics. These comparisons answer different questions, so no single small parameter decides every approximation. State what you want to predict, identify the controlling ratios, and let those estimates guide the calculation.

Try it yourself

Choose two systems from different parts of the book that obey an oscillator equation. Identify the changing quantity and restoring effect in each.

Appendix A

Constants, units, and useful identities

A.1 Common constants

Use this appendix after deciding the symbolic structure of a calculation. Carrying symbols until the final substitution makes cancellations and dimensional checks visible. Numerical constants then set the scale rather than hide the reasoning. Distinguish definitions that fix a constant exactly from measured quantities whose quoted digits are only approximate.

Exact SI-defining values are distinguished below from rounded reference values. Rounded values suffice for the calculations in this book and are not a precision metrology table.

Quantity	Symbol	Value in SI units
Vacuum light speed	c	$299\,792\,458\text{ m s}^{-1}$ (exact)
Planck constant	h	$6.62607015 \times 10^{-34}\text{ J s}$ (exact)
Elementary charge	e	$1.602176634 \times 10^{-19}\text{ C}$ (exact)
Boltzmann constant	k_{B}	$1.380649 \times 10^{-23}\text{ J K}^{-1}$ (exact)
Avogadro constant	N_A	$6.02214076 \times 10^{23}\text{ mol}^{-1}$ (exact)
Reduced Planck constant	$\hbar = h/(2\pi)$	$\simeq 1.05457 \times 10^{-34}\text{ J s}$
Gravitational constant	G	$\simeq 6.6743 \times 10^{-11}\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$
Electron mass	m_e	$\simeq 9.1094 \times 10^{-31}\text{ kg}$
Proton mass	m_p	$\simeq 1.6726 \times 10^{-27}\text{ kg}$
Vacuum permittivity	ε_0	$\simeq 8.8542 \times 10^{-12}\text{ F m}^{-1}$

Useful conversions include $1\text{ eV} = 1.602176634 \times 10^{-19}\text{ J}$, $1\text{ fm} = 10^{-15}\text{ m}$, $1\text{ angstrom} = 10^{-10}\text{ m}$, and $\hbar c \simeq 197.33\text{ MeV fm}$. The fine-structure constant is $\alpha = e^2/(4\pi\varepsilon_0\hbar c) \simeq 1/137$. In SI, $\mu_0\varepsilon_0c^2 = 1$; μ_0 is not an independently exact fixed decimal constant under the modern SI definition.

A.2 Integrals and expansions

These formulas are tools to reconstruct, not merely entries to memorize. A Gaussian integral appears whenever energy is quadratic, and differentiating with respect to its parameter generates useful moments. A Taylor expansion replaces a complicated function locally by the lowest terms that matter at the accuracy you need.

For $a > 0$,

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi/a}, \quad \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2a^{3/2}}, \quad (\text{A.1})$$

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}} \quad (n = 0, 1, \dots). \quad (\text{A.2})$$

Common small- x expansions are

$$e^x = 1 + x + x^2/2 + \dots, \quad \ln(1+x) = x - x^2/2 + \dots, \quad (1+x)^a = 1 + ax + \frac{1}{2}a(a-1)x^2 + \dots. \quad (\text{A.3})$$

An expansion is useful only within its convergence or asymptotic regime and with a sufficiently small neglected contribution.

For example, differentiate $I(a) = \int e^{-ax^2} dx = \sqrt{\pi/a}$ with respect to $a > 0$:

$$-I'(a) = \int x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2a^{3/2}}. \quad (\text{A.4})$$

Further differentiation supplies higher even moments. To justify the starting Gaussian integral, square it, regard the product as an integral over the plane, and use polar coordinates: $I(a)^2 = 2\pi \int_0^{\infty} r e^{-ar^2} dr = \pi/a$. A change of geometry turns a difficult one-dimensional antiderivative into an easy definite integral.

A.3 Natural units and restoring dimensions

Natural units reduce clutter by using constants as conversion factors between units. Setting $c = 1$ does not make a physical speed disappear; it means that length and time are measured in compatible units. Restore constants by identifying the dimensions of the physical quantity, not by placing c or $\hbar$ wherever an expression happens to look familiar.

Setting $c = \hbar = 1$ makes time and length inverse-energy quantities, while mass and momentum have energy units. Temperature can also be measured in energy if $k_B = 1$. Restore constants using physical dimensions: $E^2 = p^2 + m^2$ becomes $E^2 = p^2 c^2 + m^2 c^4$, while a Compton length $1/m$ becomes $\hbar/(mc)$. Dimensional restoration may not determine numerical factors or distinguish quantities with identical dimensions.

Appendix B

Visual models and computational experiments

Equations become easier to trust when their consequences can be seen. A visual model is not decoration: it chooses a physical representation, identifies the quantities that should change, and gives the learner an observation to test. The tables in this chapter assign one distinct visual experiment to every preceding physics chapter. They can guide figures embedded in this book, interactive applications in the repository, or both.

B.1 How to use a visual model

For each experiment, identify the state variables and adjustable parameters, make one prediction before changing a control, and compare the result with an invariant or limiting case. A good animation should reveal the mathematical structure: slopes should become rates, fluxes should cross boundaries, eigenvectors should become independent patterns, and probability should become a normalized area or count.

B.2 Mathematical and physical foundations

Chapter	Visual model	Experiment and observation
Scales, dimensions, and measurement	Scaling laboratory	Change length, time, and mass scales while preserving dimensionless ratios; check which predictions survive a change of units.
Calculus, vectors, and spatial geometry	Tangent-to-trajectory lab	Show a moving point, its tangent, velocity, acceleration, and accumulated displacement together; compare local slopes with global motion.

Chapter	Visual model	Experiment and observation
Patterns, modes, and differential equations	Standing-mode lab	Vary the boundary conditions of a string or diffusion interval; observe which modes remain allowed and how each changes in time.
Probability as a model of uncertainty	Random-walk lab	Compare many individual random walks with their average and spread; check that the typical displacement grows with the square root of the step count.

B.3 Classical mechanics

Chapter	Visual model	Experiment and observation
Newtonian motion and conservation laws	Force and momentum arena	Apply impulses and collisions to bodies with visible momentum arrows; verify that center-of-mass motion changes only through external impulse.
Lagrangian and Hamiltonian mechanics	Configuration-space paths	Draw neighboring pendulum paths and highlight the stationary-action path; switch to phase space and observe Hamiltonian contours.
Gravitation, orbits, and rigid bodies	Orbit and inertia explorer	Rotate the inertia ellipsoid and launch circular, elliptical, and escape trajectories; compare angular momentum and energy diagnostics.
Oscillations, coupled systems, and chaos	Mode and sensitivity laboratory	Start coupled oscillators in different modes, then perturb a chaotic pendulum and compare nearby trajectories.

B.4 Continuum physics, waves, and optics

Chapter	Visual model	Experiment and observation
Elasticity, fluids, and transport	Conservation-cell laboratory	Display each cell's incoming and outgoing fluxes; use a shallow-water dam break to test mass conservation, positivity, and the CFL limit.

Chapter	Visual model	Experiment and observation
Waves, sound, and interference	Wave-packet and spectrum view	Superpose pulses, show the Fourier spectrum, and vary dispersion to observe reflection, interference, and packet spreading.
Geometrical and physical optics	Ray and aperture bench	Trace rays through lenses, then replace the aperture with a slit and watch the far-field pattern emerge from Fourier structure.

B.5 Thermal physics and electromagnetism

Chapter	Visual model	Experiment and observation
Thermodynamics	State-space process map	Move a gas through isothermal, isochoric, and adiabatic paths on p - V and T - S diagrams; compare work, heat, and entropy.
Statistical mechanics	Microstate-to-macrostate simulator	Display particles, energy histograms, and the partition function together; vary temperature and watch relative fluctuations change.
Electrostatics, potentials, and matter	Field-line and equipotential map	Move charges and display field arrows and equipotential contours; verify that field lines cross equipotentials normally.
Currents, magnetism, and circuits	Charge-flow circuit bench	Animate charge conservation, voltage drops, magnetic force, and transients while plotting current and stored energy.
Maxwell's equations and electromagnetic waves	Poynting-flow field lab	Show coupled electric and magnetic fields with energy-flow arrows; change polarization and observe transversality.

B.6 Relativity, gravitation, and quantum physics

Chapter	Visual model	Experiment and observation
Special relativity	Spacetime diagram laboratory	Move between inertial frames and display worldlines, light cones, simultaneity slices, and invariant intervals.

Chapter	Visual model	Experiment and observation
Differential geometry and introductory topology	Parallel-transport surface	Carry a vector around loops on a sphere and a plane; compare transport rotation and contractible versus noncontractible loops.
General relativity	Curved-spacetime geodesic map	Launch neighboring geodesics on a metric grid and compare coordinate distortion with measurable separation and lensing.
Quantum principles and mathematical structure	State-vector projection lab	Rotate a state between measurement bases and show amplitudes, projections, and probabilities as geometric components.
Exactly solvable quantum systems	Wavefunction and boundary laboratory	Compare box, oscillator, and hydrogenic states; show nodes, probability density, phase, and quantized energy.
Quantum approximation methods and scattering	Scattering and semiclassical lab	Vary barrier height and wavelength while comparing exact, WKB, and Born-approximation predictions.
Identical particles, entanglement, and quantum information	Correlation and reduced-state display	Show a two-particle amplitude grid, marginal probabilities, and interference before and after tracing out one subsystem.
Quantum computing from bits to algorithms	Circuit and Bloch-sphere laboratory	Start with bits, apply unitary gates, display amplitude interference, and compare ideal measurement histograms with decohering qubits.

B.7 Atoms, materials, nuclei, particles, and cosmology

Chapter	Visual model	Experiment and observation
Atomic and molecular physics	Spectral-line and orbital explorer	Change quantum numbers and field strength while showing orbital density, selection rules, and line splitting.
Condensed matter and collective phenomena	Lattice and band-structure laboratory	Animate a crystal lattice, phonon modes, reciprocal space, and band dispersion; open and close a gap to change transport.
Nuclear physics	Binding-energy landscape	Plot binding energy per nucleon and animate decay or reaction channels; separate energy release from reaction probability.
Relativistic quantum theory and particle physics	Field-mode and interaction diagram	Build a particle as a field excitation, then display vertices, conservation laws, and relativistic dispersion.

Chapter	Visual model	Experiment and observation
Stars, compact objects, and the expanding universe	Stellar and cosmological scale map	Move from hydrostatic stellar balance to expanding cosmic volumes; compare pressure, density, temperature, and timescales.
Computational physics and model validation	Convergence and invariant dashboard	Run one model at several resolutions and display error, stability, conserved quantities, and the difference between visual plausibility and validity.
Connections across the subject	Physics pattern atlas	Follow conservation, oscillation, symmetry, and scale from mechanics to fields or quantum theory.

B.8 A visual experiment template

Every future figure or application should label the physical question, adjustable parameters, predicted invariant or limit, and observation that would falsify the model. For example, the shallow-water visualizer should display volume error and positivity beside the water surface, while a quantum wave-packet visualizer should display norm and regional probability beside the animation. This keeps the visual layer tied to mathematics instead of turning it into an attractive but untestable movie.

Appendix C

Review problems

Try each problem before reading its solution. State assumptions and check dimensions even when the algebra is short.

1. A particle starts from rest under linear drag and a constant force F_0 in one dimension. Find its velocity and the time at which it reaches half its terminal velocity.
2. A uniform solid cylinder rolls without slipping down a height h . Find its final center-of-mass speed and compare it with a frictionless sliding point mass.
3. For $V(r) = -\kappa/r$, find the circular-orbit radius and energy in terms of reduced mass μ and angular momentum ℓ .
4. Two equal masses have the coupled-spring system in the oscillations chapter. If $x_1(0) = A$, $x_2(0) = 0$, and both velocities vanish, find $x_1(t), x_2(t)$.
5. A string of length L is fixed at both ends. What happens to all mode frequencies if tension is multiplied by four while linear density stays fixed?
6. Two ideal polarizers are oriented at 45° to one another. Unpolarized light of intensity I_0 enters the first. Find the final intensity.
7. One mole of a monatomic ideal gas is heated reversibly at constant pressure from T to $2T$. Find Q , W_{by} , ΔU , and ΔS .
8. For N independent nondegenerate two-level systems with gap Δ , find the high- and low-temperature limits of internal energy and entropy.
9. A uniformly charged solid sphere has radius R and total charge Q . Find the electric field inside and outside, and the potential at the center with zero at infinity.
10. A capacitor initially charged to voltage V_0 discharges through resistance R . Find $V(t)$ and total heat dissipated.
11. A vacuum plane wave has peak electric field E_0 . Find average intensity and mean pressure on a perfectly reflecting normal surface.

12. A particle has speed $0.8c$. Find γ , its kinetic energy in units of mc^2 , and its momentum in units of mc .
13. An emitter is static at Schwarzschild radius-coordinate $r_e = 4GM/c^2$. What frequency is measured by a static observer at infinity relative to the emitted frequency?
14. In the infinite well $0 < x < L$, find the ground-state $\langle x \rangle$, $\langle p \rangle$, and $\langle p^2 \rangle$, interpreting the latter through kinetic energy.
15. A harmonic oscillator in its ground state is perturbed by $V' = fx$, with constant force parameter f . Find the exact energy shift by completing the square and compare with first-order perturbation theory.
16. A spin initially in $|+z\rangle$ is measured along $+x$, then along $+z$, with the first outcome recorded but not selected. What is the final probability of $+z$?
17. Trace out either qubit of $|\Phi^+\rangle$. Find the reduced purity and entropy in units of k_B .
18. A radioactive sample has two independent decay channels with rates λ_1, λ_2 . Find its half-life and the fraction of decays in channel 1.
19. Show that a spatially flat matter-only expanding universe has $a(t) \propto t^{2/3}$, taking $a = 0$ at $t = 0$.
20. For the explicit diffusion scheme, insert an error mode $u_j^n = A^n e^{ikj\Delta x}$. Derive its amplification factor and stability bound.

Appendix D

Solutions and reasoning

1. Solve $m\dot{v} = F_0 - bv$: $v = (F_0/b)(1 - e^{-bt/m})$. Thus $v_\infty = F_0/b$ and $t_{1/2} = (m/b)\ln 2$. The velocity approaches the terminal value monotonically for positive F_0, b .
2. With $I = mR^2/2$, energy conservation gives $mgh = 3mv^2/4$, hence $v = \sqrt{4gh/3}$. Sliding gives $\sqrt{2gh}$; rolling stores some energy in rotation. Enough static friction is assumed to enforce rolling.
3. Set $V'_{\text{eff}} = \kappa/r^2 - \ell^2/(\mu r^3) = 0$, yielding $r_c = \ell^2/(\mu\kappa)$. Then $E_c = -\kappa/(2r_c) = -\mu\kappa^2/(2\ell^2)$. The second derivative at r_c is positive, so the orbit is radially stable.
4. Both normal coordinates initially equal A . Therefore $x_1 = A[\cos(\omega_+t) + \cos(\omega_-t)]/2$ and $x_2 = A[\cos(\omega_+t) - \cos(\omega_-t)]/2$. A mass initially at rest can later move through exchange of mode energy contributions.
5. Since $v = \sqrt{T/\mu}$ doubles, every $f_n = nv/(2L)$ doubles. The spatial node locations for each mode do not change.
6. The first transmits $I_0/2$. The second transmits another factor $\cos^2 45^\circ = 1/2$, leaving $I_0/4$.
7. For one mole, $C_V = 3R/2$ and $C_p = 5R/2$. Thus $Q = 5RT/2$, $W_{\text{by}} = R\Delta T = RT$, $\Delta U = 3RT/2$, and $\Delta S = C_p \ln 2 = (5R/2)\ln 2$. Here the initial temperature is denoted by T .
8. At low T , all systems approach the unique ground state: $U \rightarrow 0$ and $S \rightarrow 0$. At high T , each has equal probabilities: $U \rightarrow N\Delta/2$, $S \rightarrow Nk_B \ln 2$. The bounded spectrum explains why the high-temperature energy saturates.
9. Gauss's law gives $\mathbf{E} = Qr\hat{\mathbf{r}}/(4\pi\epsilon_0 R^3)$ inside and $Q\hat{\mathbf{r}}/(4\pi\epsilon_0 r^2)$ outside. Integrating inward from infinity gives $\phi(0) = 3Q/(8\pi\epsilon_0 R)$. Inside, $\phi(r) = Q(3 - r^2/R^2)/(8\pi\epsilon_0 R)$.
10. $V = V_0 e^{-t/(RC)}$. Integrating V^2/R from zero to infinity gives heat $CV_0^2/2$, exactly the initial stored energy.
11. Since $\langle E^2 \rangle = E_0^2/2$, $I = \epsilon_0 c E_0^2/2$. Reflection reverses normal photon momentum, so $\bar{p} = 2I/c = \epsilon_0 E_0^2$. Here p is pressure, not momentum.

12. $\gamma = 1/\sqrt{1-0.64} = 5/3$. Thus $K/(mc^2) = 2/3$ and $p/(mc) = \gamma(v/c) = 4/3$. The invariant check is $(5/3)^2 - (4/3)^2 = 1$.
13. Since $r_e = 2r_s$, $\nu_\infty/\nu_e = \sqrt{1-r_s/r_e} = 1/\sqrt{2}$. This assumes a static emitter, rather than a freely falling one whose motion adds a Doppler effect.
14. Symmetry gives $\langle x \rangle = L/2$ and a real stationary wavefunction gives zero mean momentum. Since $E_1 = \pi^2 \hbar^2 / (2mL^2)$, $\langle p^2 \rangle = 2mE_1 = \pi^2 \hbar^2 / L^2$. Strict operator-domain issues for momentum in a hard-wall interval do not change this kinetic-energy result.
15. Completing the square gives

$$\frac{1}{2}m\omega^2 x^2 + fx = \frac{1}{2}m\omega^2 \left(x + \frac{f}{m\omega^2} \right)^2 - \frac{f^2}{2m\omega^2}.$$

Every level shifts by $-f^2/(2m\omega^2)$. First order vanishes because $\langle x \rangle = 0$ in an unperturbed number state; the leading shift is second order.

16. The x measurement gives either outcome with probability $1/2$. Each x eigenstate gives $+z$ with probability $1/2$, so the final probability is $1/2$. Without the intervening measurement, it would have been 1 for no intervening dynamics.
17. $\rho_A = (|0\rangle\langle 0| + |1\rangle\langle 1|)/2 = I/2$. Then $\text{Tr } \rho_A^2 = 1/2$ and $S_A/k_B = \ln 2$. The global purity is still 1.
18. Survival depends on the sum of rates: $N = N_0 e^{-(\lambda_1 + \lambda_2)t}$. Therefore $t_{1/2} = \ln 2 / (\lambda_1 + \lambda_2)$ and the channel-1 branching fraction is $\lambda_1 / (\lambda_1 + \lambda_2)$.
19. Conservation gives $\rho = C/a^3$. The Friedmann equation implies $\dot{a}^2 = (8\pi G C / 3)/a$, so $a^{1/2} da = \sqrt{8\pi G C / 3} dt$. Integration yields $a^{3/2} \propto t$ and $a \propto t^{2/3}$.
20. Substitution yields $A = 1 + r(e^{ik\Delta x} - 2 + e^{-ik\Delta x}) = 1 - 4r \sin^2(k\Delta x/2)$. Requiring $|A| \leq 1$ for all resolved modes and $r \geq 0$ gives $r \leq 1/2$.

Appendix E

Further study and a route through the material

These established textbook references provide deeper derivations and problems. They are reading pointers, not claims that this book reproduces any particular edition. No specific edition is required for the topic mapping.

Subject	References
Broad foundations	Halliday, Resnick, and Walker, <i>Fundamentals of Physics</i> ; Feynman, Leighton, and Sands, <i>The Feynman Lectures on Physics</i> .
Mathematical methods	Boas, <i>Mathematical Methods in the Physical Sciences</i> ; Arfken, Weber, and Harris, <i>Mathematical Methods for Physicists</i> .
Classical mechanics	Taylor, <i>Classical Mechanics</i> ; Goldstein, Poole, and Safko, <i>Classical Mechanics</i> .
Fluids	Acheson, <i>Elementary Fluid Dynamics</i> .
Optics	Hecht, <i>Optics</i> .
Thermal physics	Schroeder, <i>An Introduction to Thermal Physics</i> ; Reif, <i>Fundamentals of Statistical and Thermal Physics</i> .
Electromagnetism	Griffiths, <i>Introduction to Electrodynamics</i> ; Jackson, <i>Classical Electrodynamics</i> .
Relativity	Taylor and Wheeler, <i>Spacetime Physics</i> ; Carroll, <i>Spacetime and Geometry</i> .
Quantum mechanics	Griffiths and Schroeter, <i>Introduction to Quantum Mechanics</i> ; Shankar, <i>Principles of Quantum Mechanics</i> ; Sakurai and Napolitano, <i>Modern Quantum Mechanics</i> .
Condensed matter	Kittel, <i>Introduction to Solid State Physics</i> ; Ashcroft and Mermin, <i>Solid State Physics</i> .
Nuclear physics	Krane, <i>Introductory Nuclear Physics</i> .
Particles and fields	Griffiths, <i>Introduction to Elementary Particles</i> ; Peskin and Schroeder, <i>An Introduction to Quantum Field Theory</i> .
Cosmology	Ryden, <i>Introduction to Cosmology</i> .

Subject	References
Computational physics	Newman, <i>Computational Physics</i> .

Suggested sequence

First become fluent with dimensions, calculus, Newtonian conservation, oscillators, and the mathematical meaning of boundary conditions. Next connect fields and fluxes through fluid mechanics and electromagnetism, and connect microscopic probabilities to thermodynamics. Then study special relativity and basic quantum mechanics before proceeding to atoms, materials, nuclei, and fields. General relativity and cosmology benefit from all of these foundations.

For each chapter, be able to explain the physical picture, derive at least one central equation, solve a representative problem, and identify a regime in which the model fails. That combination is more useful than memorizing an isolated list of formulas.

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The source is a single self-contained file with no external figures, bibliography database, or shell-escape requirement. It uses common LaTeX packages from a standard TeX Live or MiKTeX installation. From the source directory, compile twice so the table of contents and internal references settle:

```
pdflatex -interaction=nonstopmode -halt-on-error \  
  "Physics Calc.tex"  
pdflatex -interaction=nonstopmode -halt-on-error \  
  "Physics Calc.tex"
```

Alternatively, compile with:

```
latexmk -pdf "Physics Calc.tex"
```

You can also upload the source as the main file in an online LaTeX editor with the listed packages available.